

Collaboration, Social Media, and Competitive Position

Module 2 gave you two frameworks and almost nothing to point them at. This module is the material they were built for. Every tool in this chapter — the team meeting held in four time zones, the intranet nobody reads, the review that cost a company a week, the crowd that did the work for two cents a task — changes who holds power over a firm and where inside that firm value is actually created. So you will read the chapter once, and then read it a second time through the five forces and the value chain.

1

OBJECTIVE 5.1

Understand why effective organizational communication and collaboration is a strategic matter, not an office convenience

2

OBJECTIVE 5.2

Identify the technologies that carry communication and collaboration, and evaluate what each one does to a business

3

OBJECTIVE 5.3

Apply the principles of digital collaboration to coordination, knowledge sharing and decision making inside a real organization

4

APPLICATION 5.4

Read every tool in the chapter as a competitive force acting on a firm and as an activity in its value chain

5

APPLICATION 5.5

Turn all of it into the written analysis this course keeps asking for: rate the forces, locate the value, recommend one thing

This module is built around one deliverable

The written analysis this course asks for has a fixed shape, and it is the same shape every time: apply the five competitive forces to a firm, use the value chain to find at least two places where information technology could create an advantage, recommend exactly one initiative, and write a short comparison against a company you choose. Everything below is arranged to get you there.

1. **Every section carries a mini case.** A short situation, a strip of facts, usually a small exhibit of figures, and two or three decisions to make. The debrief stays shut until you have answered all of them, so the point of the case is worked out rather than read.
2. **Read the short piece first, then work.** The prose in each section is deliberately brief. It exists to arm the activity that follows it, not to substitute for it.
3. **Every option explains itself.** Including the ones you did not choose. Knowing why a plausible answer is wrong is the part that survives into the next case.
4. **The last two sections are the frameworks.** Section 5–4 maps this chapter onto the five forces and the value chain, one tool at a time. Section 5–5 rehearses the whole written deliverable on a hypothetical firm.
5. **Finish with the final challenge.** Situations rather than definitions, scored by objective, so a weak area shows up instead of being averaged away.

Where this module sits in the course objectives

The modules follow the textbook's learning objectives, but the course is graded against a different list of eleven. This table says plainly which of them this module serves and which it leaves to other work, so you always know why you are reading something.

COURSE OBJECTIVE	WHAT THIS MODULE DOES WITH IT
5. Delineate how businesses use internet technology for electronic commerce, communicating with employees and working with business partners	This module's main job. Communicating with employees is the intranet, the employee portal, enterprise search and the internal social network. Working with business partners is the virtual team, the shared document, the videoconference and the extranet-style collaboration that replaced posting drawings back and forth.
3. Discuss how companies use information technology for competitive advantage by changing their marketplace	Also central, and the whole of section 5–4. Network effects raise entry barriers, a platform turns its own users into its product, reviews hand power to buyers, and crowdsourcing replaces a category of paid work. Each of those changes a market rather than improving a firm inside one.
2. Discuss how companies use information systems to help manage internal operations and to make decisions	Also central. Objective 5.1 is coordination: virtual teams, synchronous against asynchronous work, real-time access to one authoritative document, and employee self-service replacing a paper form that cost tens of dollars to process.

COURSE OBJECTIVE**WHAT THIS MODULE DOES WITH IT**

8. Describe the individual and organizational consequences of the use of information technology

Substantial, and unusually blunt here. A generation gap inside one workforce, a shift in what people treat as private, anonymity producing behaviour nobody would sign their name to, an algorithm deciding what a person sees, and a complaint reaching millions before the company has read it.

9. Recognize security breaches and computer crimes

Substantial. Every tool that widens sharing widens exposure: intellectual property leaving in a shared folder, location metadata a photographer forgot to strip, fake reviews, harassment and doxing, and extremist recruitment running on the same platforms marketing does.

7. Recognize a business problem which might be solved using information technology and then work with the information systems team on a solution

Practised end to end in section 5–5, which is a full pass from counted evidence to one recommended initiative with a cost and a measure attached.

1. Describe what a company's information systems department does · **4.** Apply a working knowledge of hardware, software, database technology and computer networks

Supporting. The intranet, the firewall it sits behind, the private network that reaches it from a hotel room, and the integration problem every internal tool creates are all this chapter's, but Modules 1 and 3 are where they are taught.

10. Solve a business problem by creating a database application using SQLite · **11.** Use advanced spreadsheet functions to create a decision support system

Prepared for, not completed, in section 5–5: a runnable sheet and a runnable query put this chapter's questions to a firm's own collaboration and social data. Designing a database and building an application on it belong to the course's own database and spreadsheet work.

6. Outline the life cycle of an information system

Not taught here.

COORDINATE

A team that is never in one room

The right people for a problem are rarely in the same building. Assembling them anyway, and then getting a decision out of them, is what the first half of this chapter is about.

KNOW

What the company already knows

Large organizations lose their own knowledge constantly. Intranets, search, wikis and internal social networks are all attempts to stop that, and most of them fail for the same reasons.

REACH

A public that answers back

The web stopped being a broadcast. Customers now create the content about you, and the same machinery that carries a marketing campaign carries a complaint.

POSITION

Forces and value

Every one of those changes moves a competitive force or an activity in the value chain. Naming which one is the difference between an observation and an analysis.

ONE QUESTION RUNS THROUGH THE WHOLE MODULE

For each tool you meet, ask two things and only two: **which force does it move** — rivalry, new entrants, substitutes, buyer power, supplier power — and **where in the value chain does the work actually happen?** A wiki is not a strategy. A wiki that cuts the time an engineer spends re-deriving something a colleague already knew is technology development, and it shows up as a cost the firm's rivals still pay.

SECTION 5-1A

Why collaboration became a strategic problem

Nobody needs a textbook to tell them that people at work talk to each other. What is worth a chapter is why that talking became a competitive matter: the right people for a problem are almost never in the same building, and assembling them anyway is now something a firm either does well or loses money on.

The problem is assembling, not talking

To stay competitive an organization has to bring together the right combination of people — the knowledge, the skills, the information and the authority to actually decide — quickly and repeatedly.

The traditional answer was the **task force**: a temporary work group with a finite task and a life cycle, pulled together to solve something the existing departments could not. Task forces still exist, and they are slow. Structure and logistics get in the way of people trying to get anything done.

So organizations increasingly build **virtual teams** instead: members drawn from different geographic areas, assembled as needed for a particular project. Membership is fluid. Teams form and disband, size moves up and down, and one person may sit on several at once.

Two examples from the chapter show how ordinary this has become, and neither of them is a technology company doing something clever.

Software development routinely puts the programmers in one country, the project managers in another and the testers in a third, working on one release.

Health care assembles dieticians, physicians, surgeons, pharmacists and social workers from different cities to coordinate the care of a single patient.

What both share is that the specialist a problem needs is no longer limited to whoever happens to practise or program nearby.

The tools come in three families

The chapter groups collaboration tools into three categories, and the grouping is more useful than it looks, because each family answers a different question about a team.

Electronic communication tools carry verbal and written information and move files between people — instant messaging, email, voice mail, blogs, wikis and ordinary websites.

Electronic conferencing tools support richer interaction than a message can carry, such as screen sharing and videoconferencing, where the point is watching something together.

Collaboration management tools run the group itself rather than the conversation: electronic calendars, knowledge management systems, intranets and shared online document systems.

Sort nine everyday tools into those three families, and notice how often the family you would have guessed is the wrong one.

SORT**Which family does each tool belong to?**

Put each tool into the category the chapter places it in; the reason appears as soon as you drop it, whether you were right or not.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Electronic communication tools — Carry verbal and written information, and move files and content between people.

- Email — The chapter's first example of a communication tool: it conveys written information and carries attachments, and it does nothing to manage the group sending them.
- Voice mail — A recorded verbal message waiting for its recipient. It is the oldest asynchronous tool most workplaces still run, and it belongs with the other message carriers.
- A public wiki page — The chapter lists wikis with blogs and static websites as communication tools, because their job is conveying content. That they are jointly written is a fact about how the content arrives, not about what the tool is for.

Electronic conferencing tools — Support information sharing and rich interaction, where the point is doing something together in the moment.

- Screen sharing — Nothing is conveyed that could have been written down instead. The value is that two people are looking at the same thing at the same moment and can point at it.
- Videoconferencing — The richest interaction in the chapter's list: tone, expression and the ability to interrupt, none of which survives a written message.
- A live customer chat window — Instant messaging appears in both of the first two families, and a live chat with an agent sits on the conferencing side because it is a real-time exchange rather than a message left to be collected.

Collaboration management tools — Facilitate meetings and manage the group's activity rather than carrying the conversation itself.

- A shared electronic calendar — It carries no content at all. Its whole function is arranging when the group can be in the same place, which is managing the collaboration rather than conducting it.
- A knowledge management system — It organises what the organization already knows so somebody can find it later. That is group activity management, and the chapter files it here rather than with the message carriers.
- A shared online document system — One authoritative copy that several people can work on is a way of managing an activity. The chapter groups these with intranets and calendars for exactly that reason.

Same time, or not; same place, or not

The distinction that actually decides which tool to reach for is not the family. It is time. **Synchronous** communication is coordinated in time — a call, a live chat, a videoconference — and **asynchronous** communication is not, which covers email, texting and a discussion board.

Meetings of virtual teams take the form of **virtual meetings** and can run either way. When time matters, because a deadline is close or a customer is angry, synchronous media are usually better: every hour of delay is process inefficiency or a dissatisfied customer, and both are expensive.

Cross time with place and you get the four modes collaboration software is built to serve. Step through them and read what each one costs and buys.

COMPARE

Four modes of group interaction

Step through the four combinations of time and place; each one is a real arrangement with its own tools, its own costs and its own failure mode.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Same time, same place — Everyone is in one room, and the technology is in the room with them.

Everyone free at once (And able to reach one building) → Electronic meeting room (Shared screen, shared display) → The argument happens live (Objections raised and answered) → Decision recorded (Written down before anyone leaves)

- The fastest way to close a genuine disagreement, because objections are answered in the moment rather than in a reply tomorrow.
- The most expensive mode by a wide margin once travel is counted, and the cost falls on people rather than on equipment.
- Its failure mode is the meeting that did not need to be a meeting: an hour of several salaries spent on something an email would have settled.

Same time, different places — A synchronous distributed meeting: the clock is shared, the building is not.

A decision is urgent (Waiting a day would cost something) → Videoconference or call (Everyone joins from where they are) → Live discussion (Tone and interruption survive) → Decision, then a written note (Because nobody shared the room)

- This is the mode that replaced most business travel, and the one the chapter says a virtual team lives in when a deadline is close.
- Time zones are the hidden cost. A meeting convenient in three places is unpleasant in the fourth, and that seat is where attendance quietly stops.
- Its failure mode is a decision nobody wrote down, because the shared room that used to produce minutes was never there.

Different times, same place — An electronic meeting facility used as a home base, worked on in shifts.

Work continues in shifts (Not everyone is present at once) → A shared room or board (State is left where it stands) → Each shift picks it up (Reads what the last one left) → Progress without a meeting (Nobody waited for anybody)

- The least discussed of the four and the most familiar in practice: a shift handover, a laboratory bench, a design board a team walks past.
- It buys continuity without coordination, which is exactly what a firm wants when the work is continuous and the people are not.
- Its failure mode is context that lived in somebody's head, so the next shift repeats the reasoning instead of the work.

Different times, different places — The virtual team in its purest form, and the mode most of this chapter is about.

Members across time zones (No shared hour worth using) → Discussion board, email, wiki (Everything asynchronous) → Each contributes when able (The record is the meeting) → Work moves around the clock (Handed on rather than paused)

- The only arrangement that lets a firm use expertise wherever it happens to be, which is the reason virtual teams exist at all.
- It is also the slowest to resolve a real disagreement, because every exchange costs a full turnaround rather than a sentence.
- Its failure mode is a question posted on a Friday afternoon in one time zone and answered on a Tuesday in another, with the deadline in between.

Collaboration software grew out of **groupware**, the class of software that lets people work together more effectively, and the first mainstream product of that kind appeared in 1989.

What has changed since is the price rather than the idea. Most of these capabilities are now free to anyone with a connected device, so the interesting question moved from whether a firm can afford them to whether it can get anybody to use them.

The cost of getting the mode wrong

Choosing synchronous when asynchronous would do wastes people. Choosing asynchronous when a decision is urgent wastes days. The case below puts numbers on both, for a hypothetical firm.

MINI CASE**Nine days to answer one question**

Read the brief and the exhibit, then take the three decisions in order; the firm and every figure attached to it are hypothetical and were invented for this exercise.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

A mid-sized industrial equipment maker builds one product line with an engineering group split across three countries, chosen deliberately because that is where the specialists are. A customer has asked for a change to a pump housing. The answer needs a materials engineer, a manufacturing lead and a cost analyst, none of whom share an office or a working day. The last similar question took nine days. The firm is hypothetical and its figures are invented.

The firm

A mid-sized maker of industrial pumps and their housings

Where the expertise sits

Materials in one country, manufacturing in a second, cost analysis in a third

The question

Can the housing be changed as the customer asked, and at what cost

How it is handled today

Emailed round the group, with each reply waiting for the next working day

What is at stake

The customer is holding an order open while the answer is produced

Exhibit A • How the last nine days were spent — The same question, tracked through one round of email. Every figure here is invented for practice and none of it is reported data.

STAGE	WORKING TIME USED	WAITING TIME USED
Materials engineer reads and replies	40 minutes	1 day
Manufacturing lead reads, asks a clarifying question	25 minutes	2 days
Materials engineer answers the clarification	15 minutes	1 day
Cost analyst prices the change	70 minutes	3 days
Someone assembles the three replies into an answer	35 minutes	1 day
Total	3 hours 5 minutes	8 days

1. The exhibit separates working time from waiting time. What does that separation tell the firm?

- **A.** The engineers are slow, so the fix is to set a reply deadline on internal email — Three hours of work spread over nine days is not slowness. A reply deadline would compress the waiting a little and would not touch the thing producing it, which is that every exchange costs a full turnaround.
- **B.** The delay is turnaround, not effort, so the fix is a mode change rather than more effort — **correct.** Three hours of actual work took nine days because the medium is asynchronous and the question needed several rounds. Anything that removes rounds, or removes the wait inside a round, attacks the real cause.
- **C.** The cost analyst is the bottleneck, so that role should be duplicated in each country — The cost analyst waited longest, but pricing could not begin until the first two had agreed what was being priced. Duplicating that role removes the last wait and leaves the two before it untouched.
- **D.** The question was too complex for email, so it should have gone to a formal task force — A task force is the slow traditional answer the chapter contrasts virtual teams against. Standing one up adds structure and logistics to a question that three people could settle in twenty minutes.

2. The engineering manager proposes one scheduled videoconference with all three specialists instead of the email round. What is the honest cost of that choice?

- **A.** It removes the record, since a live discussion produces nothing anybody can look up later — A live meeting does lose the automatic written trail, and that is a real risk worth managing with a written summary. It is not a cost the firm pays in money or in goodwill, which is what the question is asking about.
- **B.** It requires the dedicated room system the firm has not budgeted for this year — Desktop videoconferencing needs a webcam, a microphone, software and a connection, all of which a laptop already has. The expensive room systems the chapter prices are a separate category of purchase.
- **C.** It moves the work to a synchronous tool, which the chapter treats as the weaker choice — The chapter treats synchronous and asynchronous as suited to different situations rather than ranking them. When a deadline is close, it says synchronous media are usually the better fit.
- **D.** It needs an hour that works in three time zones, and someone pays for that in their evening — **correct.** A synchronous meeting across three time zones has no hour that is convenient everywhere, so somebody joins outside their working day. That cost is real, it is unevenly distributed, and it is why attendance quietly decays.

3. Suppose the change turns out to be routine and this same question arrives roughly twice a month. What does that change about the recommendation?

- **A.** Nothing, because the mode that suits a question does not depend on how often it arrives — Frequency changes the arithmetic completely. A one-off question is worth interrupting people for; a question that recurs twice a month is worth engineering out of existence.

- **B.** It argues for a recurring meeting, so the three specialists are already booked together — A standing meeting is a real improvement over waiting for email, and it still spends three salaries on every instance of a question that has already been answered before.
- **C.** It argues for writing down the answer once, so most future questions skip the specialists — **correct.** A recurring question is a knowledge problem wearing a coordination problem's clothes. Recording the reasoning where the next person can find it is what the chapter's collaboration management tools are for, and it is the answer the next section develops.
- **D.** It argues for hiring a fourth specialist who can answer all three parts alone — One person who covers materials, manufacturing and costing is not a realistic hire, and the whole reason this team is distributed is that each specialism sits where its specialists are.

What this case was really about. Three hours of work took nine days, and no individual was slow. The delay came from the mode: an asynchronous medium multiplied by several rounds of clarification. That is the chapter's point about time being the distinction that matters. Notice also where the case ended up. A synchronous meeting fixes one instance and costs somebody their evening; writing the answer down once fixes every future instance and costs nothing after the first. Coordination problems that recur are almost always knowledge problems, which is exactly why the next section is about intranets and search rather than about meetings.

Videoconferencing, from a theme park to a laptop

Video calling was demonstrated to theme-park audiences in the 1960s and then took about thirty years to matter. What changed was the shopping list.

Desktop videoconferencing needs four things: a **webcam**, a microphone and speakers, the software, and a connection fast enough to carry it.

Most laptops arrive with three of the four, and the software is generally free, so the marginal cost of one more meeting is close to nothing.

Dedicated room systems still exist, and the chapter prices high-end installations in the hundreds of thousands of dollars, which is a capital purchase rather than a download.

That gap is this chapter in one line. A capability that used to require a budget request now requires only a decision.

Collaboration became strategic when it stopped being expensive. Once anybody can assemble the right people from anywhere, the firms that still cannot are losing to the ones that can, and losing on speed rather than on cost.

Three questions on the vocabulary and the judgement before you move on.

CHECK**Teams, modes, and tools**

Three questions on the vocabulary and the judgement; every option explains itself, including the ones you did not pick.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. What distinguishes a virtual team from a traditional task force, in the chapter's terms?

- **A.** A virtual team has no manager, so decisions are made by the members themselves — Nothing in the chapter removes the manager. Virtual teams have the same authority structures as anything else; what changes is where the members physically are.
- **B.** A virtual team is drawn from different locations and its membership stays fluid — **correct.** The chapter defines a virtual team by geographic spread and by fluid membership: teams form and disband as needed, size fluctuates, and people join and leave as the work requires.
- **C.** A virtual team is permanent, while a task force disbands once its task is finished — This inverts the definition. The task force is the temporary form with a finite life cycle, and virtual teams are described as even more dynamic than that, not less.
- **D.** A virtual team uses video, while a task force meets in a conference room — The medium follows from the spread rather than defining it. A distributed team could work almost entirely by discussion board and would still be a virtual team.

2. A support team must resolve an outage affecting a customer right now. Which mode fits, and why?

- **A.** Asynchronous, because a written trail matters most when a customer is affected — The written record is genuinely valuable and can be produced afterwards in a few minutes. Choosing the slower medium to obtain it means the customer pays for the paperwork.
- **B.** Asynchronous, because it lets each specialist contribute without waiting for the others — Letting people contribute independently is exactly what asynchronous work is good for, and it is the wrong strength here, because every round of clarification costs a full turnaround.
- **C.** Synchronous, because delay itself is the cost when a deadline or a customer is waiting — **correct.** The chapter's rule is about the cost of delay: when time is of the essence, delays create process inefficiencies or dissatisfied customers, so synchronous media are better suited.
- **D.** Synchronous, because rich media convey more information per minute than writing does — Richness is a real advantage of conferencing tools and it is not the deciding factor here. A very rich discussion held tomorrow is still held tomorrow.

3. A firm buys a room videoconferencing system for each of its four offices and reports that collaboration has not improved. What does the chapter suggest went wrong?

- **A.** The equipment answers only one of the four modes, and most of the work sits in the others — **correct.** A room system serves a scheduled synchronous meeting between fixed locations. The work of a distributed team is spread across all four modes, and much of it is asynchronous, which no amount of video equipment touches.
- **B.** Room systems are outdated, and desktop videoconferencing has replaced them entirely — Desktop videoconferencing is cheaper and adequate for most meetings, and dedicated systems still exist for reasons of quality and reliability. The problem here is what the purchase covers rather than which kind it was.
- **C.** Four offices are too few for a network effect, so the systems went unused — Network effects matter for social tools whose value depends on how many colleagues participate. A meeting between two offices works perfectly well with two offices.
- **D.** Videoconferencing is a conferencing tool, and collaboration needs communication tools — The three families are not ranked, and collaboration draws on all of them. Buying only from one family is closer to the problem, but the mode analysis is what explains the disappointment.

Inside the firewall: intranets, portals, and search

The previous section ended on a coordination problem that turned out to be a knowledge problem. This section is about the machinery firms build for that: a private web, a search engine that can read it, and a portal that lets an employee serve themselves.

A private web, behind the firewall

An **intranet** is a private network built with web technologies, used to move proprietary information securely inside one organization. It looks and behaves like an ordinary website. The difference is that its pages sit behind the company firewall, so only authorised people reach them with a browser.

Modern intranets build each employee a page rather than serving everyone the same one, which turns one site into several thousand.

The look and the data are shared, so every page is drawn from the same underlying systems and nobody maintains a separate copy for each department.

What differs is entitlement, following the principles of least permission and least privilege: somebody in human resources sees payroll and hiring figures, and somebody in the workshop does not.

The filtering happens at assembly, so the restricted data is absent from the page rather than present and hidden, which is a meaningful difference when a laptop is lost.

That worked while everyone was in the building. Since they are not, most firms let employees reach the intranet through a **virtual private network**, which carries the connection across the public internet inside an encrypted tunnel and delivers it through the firewall. Once connected, a laptop in a hotel behaves as though it were plugged in at the office.

Put the steps of that journey in order, because the order is what makes the arrangement secure rather than merely convenient.

SEQUENCE

How an employee reaches the intranet from a hotel room

Drag the steps into order; each one explains why it sits where it does, and the security of the whole arrangement depends on the sequence.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

A salesperson opens their laptop in a hotel and loads the company's benefits page. Put the steps in the order they actually happen.

1. **The laptop opens a virtual private network connection** — Nothing is requested yet. The first move is establishing the tunnel, because the intranet is not reachable from the public internet at all.
2. **The employee authenticates, and an encrypted tunnel forms across the internet** — The public internet still carries the traffic. What changes is that its contents are unreadable to anything between the hotel and the company.
3. **The tunnelled traffic arrives at the corporate firewall** — The firewall is the boundary. Traffic arriving through the tunnel is treated as internal; the same request arriving without one would be refused here.
4. **The request reaches the intranet server inside the corporate network** — Only now is the employee, in effect, inside the building. The server sees a request from the internal network rather than from a hotel.
5. **The server identifies the employee and their job function** — Modern intranets build a page per person rather than serving one page to everyone, so it has to know who is asking before it can decide what to assemble.
6. **A page is assembled containing only the data that person may see** — Least permission and least privilege applied in practice: the underlying data is shared, and the payroll figures a human resources employee sees are simply absent from a warehouse supervisor's copy.

Three things a corporate intranet is actually for

The chapter names three benefits, and each one replaces a specific older practice rather than adding a new capability.

Real-time access to information replaces the paper document that had to be updated, reprinted and redistributed every time anything changed, which meant several versions were always in circulation at once.

Enterprise search replaces knowing who to ask, by retrieving content across internal documents, databases, applications and cloud storage that a public search engine cannot see at all.

Collaboration replaces sending work back and forth, by letting one person revise a shared drawing or document and put the updated version where the next person will look for it.

The chapter's example of the first is an aircraft manufacturer distributing company news as multimedia over its own intranet, so employees see releases as they happen instead of waiting for a printed

bulletin. Its example of the third is the same firm, sharing three-dimensional models of aircraft components between engineers at different sites and cutting product development time as a result.

Why enterprise search is a different problem

Searching the public web and searching a company are not the same task, and the chapter is explicit about why. Internal content sits in different languages, on different servers, in databases and applications rather than on pages, and increasingly in cloud storage a firm does not own.

A public search engine also has millions of pages linking to each other, which is most of how it decides what is important. A company has no such signal. The result is that as an intranet grows, the information in it becomes harder to find rather than easier, which is precisely the point at which employees give up and ask a colleague instead.

The portal, and the cost of a paper form

Intranets also carry **employee portals**, which support **employee self-service**: administering benefits, managing a retirement plan, submitting the ordinary human resources paperwork of a working life.

The chapter's figures for one form are worth carrying around, because each channel costs roughly an order of magnitude less than the one before it.

On paper the chapter puts processing at roughly twenty to thirty dollars a form, most of it the time of somebody re-keying what an employee already wrote.

Through a telephone system that falls to a few dollars, which is a large saving and still involves a conversation somebody has to be paid to have.

Through intranet self-service it falls to a few cents, at which point an employee filing about fifteen transactions a year stops being an administrative cost at all.

Errors fall as well, because templates are managed centrally so nobody submits last year's version, and entries are checked as they are typed rather than after filing.

MINI CASE

Sixty thousand forms a year

Read the brief and the exhibit, then take the three decisions in order; the employer and every figure attached to it are hypothetical and were invented for this exercise.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

A hospital group employs about four thousand people across six sites. Its human resources paperwork is still filed on paper and re-keyed by a central team, and the group is considering an employee portal that would let staff file it themselves. A consultant has produced one page of figures. The board wants to know whether this is an efficiency project or something larger. The employer is hypothetical and every figure attached to it is invented.

The employer

A hospital group of roughly four thousand staff across six sites

How paperwork is filed now

On paper, posted internally, and re-keyed by a central administration team

What is proposed

An employee portal on the existing intranet, with self-service forms

Who objects

Ward managers, who say clinical staff will not use a computer for paperwork

The decision

Whether the portal is worth funding, and on what grounds

Exhibit A · One year of human resources paperwork — What the consultant counted, with the group's own figures beside the chapter's cost illustration. Every figure here is invented for practice and none of it is reported data.

LINE	TODAY, ON PAPER	WITH AN EMPLOYEE PORTAL
Transactions filed per employee per year	15	15
Total transactions per year	60,000	60,000
Processing cost per transaction	US\$24	US\$0.15
Forms returned because the wrong version was used	3,100	None; templates are held centrally
Forms returned for an entry error	5,400	Checked as the employee types
Average days from filing to confirmation	9	Under 1

1. The finance director says the case rests on the cost per transaction. What does the rest of the exhibit add that the cost line does not?

- **A.** Nothing material; the other lines are consequences of the same processing cost — Treating the other lines as consequences of the cost line is how a business case ends up too small. Rework and delay are separate quantities, and they fall on different people.

- **B.** The returned forms, which are rework the cost line has already counted once — The returned forms are indeed rework, and the exhibit prices processing per transaction rather than per attempt, so a form filed three times is not counted three times anywhere in that line.
- **C.** Eight and a half thousand returned forms and eight lost days, none of it in that cost — **correct.** Roughly eight and a half thousand forms a year come back for a wrong version or a mistyped entry, and confirmation takes nine days. Neither appears in the per-transaction cost, and both land on staff rather than on the administration budget.
- **D.** The transaction count, which is what makes the per-transaction saving worth having — The volume does matter, and multiplying it by the cost is exactly the calculation the finance director already made. The question is what that calculation leaves out.

2. The ward managers say clinical staff will not use a computer for paperwork. What does the chapter suggest about that objection?

- **A.** It is a culture and access question, so it is settled before the build — **correct.** The chapter is blunt that providing a tool and hoping people use it is not enough: the conditions for adoption decide the outcome, and they are a management problem rather than a technical one. Where a nurse can reach a machine mid-shift is part of that.
- **B.** It is resistance to change, and the portal should be made mandatory on a fixed date — A mandate produces filed forms and does not produce adoption. It also converts a solvable design question into a grievance, which is the most expensive way to learn what the objection actually was.
- **C.** It is unfounded, because self-service is easier than paper for everybody involved — Self-service is easier for somebody at a desk with a browser open. Whether it is easier for somebody who has no desk is precisely the thing the ward managers are reporting, and dismissing it discards the best information available.
- **D.** It is a training issue, so the budget should carry a training allowance and proceed — Training helps and it is not the objection. The managers are describing access and working conditions, and a training budget aimed at a device availability problem buys nothing.

3. The board asks whether this is only an efficiency project. What is the strongest case that it is not?

- **A.** It reduces the administration team, which changes the shape of the organization permanently — Headcount reduction is the most visible consequence and still an efficiency argument, only expressed as people rather than as dollars. It is also the version of the case most likely to make adoption harder.
- **B.** It gives every employee a personalised entry point the firm controls and can extend — **correct.** The portal is a per-person, permission-aware entry point that the firm owns. Once it exists, real-time access to information, enterprise search and internal collaboration all have somewhere to live, which is a capability rather than a saving.
- **C.** It moves the group onto web technology, which every competing employer already uses — Matching what other employers have is not a benefit. It is the sort of reasoning the chapter warns against, where a feature list quietly becomes the justification.

- **D.** It removes nine days of delay, which staff will notice more than the money saved — The delay reduction is genuine and welcome, and it is still the same efficiency argument measured in days rather than dollars. It describes the current process running faster, not the group being able to do something new.

What this case was really about. The per-transaction saving is the easiest number to produce and the least interesting. The rework and the nine-day wait are costs the finance line never counted, and they fall on clinical staff rather than on the administration budget. The ward managers' objection is not noise either: it is the adoption condition the chapter spends its third objective on, arriving early enough to design around. And the strategic case is not the money at all. It is that a permission-aware personal entry point, once built, is where search, news and internal collaboration can then be put, which is why the chapter treats an intranet as infrastructure rather than as a project.

The social intranet

Traditional intranets let a small group with editorial rights publish, and everyone else read. **Social intranets** let every user create and update content and connect to whoever wrote it, which is the same shift the next section describes happening on the public web.

Match five ordinary complaints to the capability that actually answers each one, and notice how often the complaint names a symptom rather than the thing that would fix it.

MATCH

The complaint, and the capability that answers it

Pair each complaint with the capability that addresses its actual cause rather than its symptom; every pair explains itself once matched.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Three versions of the safety procedure are in circulation and nobody knows which is current

Real-time access to one authoritative document The chapter's first intranet benefit exists for exactly this. A paper document has to be updated, reprinted and redistributed, so older copies keep working; one page updated in place has no older copies.

We know somebody here has solved this before, but not who or where they wrote it down

Enterprise search across documents, databases and applications This is the problem enterprise search is built for, and why it differs from web search: the content sits in internal systems and cloud storage rather than on linked public pages.

Two engineers are emailing revisions of the same drawing back and forth

A shared workspace where the revised file replaces the old one The chapter's collaboration example: one engineer revises and uploads, the next works from what is there, and the round trip disappears along with the question of whose copy is current.

Staff cannot reach any of this while working away from the office

A virtual private network into the corporate network Availability, not content. The private network carries an encrypted connection through the firewall so a remote laptop can use resources as though it were inside the building.

Only the communications team can publish, so most of what we know never gets written down

A social intranet where any employee can create and update content A traditional intranet grants editorial rights to a few people, which caps how much of the organization's knowledge can ever reach it. A social intranet removes the cap and connects readers to whoever wrote the page.

It takes nine days to confirm a change of hours, and half the forms come back

An employee portal with self-service forms Self-service moves the filing to the person who has the facts, checks entries as they are typed, and holds one central template so the wrong version cannot be submitted.

An intranet is not a website with a password. It is the firm's attempt to hold its own knowledge in one authoritative place, reachable from anywhere, visible only to the people entitled to it, and findable once it is there. Every one of those four requirements is where a real deployment fails.

CHECK

Intranets, portals, and search

Three questions on what this machinery is and is not; every option explains itself, including the ones you did not pick.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. What actually makes an intranet an intranet, rather than a website with a login?

- **A.** It runs on hardware the company owns rather than on rented cloud infrastructure — Ownership of the hardware is not the definition. The chapter describes intranets integrating cloud resources, and a private network can perfectly well run on infrastructure the firm rents.
- **B.** It carries only internal information, and never anything a customer could also see — What the content is about does not decide the category. A firm may publish the same product information internally and externally; where it sits relative to the firewall is what differs.
- **C.** It is written by employees rather than by a communications department — Who writes the content distinguishes a social intranet from a traditional one. Both are intranets, so this cannot be what makes something an intranet in the first place.
- **D.** It is built with web technologies and sits behind the company firewall — **correct.** The chapter defines an intranet as a private network using web technologies for the secure transmission of proprietary information, with the pages behind the firewall so only authorised users reach them.

2. Why can a company not simply run a public web search engine over its own content?

- **A.** Public engines refuse to index content that sits behind a firewall for legal reasons — The obstacle is practical rather than legal: content behind a firewall is not reachable by an outside crawler at all, and there is no refusal involved.
- **B.** Internal content is in too many languages for a public engine's ranking to work — Multiple languages are one of the difficulties the chapter names, and a public engine handles many languages routinely. It is not what makes the problem different in kind.
- **C.** Internal content sits in databases and applications, with no links to rank it by — **correct.** The chapter contrasts the two directly: enterprise search must retrieve from documents, databases, applications and cloud storage, none of which is a linked page, so the signal a web engine ranks by does not exist.
- **D.** Public engines update too slowly for information that changes several times a day — Freshness matters and it is not the distinguishing difficulty. A public engine can recrawl frequently; what it cannot do is see inside a database or an internal application.

3. A firm computes its portal business case as fifteen transactions times four thousand staff times the saving per transaction. What is the most important thing that calculation leaves out?

- **A.** The cost of building and running the portal, which offsets the annual saving — A business case that omits the build cost is incomplete, and every reader already knows to ask for it. It is the expected omission rather than the interesting one.
- **B.** The rework and the delay, which fall on employees — **correct.** Forms returned for a wrong version or a mistyped entry are counted nowhere in a per-transaction figure, and neither is the wait for confirmation. Both are real costs, and both land on staff rather than on the department doing the arithmetic.
- **C.** The employees who will not use it, who reduce the volume the saving applies to — Partial adoption does reduce the realised saving, and it is a reason to discount the estimate rather than something the estimate structurally cannot see.
- **D.** The training and change work, which the chapter says decides whether it is adopted — Change work decides whether the saving arrives at all, which makes it a risk to the whole case. The question asks what the calculation leaves out of the benefit, and rework and delay are benefits it never counted.

The web that answers back

Everything in the first half of this chapter is a firm talking to itself. This half is about what happened when the web stopped being a broadcast, and the people a company used to talk at started producing the content about it.

One change, and everything downstream of it

Until the early 2000s the web was treated as a one-way medium. Some organizations made content; everybody else consumed it. The chapter calls that arrangement Web 1.0, and it holds a strict line between the two roles.

Web 2.0 names the dynamic applications that broke the line. They let people collaborate and share online, which turns a user from a consumer of content into a creator of it. The chapter's own comparison is the sharpest one available: a television network's site carries what the network made, while a video-sharing site carries **user-generated content** and would be empty without it.

The same contrast holds for reference material, and businesses adopted the pattern almost immediately once they saw what it bought them.

Reference works changed hands: a traditional encyclopaedia commissions professionally researched articles, while an online one is written and edited by its own readers, and the chapter notes the older publisher stopped printing in 2012.

Retailers took the cheapest version of the deal, adding customer reviews to product pages and gaining a channel for customers, better decisions for the next shopper, and more traffic, without writing any of it.

Platform businesses took the whole premise, matching people who have something with people who want it, which leaves the platform owning no inventory and producing no content.

Sort eight postures into the era each belongs to, using the shift the chapter tabulates.

SORT

Which era does each posture belong to?

Sort each posture into the era the chapter's comparison places it in; the reason appears as soon as you drop it.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Web 1.0 posture — A strict line between the few who create content and the many who consume it.

- A broadcaster's site carrying only what the broadcaster produced — The chapter's own example of the older arrangement: the organization supplies the content and the audience receives it, which is the whole of the relationship.
- Professionally researched articles commissioned and paid for by a publisher — Expertise bought in advance and published outward. The chapter contrasts this directly with an encyclopaedia its own readers write.
- Finding information by searching for keywords yourself — The chapter's shift table pairs search with recommendation. Searching alone is the earlier posture, where the user does all the locating.
- A site whose content only its technical staff can change — The chapter summarises the older era as one where technically skilled people ruled, because publishing required them.

Web 2.0 posture — The line is gone: users create, share and connect, and the site depends on them doing so.

- A video site whose catalogue was uploaded by its viewers — User-generated content in its purest form. Remove the users and there is no product left, which is exactly what distinguishes this from a broadcaster's site.
- An encyclopaedia article edited by anyone who reads it — The community both produces and corrects the content, which is peer production and the reason the chapter treats it as the defining Web 2.0 case.
- Finding a restaurant because people you know recommended it — Receiving and giving recommendations is the later half of that same pair, and it makes other people rather than an index the route to information.
- A retailer's product page carrying reviews written by its customers — The firm supplies the catalogue and its customers supply the judgement. The chapter names this as a business deliberately incorporating Web 2.0 into its model.

Social media is the part of Web 2.0 that matters here

Applications that embody the collaborative and sharing values of Web 2.0 are classed as **social media**, or social software: they let people communicate, interact and collaborate. That is the whole category, and it is broader than the handful of large sites people picture.

What follows from that is societal rather than technical, and the chapter names three shifts.

How people look for information changed: a network of acquaintances is now consulted where a reference work used to be, which is a change in whose judgement is trusted.

What people treat as private shifted markedly, and the chapter is unsentimental about the consequences of publishing without considering who eventually reads it.

Every conversation now generates data, which the platforms use and which the organizations operating on those platforms increasingly use as well.

The workplace inherited the expectations

The people who grew up on these tools now work. The chapter reports a set of changed expectations that has more to do with employment than with software.

A portfolio career instead of a single long job, which changes how long a firm has to capture what an employee knows before they take it somewhere else.

Loaning talent rather than being a resource, a shift in self-description that shows up in what people expect to be consulted about.

Working across the old boundaries, so that a firm collaborates with customers and partners with rivals rather than treating industry lines as walls.

The managerial consequence is concrete. A firm that blocks social tools at work will find that people route around the block or take a job somewhere that does not have one. Whether that is fair is a separate question from whether it is true.

MINI CASE

Should the customers be allowed to write?

Read the brief and the exhibit, then take the three decisions in order; the firm and every figure attached to it are hypothetical and were invented for this exercise.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

A specialist outdoor equipment retailer sells through a catalogue site it writes entirely itself. Its product descriptions are careful and its returns rate is high, because customers buy boots and tents that turn out to be wrong for them. A proposal on the table would add customer reviews and photographs to every product page. The marketing director is against it. The firm is hypothetical and every figure attached to it is invented.

The firm

A specialist retailer of outdoor clothing and camping equipment

How its pages are written now

Entirely in-house, by two staff writers working from supplier data

The problem

A high returns rate, concentrated in fit and suitability rather than in faults

The proposal

Customer reviews and photographs on every product page

The objection

The marketing director expects negative reviews to cost sales

Exhibit A · What the returns log says — Twelve months of returns, grouped by the reason the customer gave. Every figure here is invented for practice and none of it is reported data.

REASON GIVEN FOR THE RETURN	SHARE OF RETURNS	WHAT THE PRODUCT PAGE SA
Sizing ran small or large for this brand	34 percent	The supplier's size chart, unedited
Not suitable for the conditions the buyer had in mind	26 percent	A general description of the product's purpose
Heavier or bulkier than expected	17 percent	A weight in grams, with nothing to compare it to
Colour differed from the photograph	12 percent	One supplier photograph on a white background
Faulty or damaged	6 percent	Not a description problem
Changed their mind	5 percent	Not a description problem

1. The exhibit is a returns log. What is the argument for reading it as an argument about reviews?

- **A.** Returns are expensive, and reviews are the cheapest way to reduce any cost — Cost alone does not point at a solution. Plenty of expensive problems are not fixed by reviews, and the exhibit is doing more work than establishing that returns cost money.
- **B.** Nearly nine in ten returns come from information the firm's own writers cannot supply — **correct.** Sizing against other brands, suitability for particular conditions, real bulk and real colour are all things an owner knows and a staff writer working from supplier data does not. That is eighty-nine percent of returns.
- **C.** Customers who return an item are the most likely to write a review afterwards — There may well be a relationship, and it cuts the wrong way for this argument: reviews written mostly by disappointed buyers would be the marketing director's case rather than the proposal's.
- **D.** A high returns rate signals poor products, which reviews would help the firm identify — Only six percent of returns are faults. The log describes a description problem rather than a quality problem, and reading it as the latter misses what it is showing.

2. The marketing director says negative reviews will cost sales. What is the strongest response the chapter supports?

- **A.** Negative reviews can be removed, so the risk is manageable with a moderation policy — Removing unfavourable reviews is the practice the chapter treats as undermining trust in review systems altogether, and it destroys the credibility that makes the good reviews worth anything.
- **B.** Negative reviews are rare enough on specialist products that the risk is small — Nothing supports this, and a firm whose returns are driven by fit and suitability should expect a substantial minority of disappointed owners to say so.
- **C.** A larger body of reviews helps buyers choose better, and that is what draws visitors — **correct.** The chapter's account of a retailer adding reviews is that a larger number of reviews helps other customers make better decisions and attracts more visitors to the site, which is the return the firm is buying.
- **D.** Buyers discount reviews heavily, so their effect on sales in either direction is slight — Treating reviews as ignored would remove the benefit along with the risk. The whole proposal rests on buyers acting on what other buyers report.

3. What should the firm expect to have to do that it does not do today?

- **A.** Monitor and respond to what appears on its own pages, as a continuing cost — **correct.** The chapter is explicit that a page carrying user content is not free to run: it has to be watched and answered, and the firm has to decide where the line between unwelcome and unacceptable sits before anything appears on it.
- **B.** Employ more staff writers, because review pages need more editorial oversight — More writers is the opposite of what the proposal does. The point is that owners supply the information the writers cannot, so the writing load falls rather than rises.
- **C.** Verify each reviewer's purchase, since unverified reviews carry no weight with buyers — Verification raises trust and is a design choice rather than a requirement. Plenty of review systems run without it, and the chapter's concern is with reviews that are deliberately fabricated rather than merely unverified.
- **D.** Reduce the returns window, since better information should mean fewer legitimate returns — Shortening the returns window punishes customers for a problem the firm created and would not follow from better information. It also has nothing to do with reviews.

What this case was really about. The returns log is the evidence, and it says the firm's own writers cannot produce the information its buyers need: how a brand's sizing compares, whether a tent suits particular conditions, how heavy something really feels. Owners know all of it. That is the trade Web 2.0 offers a business, stated exactly: you get content, judgement and traffic you did not pay for, and you give up control over what is said on your own pages. The marketing director's fear is real and is priced into the deal rather than avoidable, and the continuing cost of the arrangement is not the reviews themselves but the watching and answering they oblige the firm to do.

What is still a forecast

Two ideas in this chapter have not arrived, and it is worth being precise about which parts have.

The **semantic web** is a set of design principles that would let computers index pages by meaning rather than by keywords, so a search for what eats penguins returns an answer rather than pages containing three words.

Search engines ask **webmasters** to add structured markup for exactly this reason, and one large engine's knowledge panels are a visible result. The full vision remains largely unrealized, and may stay that way on grounds of complexity, feasibility and privacy.

Web 3.0 is contested even as a term. Some frame it around mobility, some around context — content filtered by time, location and activity — and some around artificial intelligence extending what the web can do. Treat it as a set of competing forecasts rather than a description of anything.

The shift that matters is not technical. When the audience became the authors, a firm lost its monopoly on what is said about it, and gained a source of labour, content and judgement it does not employ. Every remaining section of this chapter is a consequence of that one trade.

COMPLETE

The vocabulary of the evolving web

Choose the term that completes each sentence; each blank explains why the other options do not fit.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

- Content created and uploaded by a site's own visitors, rather than by the organization running the site, is **user-generated content**, and a video-sharing site would have nothing to show without it. Peer production and collective intelligence describe how a community produces something jointly, and crowdsourcing is work a firm distributes and usually pays for. This term is simply about who created the content.
- Applications that let people communicate, interact and collaborate, embodying the collaborative and sharing values of the later web, are called **social media**, which is a whole category rather than a handful of large sites. The other three are specific tools from earlier in the chapter. This term is the category that covers them when their defining feature is social participation.
- A set of design principles intended to let computers index pages by meaning rather than by keywords is the **semantic web**, which the chapter describes as largely unrealized. A knowledge graph is one search engine's partial implementation of the idea, and the contextual web is one proposed reading of what comes next. The design principles themselves have their own name.

- The person responsible for creating and maintaining a site's pages, and the person search engines ask to add structured markup, is the **webmaster**, which is a role rather than a technology. The chapter names this role specifically when it explains how the semantic web would have to be built: page by page, by the people who maintain them.
- The label given to the next wave of internet technologies, variously framed around mobility, context or artificial intelligence, is **Web 3.0**, and the chapter treats it as a forecast rather than a description. Web 2.0 has already happened and is the era this section describes. The other two name single aspects of what various people expect the next wave to be about.

Three questions on the era, the vocabulary and the forecasts before you move on.

CHECK

Eras, categories, and forecasts

Three questions on what actually changed and what has not yet; every option explains itself, including the ones you did not pick.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. A firm says it has adopted Web 2.0 because it redesigned its site and added a news feed it writes itself. What is wrong with that claim?

- **A.** Nothing, provided the feed is updated frequently enough to count as dynamic — Frequency of publication was never the distinction. A daily bulletin and a monthly one are both the organization talking and the audience listening.
- **B.** The site is still one-way, since the firm creates and the visitors consume — **correct.** The shift the chapter describes is in who creates the content. A redesigned site with a firm-written feed keeps the strict line between creator and consumer that defines the earlier era.
- **C.** A news feed is a communication tool, and Web 2.0 requires conferencing tools — The three families of collaboration tools belong to the first half of the chapter and do not classify eras. A conferencing tool used one-way would be just as one-way.
- **D.** The claim needs a social network account before the firm can make it — Presence on a particular platform is not the test. A firm could run reviews, a community forum and a wiki entirely on its own site and have changed the relationship completely.

2. Which statement about the semantic web is accurate as the chapter describes it?

- **A.** It has replaced keyword search at the major engines, which now index by meaning — Keyword matching remains the foundation of web search. Meaning-based indexing is the ambition, and the chapter is careful to say the vision has not been reached.
- **B.** It was abandoned because privacy regulation made structured markup unworkable — Privacy is named as one of the reasons the full vision may never be realized, alongside complexity and feasibility. Nothing was abandoned, and markup is in wide use.

- **C.** It is largely unrealized, though structured markup has produced visible results — **correct.** The chapter calls the semantic web largely unrealized while pointing at what has worked: engines encourage structured markup, and one engine's knowledge panels show computers getting closer to the meaning of a page.
- **D.** It is a proprietary standard that one search engine developed and then published — The idea was proposed by one of the inventors of the web as a set of open design principles, not developed inside a company as a proprietary standard.

3. A manufacturer bans social media on company devices to protect productivity. What does the chapter suggest will follow?

- **A.** Productivity will rise, since the chapter treats social tools as a distraction at work — The chapter makes no such claim. It treats these tools as a source of culture, customer relationships, recruitment and competitive insight, which is why banning them is not a free choice.
- **B.** Staff will route around the ban or leave, and the firm loses the tools' benefits too — **correct.** The chapter says people will find creative workarounds or move to an employer that does not have the ban, and that embracing these tools is argued to be good for business as well as for retention.
- **C.** Nothing will change, because work accounts are separate from personal accounts — The chapter's concern is precisely that the separation does not hold: people use whichever tools they are used to, which is the consumerization the later sections describe.
- **D.** The firm will need a written policy before the ban can be enforced at all — A written policy is good practice and is a matter of how a ban is implemented. It says nothing about what happens once it is.

Communicating and cooperating in public

The chapter sorts social media by the job it does: communicate, cooperate, collaborate, connect. This section takes the first two. Communication is a firm putting something out and hearing back. Cooperation is the arrangement where one person's contribution improves everybody else's outcome, including the contributor's.

Three ways a firm talks in public

Three tools carry almost all of it, and they differ in length, in audience and in how long the exchange stays open.

A blog is an online diary of dated entries, long enough to explain something, published on a site the author controls; firms use one because it reads as more approachable than a press release.

Microblogging broadcasts short status updates in near real time to anyone who chooses to follow, with topics marked by a **hashtag** and a topic tagged faster than others said to be **trending**.

Instant messaging is real-time written conversation, and it has absorbed internal team channels, customer-facing live chat, and automated assistants that gather details before passing somebody to a human.

The chapter takes one criticism of blogging seriously: publishing without editorial process brings news out fast and produces what one critic calls the amateurization of journalism, with unverified sources and visible bias. Answering customers on a microblog has the mirror-image cost, since the channel that signals attentiveness hands critics the same megaphone.

EXPLORE

Six public tools, four questions each

Open each card and read all four panels; the same four questions are asked of every tool, which is what makes them comparable, and no real company is named.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Blogs · Long-form, chronological, owned

- **What it does:** Publishes dated entries at whatever length the subject needs, on a site the author controls, usually with comments or a linked discussion.

- **What a firm gets from it:** A voice that reads as a person rather than as a communications department, and a place to explain a decision at length before anybody else frames it.
- **What it does not solve:** It does not reach anybody by itself. A blog with no audience is a filing cabinet, and building the audience is a separate and much harder job.
- **Where it goes wrong:** Speed at the expense of verification. The chapter's charge of amateurization is that corners get cut to publish first, and a company blog carries the firm's name when they are.

Microblogging · Short updates, public by default

- **What it does:** Broadcasts brief status updates in near real time to anyone who has chosen to follow, with topics marked by hashtags and repostable by recipients.
- **What a firm gets from it:** A public channel for answering customers where other customers can see it happening, which is a signal about the firm as much as an answer to one person.
- **What it does not solve:** It cannot hold an explanation. Anything requiring context has to live somewhere else, with the post serving only as a pointer to it.
- **Where it goes wrong:** The channel belongs to everybody. A hashtag a firm creates can be taken over by its critics within hours, and the chapter's example of exactly that involved a police department.

Instant messaging · Real-time written conversation

- **What it does:** Carries synchronous written exchanges between two people or a group, now including team platforms organised into channels and customer-facing live chat.
- **What a firm gets from it:** Immediate feedback without booking anybody's time, and a customer service channel that costs far less per conversation than a telephone line.
- **What it does not solve:** It produces no durable record anybody will find later. Decisions made in a channel are lost to the next person unless somebody writes them somewhere else.
- **Where it goes wrong:** Everything becomes urgent. A medium built for immediate answers tends to be used for questions that did not need one, which is the interruption cost distributed teams complain about.

Media sharing and webcasting · Host once, consumed anywhere

- **What it does:** Publishes images, video, audio or presentations on a hosting service, where others can view them in place or embed them in their own pages.
- **What a firm gets from it:** Reach the firm did not pay for. An embedded video improves somebody else's page and gains an audience in the same movement, which is cooperation working as advertised.
- **What it does not solve:** It does not give the firm control of the surrounding context. Content is consumed next to whatever the embedding site put there, including a competitor.
- **Where it goes wrong:** Live video in particular. Content appears faster than anyone can review it, and the chapter is clear that continuously monitoring a live broadcast is close to impossible.

Tagging and geotagging · Metadata added by hand and by machine

- **What it does:** Attaches descriptive keywords, and sometimes coordinates, to content so it can be categorised, searched and placed on a map.
- **What a firm gets from it:** Findability. Photographs, videos and documents are almost unsearchable without it, and a tag cloud shows the shape of a collection without anyone reading it.
- **What it does not solve:** It does not make tags consistent. Ten people tag the same thing ten ways, which is why an untended tag set degrades rather than improves.
- **Where it goes wrong:** Location metadata attached without anybody deciding to attach it. The chapter's case is a photograph that revealed where a man in hiding was, because nobody thought to strip it.

Social bookmarking and cataloging · Shared collections, shared categories

- **What it does:** Lets people save and categorise links or items publicly, building shared categorisation systems that grow more useful as more people contribute.
- **What a firm gets from it:** Inside a firm, a map of who knows what. The chapter's point is that shared bookmarks expose the islands of expertise an organization chart does not show.
- **What it does not solve:** It does not evaluate anything. A heavily bookmarked page is popular rather than correct, and the two are easy to confuse when you are in a hurry.
- **Where it goes wrong:** It depends entirely on participation. Below a certain number of contributors a shared collection is worse than one person's own bookmarks, and nobody returns to it.

Cooperation, and why it is not collaboration

Cooperation creates a win-win: one participant's success improves the chances of the others. Nobody has to work on the same thing at the same time, which is what distinguishes it from collaboration.

Media sharing is the clearest instance of it working.

An embedded video hosted on a sharing site makes somebody else's article better and brings the video an audience, and neither party had to negotiate anything at all.

Webcasts extend the same logic to on-demand audio and video, which organizations use for shareholder meetings, training, road shows and recorded lectures.

Subscription feeds, such as **RSS**, close the loop by notifying readers of new content, so nobody has to remember to check forty sources by hand.

Live video is the newest form and the hardest to govern. Broadcasting as it happens lets a firm answer questions in the moment, and it also removes every check between something occurring and an audience seeing it.

The chapter is blunt that continuously monitoring live broadcasts is close to impossible for the platforms themselves, and it cites a murder streamed live in 2016 to make the point. A firm going live is accepting the same asymmetry at a smaller scale.

Metadata is the quiet part

Metadata is data about data: who, where, when, why. Some is captured automatically — a document's author and save time, a photograph's date, shutter speed and coordinates — and some has to be added by a person.

The added kind is a **tag**, and tags are what make user-generated content findable at all, because photographs and videos categorise themselves no better than they title themselves. A **tag cloud** displays them with size standing in for frequency, so the shape of a collection is visible at a glance.

Geospatial metadata is called a **geotag**, and it is where this gets serious. Coordinates attached to a photograph put it on a map, which is how a mapping service can show pictures, reviews and landmarks it never produced.

It is also how a photograph of a man in hiding, posted by a journalist who forgot to strip the location, led police to the place it was taken.

MINI CASE

The photographs that gave away a site

Read the brief and the exhibit, then take the three decisions in order; the firm and every detail attached to it are hypothetical and were invented for this exercise.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

A civil engineering consultancy posts progress photographs from its projects on a public media-sharing account, which its marketing team says has been the single most effective thing it does. A client's security officer has just called to say the photographs reveal the location and layout of a site the client had asked to keep unpublicised. Nobody at the consultancy typed a location anywhere. The consultancy is hypothetical and every detail here is invented.

The firm

A civil engineering consultancy of about ninety staff

What it posts

Progress photographs taken on site with ordinary smartphones

Why it posts them

The account is its main source of enquiries from new clients

What the client says

The photographs disclose the location and layout of a site meant to stay unpublicised

What nobody did

Nobody typed a location, a client name, or an address into any post

Exhibit A · What was in the files, and where it came from — The information the client's officer was able to recover, and how each piece got there. Every detail here is invented for practice.

WHAT WAS RECOVERABLE	HOW IT GOT INTO THE FILE	DID ANYONE CHOOSE TO PUB
Latitude and longitude of each photograph	Attached by the phone when the picture was taken	No
Date and time of each visit	Attached by the phone when the picture was taken	No
Phone model, and therefore which employee	Attached by the phone when the picture was taken	No
Site layout and equipment on the ground	Visible in the photographs themselves	Yes, by choosing the shot
Sequence of construction across four months	Reconstructed from the dates across the set	No
A hashtag naming the type of structure	Typed by the marketing team	Yes

1. Four of the six lines were published by nobody. What is the general lesson the chapter draws from that?

- **A.** Photographs should not be published from any site a client has asked to keep private — That would resolve this case and states a policy rather than a lesson. It also gives up the firm's best source of enquiries to avoid a problem that has a narrower fix.
- **B.** Some metadata is captured automatically, so publishing content publishes more than the content — **correct**. The chapter distinguishes metadata captured automatically from metadata a person adds, and geotags are its example of the former becoming a disclosure nobody intended.
- **C.** Smartphones are unsuitable for professional photography and a camera should be used instead — A dedicated camera records the same categories of metadata, and many record coordinates too. The device is not what produced the exposure.
- **D.** Hashtags are the real risk, since they are the part a person deliberately typed — The hashtag was chosen deliberately and is the one line the firm can already see. The difficulty is with the five it could not.

2. The marketing team proposes deleting the account. What is the strongest objection?

- **A.** Deletion does not remove copies already downloaded, so the exposure persists regardless — This is true and it is an argument about what deletion achieves rather than about whether to delete. The exposure has to be handled either way.
- **B.** It removes the firm's main source of enquiries to solve a problem with a narrower fix — **correct.** The facts say the account is the firm's main source of enquiries. Closing it converts a fixable metadata problem into a permanent loss of the channel, which is the cooperation benefit the chapter describes being thrown away.
- **C.** The client asked for the site to be unpublicised, not for the whole account to be closed — The client's request does define the scope of the harm, and staying inside a client's request is not by itself a reason to reject a firm's own precaution.
- **D.** The photographs are the firm's own intellectual property and deleting them wastes an asset — The photographs are indeed the firm's, and no asset argument outweighs a client disclosure. This reason would not survive the meeting.

3. What should the consultancy change so that this cannot happen again?

- **A.** Ask each client at the start whether photographs may be published from their site — Asking permission is necessary and covers only the part somebody chose to publish. It does nothing about coordinates a phone attached without anybody knowing.
- **B.** Have a second person review every photograph before it is posted publicly — A second reviewer looking at the picture sees what the first reviewer saw. The disclosure was in the file rather than in the image, so another pair of eyes on the image catches nothing.
- **C.** Strip location metadata before publication, and confirm publication rules per client — **correct.** One change removes the automatic metadata, and the other governs the deliberate content. The case needs both, because the two lines of the exhibit have entirely different causes.
- **D.** Post photographs only after a project has finished and the site has been handed over — Delaying publication reduces the operational risk during construction and leaves the location, the dates and the sequence just as recoverable afterwards.

What this case was really about. This is the chapter's geotag example with a firm attached to it. The important distinction is the one the exhibit is built around: metadata a person typed against metadata a device attached. The firm can review the first by reading its own posts, and it cannot see the second at all without going looking. Notice too what the case refuses to do. Closing the account would solve the disclosure and destroy the cooperation benefit that made the account worth having, which is nearly always the wrong trade, and the fix that survives is narrow, mechanical and applied before publication rather than after.

Sharing what you found, and what you own

Two more cooperative forms round out the section, and both work on the same principle: individually useful, collectively far more useful.

Social bookmarking lets people share the pages they found worth keeping, and the categories they build together are called a folksonomy, so each additional participant makes the shared set more complete for everybody.

Social cataloging does the same for collections rather than links — books, music, academic citations — which is why researchers use shared citation tools to assemble reference lists they would otherwise each build alone.

Both have a use inside a firm that the chapter is specific about. Bookmarking maps the islands of knowledge in an organization, which makes it possible to find the person who already knows something. Cataloging structures the volume of information a firm holds about suppliers, complaints and products so it can be reached later.

MATCH

Which kind of metadata is this?

Pair each example with the term the chapter gives it; every pair explains itself once matched.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

The author, creation time and last-saved time stored inside a document file

Metadata captured automatically The chapter's own example of metadata a person never enters. It is written by the software as a by-product of ordinary use, which is why nobody remembers it is there.

A keyword a person types onto a photograph so it can be found later

A tag The manually added kind. The chapter's point is that content which cannot describe itself needs somebody to describe it, and that this is what makes user-generated content searchable at all.

The latitude and longitude recorded when a picture was taken

A geotag Geospatial metadata, which is what allows content to be placed on a map, and what allowed a mapping service to show photographs and reviews it never produced.

A display where a word's size shows how often it was used

A tag cloud A way of visualising a set of tags so the important or frequent ones are immediately visible, without anybody reading the collection they came from.

The shared categorisation that emerges as many people bookmark and label pages

A folksonomy Categories built by users rather than imposed by a librarian. Its value grows with participation, which is why the chapter treats social bookmarking as a network-effect application.

A shared catalogue of academic citations that researchers build and reuse

Social cataloging The same idea applied to collections rather than links. Researchers assemble reference lists from a catalogue many people contributed to instead of each building one alone.

Communication tools change what a firm can say and who can answer. Cooperation tools change what a firm gets for free. Metadata is the connective tissue under both, and it is the part that leaks, because most of it is attached without anybody deciding to attach it.

CHECK

Communication, cooperation, and metadata

Three questions on the categories and their consequences; every option explains itself, including the ones you did not pick.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. What makes an arrangement cooperative rather than collaborative, in the chapter's sense?

- **A.** The participants work on the same artefact, but at different times rather than together — Working on one artefact at different times is asynchronous collaboration, which the chapter treats separately. Cooperation does not require a shared artefact at all.
- **B.** The participants are outside the firm rather than employed by it — Cooperation happens inside organizations as readily as outside them, and shared bookmarking among colleagues is one of the chapter's own internal examples.
- **C.** The work is unpaid, which is what distinguishes it from a commercial arrangement — Payment is irrelevant to the distinction. Plenty of cooperative arrangements are commercial, and the mutual benefit is what defines them.
- **D.** Each participant pursues their own aim, and one participant's success improves the others' — **correct.** The chapter defines cooperation as a win-win in which one participant's success improves the chances of the others, which is why an embedded video helps both the host page and the video.

2. A marketing team creates a hashtag and invites the public to post with it. What does the chapter say to expect?

- **A.** The topic will trend if enough of the firm's own followers post with it early — Trending happens when a topic is tagged faster than others, and a firm's own followers are a small input to that. The chapter's caution is about direction rather than volume.
- **B.** Whoever posts with it decides what it means, and critics can take it over — **correct.** The chapter's example is a police department that invited photographs with its hashtag and received pictures of alleged brutality instead. The tag is a public channel, and nobody owns what enters it.
- **C.** The firm can retire the hashtag if the campaign is not going the way it hoped — A firm can stop using a tag and cannot stop anyone else from using it. That asymmetry is exactly the risk the example illustrates.

- **D.** Reach will be limited to followers, since a hashtag is visible only to them — Hashtags exist to make posts findable beyond a follower list. Limited visibility would remove both the benefit and the risk.

3. Which of these is a genuine reason a firm might run social bookmarking internally?

- **A.** It maps where expertise sits, so somebody can find who already knows a topic — **correct.**
The chapter says enterprise-oriented bookmarking makes it easy to map islands of knowledge within an organization, which is a way of finding the expert rather than the document.
- **B.** It replaces enterprise search, since bookmarks index everything employees have read —
Bookmarks cover the fraction of content somebody bothered to save. Enterprise search exists to reach documents, databases and applications nobody bookmarked at all.
- **C.** It provides a permanent archive of external pages the firm relies on — A bookmark is a pointer. If the page changes or disappears the bookmark records only that it once existed, which is the opposite of an archive.
- **D.** It reduces browsing, because employees stop looking for pages colleagues already found —
Reducing effort is a plausible side effect, and the value the chapter identifies is finding the person, not saving the minutes.

Collaborating at scale

The last two jobs on the chapter's list are the ones with the largest strategic consequences. Collaboration is several people working the same problem at once. Connection is what happens when a firm can reach a crowd, and the crowd can reach back.

The cloud made collaboration ordinary

Sharing a document used to mean emailing it and hoping. **Cloud-based collaboration tools** removed the round trip: a file lives in one place, syncs to every device, and several people work the current version rather than their own copy. Cloud productivity suites go further and put the software itself in the browser.

That sounds like an unmixed good, and the chapter is careful to price both sides. The benefits and the risks fall in four places, and a firm rarely notices a risk until the day it arrives.

One of the four deserves saying plainly before you sort them. A cloud tool needs a live connection, so the work stops when the connection does, and the firm has traded a problem it could fix for one it can only wait out.

SORT

Where does each consequence of cloud collaboration land?

Each statement is a benefit or a risk of cloud-based collaboration tools; put it in the domain where it lands, using the chapter's four categories.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Information technology — Consequences for the technology function itself: what it has to build, run and control.

- Costs and risks fall, because the tools already exist and are cheap to deploy — The chapter's benefit for the technology function: a preexisting web-based tool avoids the cost and the risk of building an equivalent in house.
- The firm loses control of its data and of service quality to the provider — The matching risk. Data and tools sit on somebody else's servers, so both availability and the handling of the data become somebody else's decision.

Organization — Consequences for how the tools spread through a workforce and what happens when they are misunderstood.

- The tools are easy enough to use that they spread through the workforce quickly — The chapter's organizational benefit, and the reason these tools arrive without a project: people adopt them because they already know how.
- There is little documentation, training or support when something goes wrong — The matching organizational risk. Tools that arrived without a project also arrived without anybody responsible for them when they fail.

Competition — Consequences for how fast the firm can move against rivals, and what rivals might obtain.

- Product cycles speed up and the firm can answer a rival's move quickly — The competitive benefit: these tools are more efficient than email and legacy collaboration, which shortens the time between deciding and doing.
- Security and compliance rules are hard to enforce, raising the risk of exposure — The competitive risk, and the one the chapter puts most sharply: sensitive material moves outside the firm's controls, and industrial espionage becomes easier.

Upgrade cycles — Consequences of the fact that somebody else decides when the software changes.

- Nobody has to buy or install a software upgrade ever again — The benefit of somebody else running the software: upgrades stop being a purchase, a project, and a compatibility problem.
- Features can change without notice, breaking how people worked yesterday — The matching risk, and the one firms consistently underestimate. The provider's roadmap is now part of the firm's technology strategy whether it agreed to that or not.

The category is wider than a shared folder. The chapter's own list runs from single-purpose tools to suites that mirror an offline office.

Single-purpose services cover one job well — a spreadsheet, a presentation, a project board, a notes and task list, or a place to store and share files.

Integrated suites pull document sharing, editing and videoconferencing into one product, which is how a firm ends up with a technology strategy it did not write.

A content management system lets people publish, edit, version and retrieve digital content without a technical department, which is why a product manager can correct a description directly.

A learning management system applies the same idea to teaching: content, assessment, enrolment and grades, and increasingly discussion and group work as well.

What makes a content management system more than a publishing tool is the permission model. Roles decide who may create, who may edit and who may only read, which is how a firm lets a hundred people maintain its content without any of them being able to break it.

The same platforms then carry intranet and extranet sites, shared workspaces, to-do lists, discussion boards and alerts, which is how a system bought to run a website ends up running the collaboration around it.

Work done by people a firm does not employ

Collective intelligence is the idea that a distributed group holding divergent information and expertise can outperform individual experts. The chapter takes it from a book on the wisdom of crowds, and the word doing the work in that definition is divergent: a group that all knows the same thing is just one expert repeated.

Peer production is what collective intelligence looks like when it makes something: goods or services created by self-organizing communities, where the outcome depends on incremental contributions and anyone can improve it.

Firms reach for this deliberately as well. **Open innovation** is the practice of pulling external stakeholders into a firm's own research and development, forming ad hoc networks that reach past the people on its payroll.

Three arrangements in the chapter run on that principle, and they are not interchangeable.

Open source software is built and maintained by volunteers worldwide, and firms participate to influence its direction, to audit code they depend on, and to build goodwill in a community they draw from.

Wikis let anyone create, edit or delete content, with every prior version retained so vandalism can be reverted and disputes can be argued in the open rather than in private.

Crowdsourcing distributes small, self-contained pieces of work called **human intelligence tasks** — tagging an image, fixing a title, transcribing audio — to whoever will take them, usually for a few cents each.

Compare all three side by side, because the differences decide which one a firm can actually use for a given problem.

COMPARE**Three ways to get work from people you do not employ**

Step through the three arrangements; they run on the same principle and are not interchangeable, and the notes under each say when it can actually be used.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Open source software — Volunteers build and maintain something everyone can use, including firms that contributed nothing.

Someone publishes a first version (Usually to solve their own problem) → Developers join the project (Looking for interesting work) → Contributions are reviewed (Often only committers may merge) → Anyone may use the result (Including firms that gave nothing)

- A firm participates to steer the direction of software it depends on, to audit source code it is trusting, and to build standing in a community it recruits from.
- It is the least open of the three in one specific way: many projects let anyone propose a change and let only a few actually apply it.
- It cannot be commissioned. A firm can contribute and cannot decide what the project will do, which makes it unusable for anything the firm must control.

Wikis and peer production — A community writes and corrects a shared body of content, with every version kept.

Anyone may create or edit (No editorial gatekeeper) → Every version is retained (In a database behind the pages) → Bad edits are reverted (And argued over in the open) → Quality rises with participation (And inherits who participates)

- The retained history is the whole mechanism. Vandalism is cheap to undo, which is why open editing survives contact with the public at all.
- Inside a firm this is how an internal knowledge repository gets written by the people who know things rather than by whoever was assigned to document them.
- It inherits the composition of its contributors, so uneven coverage and systematic bias are structural rather than accidental, and both need checking against other sources.

Crowdsourcing micro-tasks — A firm splits work into small self-contained tasks and pays whoever completes them.

Work is split into small tasks (Each one easy for a person) → Tasks are posted to a marketplace (With a price attached, often cents) → Anyone may take one (Using time they were not selling) → Results return, and are paid for (Quality checked by the requester)

- This is the one a firm can actually commission, because it is paid work with a specification, and it is the one that displaces a category of employment.
- It only fits tasks a person finds easy and a computer finds hard, and that are small enough to describe completely: tagging an image, fixing a title, transcribing a clip.

- Quality control is the requester's problem and the largest hidden cost. Work arrives from people with no context, so the checking has to be designed before the tasks are posted.

The chapter is honest about what peer production costs, and names three failures that are structural rather than accidental.

Coverage inherits its contributors, so an open encyclopaedia is uneven in exactly the places its volunteer base is thin, and its known bias runs in a particular direction.

Interested parties edit their own entries, which means politicians, companies and other subjects of interest quietly improve the record about themselves.

Editing wars break out, where contributors repeatedly undo one another, which is the visible form of a disagreement no editor is empowered to settle.

The chapter's advice is to treat such a source as a starting point and check what it cites, which is advice about every peer-produced source rather than about one site.

MINI CASE

Forty thousand untagged photographs

Read the brief and the exhibit, then take the three decisions in order; the archive and every figure attached to it are hypothetical and were invented for this exercise.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

A regional newspaper group has digitised forty thousand photographs from its own archive. None of them carries a usable description, so none can be found, and the group cannot license what it cannot find. Three routes have been proposed: two staff researchers, a crowdsourcing marketplace, or opening the archive to readers on a wiki. The group is hypothetical and every figure attached to it is invented.

The holding

Forty thousand digitised photographs from a regional newspaper archive

The problem

No usable descriptions, so nothing in the archive can be found or licensed

What a good record needs

Subject, place, approximate date, and the names of anyone identifiable

Who knows the answers

Local readers, for the places and many of the people; nobody at the group

The three routes

Two staff researchers, a crowdsourcing marketplace, or a public wiki

Exhibit A • The three routes, costed — What each route would produce on the group's own estimates. Every figure here is invented for practice and none of it is reported data.

LINE	TWO STAFF RESEARCHERS	CROWDSOURCING MARKETPLACE	PUBLIC
Cost to describe all forty thousand	US\$96,000 in salary	US\$4,400 in task fees	US\$9,000 in task fees and more
Time to complete	About 14 months	About 6 weeks	Open-ended; finishes when done
Can identify the subject in the picture	Sometimes, from the caption files	Rarely; no local knowledge	Often; there are many people who know
Can supply place and approximate date	Usually	From visual clues only	Usually errors
Errors found in a sample of 200	6	31	22 at first; other routes
Who owns the resulting descriptions	The group	The group	Negotiated; automated

1. The crowdsourcing route is cheapest by a wide margin and has the worst error rate. What explains both at once?

- **A.** Marketplace workers are less careful, because the payment per task is very small — Payment shapes effort at the margin, and it cannot buy something the worker does not have. Doubling the fee does not make a distant worker recognise a street they have never seen.
- **B.** The task needs local knowledge, which the marketplace cannot supply at any price — **correct.** Crowdsourcing fits tasks that are easy for any person and hard for a computer. Naming a local street in 1974 is easy only for a person who was there, which is why the price and the error rate move together.

- **C.** The photographs are too old for anybody to identify without the original caption files — Readers manage it often, according to the same exhibit, so age is not what is stopping identification. What differs between the routes is who is looking.
- **D.** The group specified the task badly, and a better specification would fix the errors — A sharper specification helps with what is describable from the image and cannot conjure knowledge the worker does not hold. The mismatch is in the task, not the wording.

2. The wiki route shows errors falling from twenty-two to four once other readers had edited. Which mechanism is that?

- **A.** Moderation, since paid moderators removed the incorrect descriptions — Moderators handle abuse and vandalism, and the exhibit describes readers correcting readers. The cost line budgets for moderation without making it the mechanism.
- **B.** The network effect, since more readers made the archive more valuable to each of them — The network effect describes value rising with participation and would explain why readers keep coming. It does not explain why a specific wrong date became a right one.
- **C.** Peer production, where incremental contributions correct each other over time — **correct.** Peer production is exactly this: the result depends on incremental contributions, and anyone can improve what somebody else produced, which is how open editing converges rather than degrades.
- **D.** Version history, which let the group roll back to a description that was right — Retained history makes correction safe, so a bad edit can be undone. Rolling back does not create the better description that replaced the wrong one.

3. The board wants the archive described and owned outright within the financial year. What does the exhibit imply?

- **A.** The wiki route, since it alone produces descriptions with accurate local detail — The wiki produces the best local descriptions, never finishes, and leaves ownership to be negotiated. It fails both of the board's constraints, however good its output is.
- **B.** The crowdsourcing route, since only it completes inside the year at a defensible cost — Crowdsourcing does finish inside the year, and it fails on quality for the fields that matter: subject, place and date are precisely what it cannot supply.
- **C.** A combination, with crowdsourcing for what an image shows and readers for local detail — A combination is the strongest answer to the underlying problem and does not satisfy the board's own constraints, because the reader half neither completes nor transfers ownership.
- **D.** The staff route, since it is the only one that meets both requirements as stated — **correct.** Fourteen months is uncomfortable and the other two routes fail a stated requirement outright. The honest conclusion is that the board's constraints, not the routes, are what needs revisiting.

What this case was really about. The exhibit is arranged so the cheapest option is the worst fit, and the reason is the chapter's own rule about crowdsourcing: it suits tasks that are easy for any person and hard for a computer. Identifying a local street in an old photograph is easy only for somebody who was there, so the marketplace cannot do it at any price. The wiki reaches exactly those people and gives up completion and ownership to do it, which peer production always does. The last decision matters most: two of the three routes fail a constraint the board set rather than a test the analysis applied, and saying so is part of the analysis.

Peer production is the cheapest labour a firm will ever find and the least controllable. It cannot be commissioned, it cannot be hurried, and what it produces belongs to the composition of whoever turned up — which is why the next section is about the arrangements where a firm keeps more of the say.

CHECK

Cloud tools, crowds, and what they can be asked for

Three questions on the mechanisms behind collaboration at scale; every option explains itself, including the ones you did not pick.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. A firm standardises on a cloud productivity suite and cancels its own software licences. Which of the chapter's risks has it accepted along with the saving?

- **A.** That the provider decides when features change — **correct.** The chapter's upgrade-cycle entry has a benefit and a matching risk: nobody buys upgrades any more, and in exchange the provider's roadmap becomes part of the firm's technology strategy, changing without notice whether it agreed or not.
- **B.** That the tools will be harder to use than the software they replaced — Ease of use is the chapter's organizational benefit rather than a risk, and it is why these tools spread through a workforce without a project behind them.
- **C.** That staff will need more training than the previous suite required — The chapter's organizational risk is the opposite: little documentation, training or support exists, precisely because the tools are easy enough that nobody planned for any.
- **D.** That the annual cost will rise once the introductory pricing period ends — Price increases are a genuine commercial risk of any subscription and are not among the four consequences the chapter tabulates.

2. What does collective intelligence actually require, in the chapter's account of it?

- **A.** A large enough group that individual mistakes cancel out across the whole — Size helps and is not the condition. A very large group that all knows the same thing is one expert repeated many times, and it makes the same mistake in unison.

- **B.** A group whose members hold divergent information and expertise — **correct.** The chapter's definition turns on divergence: distributed groups with a divergent range of information and expertise can outperform individual experts, which is where the advantage comes from.
- **C.** A moderator able to settle disagreements the group cannot resolve itself — Wikis work with no such authority, and the chapter describes editing wars as what happens when nobody is empowered to settle a dispute. It is a known cost rather than a requirement.
- **D.** Members who are volunteers, since paid contributors distort the outcome — Crowdsourcing pays its contributors and is one of the chapter's examples of harnessing the crowd. Payment changes who turns up rather than whether the mechanism works.

3. A publisher wants a crowdsourcing marketplace to write summaries of its specialist legal titles. Why is that a poor fit?

- **A.** Summarising is creative work, and marketplaces handle only mechanical tasks — The line is not creative against mechanical. Plenty of judgement-based micro-tasks work well, and plenty of mechanical ones do not fit for other reasons.
- **B.** The task is too large to be paid for at the rates these marketplaces use — Cost would follow from the size of the job and the chapter's constraint is about the nature of the task, not its price. A well-matched task stays cheap at volume.
- **C.** The task is not one a person can do easily without expertise the crowd lacks — **correct.** The chapter's rule is that these are tasks humans solve easily and computers find hard. Summarising specialist legal writing is hard for almost every human too, so the crowd cannot supply it at any rate.
- **D.** The publisher would lose copyright in summaries written by outside contributors — Ownership is set by the marketplace's terms and is a contractual matter rather than the reason the work would come back unusable.

Connecting at scale

The last of the chapter's four jobs is connection, and it is the one with the largest strategic consequences. Collaboration changes what a firm's own people can do together. Connection changes who counts as the firm's people at all, and it introduces the single mechanism that decides who wins these markets.

Networks, professional and internal

Social networking builds **social online communities** where people with a broad and diverse set of interests meet, communicate and collaborate. Professional networks do the same for careers, letting people post a profile, ask for expert advice and be approached about work.

Enterprise social networks bring the format inside a company, and the chapter names why they look so familiar. **Consumerization of information technology** is the pattern where a product designed for the consumer market arrives at work, because that is what people already know how to use.

The three kinds are worth keeping apart, because a firm's reason for being on each one differs.

A general-interest network is where a firm meets its customers and the wider public, gets feedback on what it sells, and maintains a brand image it does not fully control.

A professional network is a labour market: profiles, expert advice and approaches about work, which makes it a recruiting channel before it is anything else.

An enterprise network is private to one organization, and it exists to connect employees to each other and to what colleagues already know.

The effect that decides everything

What makes these markets behave strangely is the **network effect**. Each additional member makes the service more valuable to every existing member, so growth feeds itself and a leader becomes very hard to dislodge.

The chapter's counter-example matters just as much. The network that was largest before the current one declined until it barely ranked, which is the same effect running in reverse: value falls with every departure, and the falling value causes the next one.

Sort eight situations by whether the effect is working for the firm, against it, or is not the mechanism at all.

SORT**Is the network effect the mechanism here?**

Sort each situation by what the network effect is doing in it; two of these look like the effect and are something else entirely.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Working for the firm — Each additional participant is making the firm's own service more valuable.

- Every new reviewer makes a firm's product pages more useful to the next shopper — The definition applied directly: each additional participant raises the value of the service for everybody already using it, and the firm did not write any of it.
- A user forum answers most support questions, and answers accumulate year on year — Participation compounds into an asset. A competitor starting today has the same software and none of the nine years of answers, which is what makes this hard to copy.

Working against the firm — The same mechanism running in reverse, or running in a rival's favour.

- Members leave a declining network, which makes it less worth staying on — The chapter's counter-example: value falls with every departure, and the falling value causes the next one. The mechanism is identical and the sign is reversed.
- A rival platform's community grows, so your customers increasingly compare you there — The effect is real and it is happening to somebody else's asset. Growth on a rival's platform moves buyers to a venue where the firm does not set the terms.
- An internal directory has four completed profiles, so nobody bothers to complete a fifth — Below critical mass the effect works against the tool: too little participation to be worth consulting, so it never gains the participation that would make it worth consulting.

Not the network effect — Something real is happening and participation is not what is causing it.

- A firm's advertising budget doubles and its brand awareness rises — Awareness bought directly is not participation compounding. Stop paying and it decays, which is exactly what a network-effect asset does not do.
- A supplier raises its prices, so the firm's margin falls across every channel — This is bargaining power acting on cost, and no user's participation is involved anywhere in it. The frameworks section is where this one belongs.
- A firm's product is cheaper to make this year because component prices fell — A cost movement in a supply market. It changes the firm's economics and has nothing to do with how many people use anything.

Making something spread on purpose

Viral marketing uses the same mechanism to spread brand awareness, by making content people choose to pass on. The term was coined to describe marketing driven by word of mouth, in the way a real virus moves through the contacts somebody already has.

The chapter's principles for it are unglamorous, and each one has a matching way of being broken.

Do something unexpected, because a viewer who can predict the rest of the piece from its first seconds has no reason to show it to anybody.

Make people feel something, since a purely informative piece is passed on by nobody, however accurate and however well made it is.

Make sequels, because an audience assembled once and never given anything else is an audience the firm has spent and thrown away.

Allow sharing and easy distribution, and **never restrict access**, because a registration wall stops the mechanism at the first hand-off and leaves nothing to spread.

The chapter's own founding example is an email service that added a single line advertising itself to the bottom of every message its users sent, and bought twelve million subscribers for a fraction of what advertising would have cost.

MATCH

The viral principle, and the failure that breaks it

Pair each of the chapter's principles with the mistake that defeats it; every pair explains itself once matched.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

The video is competent, on-message, and exactly what a viewer expected from the first frame

Do something unexpected Predictability is fatal to sharing. Passing something on is a small social risk, and nobody takes it for material the recipient could have guessed.

The piece is accurate, informative, and leaves the viewer feeling nothing at all

Make people feel something Information travels through search; feeling travels through people. A piece that provokes nothing gets watched and not forwarded, which ends the chain.

One excellent piece runs, gathers an audience, and is never followed by anything

Make sequels An audience assembled once and then abandoned is a cost the firm paid and did not collect on. The chapter lists sequels precisely because campaigns stop here.

Watching requires an email address and a confirmation link first

Never restrict access to the content The gate breaks the mechanism at the first hand-off: the person who would have passed it on now has nothing to pass except a form.

The content lives on a platform with no way to embed it or send it to anybody

Allow sharing and easy distribution Every extra step between wanting to share and having shared costs a proportion of the people who intended to, which compounds against you at each hop.

The campaign runs as paid advertisements sent to a purchased list of addresses

Not viral marketing at all Reach the firm buys directly is advertising. Viral marketing spends the network's own connections instead, which is why its cost per person reached keeps falling.

Social, local, mobile

Where connection meets a phone that knows where it is, the chapter uses the shorthand **social-local-mobile**: reviews written by other customers, a device aware of its own position, and a local business willing to pay to reach somebody standing nearby and hungry.

The review platform in that example never wrote a review or took a photograph. It holds what its users wrote, and that is enough to make it the thing people consult before choosing where to eat, and enough to sell an advertisement to the restaurant next door.

The chapter's closing industry shows the same shape at a larger scale. Travel was booked through a few large online agencies, which had themselves displaced the high-street agent. A newer set of platforms then let people rent out their own homes and drive their own cars, and both rest on ratings that make a stranger worth trusting.

Neither of those platforms owns a room or a vehicle. Each owns the review corpus and the audience, which turns out to be worth more than the assets, and that is the observation the strategy supplement is built on.

MINI CASE

The platform that owns nothing

Read the brief and the exhibit, then take the three decisions in order; both firms and every figure attached to them are hypothetical and were invented for this exercise.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

A regional serviced-apartment company owns and lets ninety furnished flats in one city. A platform that owns no property at all now lists about two thousand rooms in the same city, taken from people letting their own homes, and its listings are ranked by guest ratings. The company's board wants to know what it is actually competing against. Both firms are hypothetical and every figure attached to them is invented.

The company

A regional serviced-apartment operator with ninety furnished flats in one city

What it owns

The buildings, the furniture, the cleaning contracts and the staff

The platform

Lists around two thousand rooms in the same city and owns none of them

What the platform owns

The listings, the guest ratings and the audience that reads them

The question

What the company is actually competing against, and what it should do

Exhibit A · The two ways to add a room in this city — What it takes each business to make one more room available, and what happens to it afterwards. Every figure here is invented for practice.

LINE	THE SERVICED-APARTMENT COMPANY	THE PLATFORM
Cost to add one more room	About US\$140,000 to buy and furnish	None; somebody else lists their own
Time to add one more room	Six to nine months	About twenty minutes
Who carries the room if it stays empty	The company	The person who listed it
Guest reviews held about the city's stock	410, about its own flats only	31,000, about everybody's
What a guest compares before booking	One operator's photographs	Two thousand listings ranked by rating
Share of the company's bookings arriving via the platform	38 percent, up from nothing three years ago	-

1. The board describes the platform as a competitor with two thousand rooms. What is wrong with that description?

- **A.** The platform holds no rooms; it is a venue, and the company is already selling inside it — **correct.** Two thousand independent hosts compete with the company; the platform competes with nobody and takes a share of everybody. Thirty-eight percent of the company's own bookings already arrive through it, which a description of it as a rival cannot account for.
- **B.** The platform's rooms are lower quality, so the two are not competing for the same guest — Quality varies in both directions and nothing here measures it. Assuming the platform's stock is worse is the comfortable answer rather than the supported one.

- **C.** The platform's rooms come and go, so counting them overstates how large it really is — Availability fluctuates for both, and it changes the effective size of the venue rather than what kind of thing the venue is.
- **D.** The platform is larger than two thousand rooms once other cities are counted in — Scale in other cities is irrelevant to a board deciding about this one, and it makes the same category error at a larger number.

2. Adding a room costs the company about a hundred and forty thousand dollars and the platform nothing. What does that difference mean strategically?

- **A.** The company should sell flats and become a platform, since the economics are better — Becoming a platform means starting a network from zero against an incumbent with thirty-one thousand ratings, which is the hardest possible entry. The economics are better for whoever already has the network.
- **B.** The company is at a permanent cost disadvantage and cannot compete on price — The company owns assets that produce rent and a service standard it controls, and neither is a disadvantage. Cost per room is only a disadvantage if price is the dimension it chooses to fight on.
- **C.** Supply on the platform can expand faster than demand, so the company cannot outbuild it — **correct.** The constraint on the platform's supply is how many people feel like listing, not capital or construction, so it can add a thousand rooms in a season. No amount of building keeps pace, which rules out capacity as a response.
- **D.** The platform is more profitable per room, since it has no capital tied up in property — Profitability per room is plausible and unmeasured here, and it describes the platform's business rather than telling the company what to do about it.

3. Thirty-one thousand guest ratings sit on the platform and four hundred and ten sit with the company. What follows?

- **A.** The company should ask its own guests to leave ratings on the platform instead — This grows the platform's asset with the company's own guests, deepening exactly the dependence the exhibit is describing. It is the most common version of this mistake.
- **B.** The company should not attempt to build a rival review corpus from ninety flats — **correct.** Ninety flats cannot generate a corpus that competes with everybody's, so a rival review site is a fight the company loses by arithmetic. The useful moves are elsewhere: what it can promise that an individual host cannot.
- **C.** The company should dispute unfair ratings, since each one carries more weight at its scale — Disputing individual ratings is housekeeping. It matters at the margin and does not touch a difference of two orders of magnitude.
- **D.** The company should list every flat on the platform, since that is where guests are comparing — Listing everywhere raises occupancy and raises the platform's share of the company's bookings, so it is a revenue decision that makes the strategic position worse rather than better.

What this case was really about. The board's framing was the error, and every decision followed from correcting it. A platform that owns nothing is not a competitor with two thousand rooms; it is the venue in which the company is already selling thirty-eight percent of its own nights. Its supply expands at the speed people feel like listing, so capacity is not a defence, and its ratings corpus is two orders of magnitude larger, so review volume is not one either. What that leaves is what a network-effect asset cannot supply: a standard a guest can rely on without reading anything, which is the argument the strategy supplement will ask you to make properly.

What connection costs

Two consequences are worth naming before the next section, because both land on people rather than on firms.

Feeds are assembled by algorithms predicting what will hold attention, which quietly filters out material conflicting with a user's existing views and produces what critics call a filter bubble.

Clickbait exists because attention is the product, so a headline is written to be clicked rather than to be accurate, and the article behind it routinely fails to deliver what the headline promised.

Connection changes who counts as the firm's people: the volunteers maintaining code it runs on, the crowd labelling its data, the customers writing its product pages, the strangers whose ratings make its platform usable. None of them is employed, and all of them are load-bearing.

CHECK

Networks, spread, and what they cost

Three questions on the mechanism behind connection; every option explains itself, including the ones you did not pick.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. A firm rolls out an internal social network and staff complain it is clumsy compared with what they use at home. What does the chapter call that pressure?

- **A.** Technological inertia, since people will not adopt a tool without a tangible benefit — Inertia is refusing to move at all. These staff have moved and are reporting that the destination compares badly, which is a different complaint with a different remedy.
- **B.** A generation gap, since younger staff hold higher expectations than older colleagues — A gap describes differing comfort with these tools across a workforce. Here everybody agrees on the comparison, which is the opposite of a gap.

- **C.** A critical mass problem, since too few colleagues have joined for it to feel useful — Thin participation makes a social tool feel useless rather than clumsy, and the complaint described is about the tool itself.
- **D.** Consumerization, where consumer products set the benchmark for internal tools — **correct.** The chapter names this directly: applications designed for the consumer marketplace make inroads into organizational settings, and users judge an internal tool's usability against what they already use.

2. Why is a large social network so hard for a well-funded new competitor to displace?

- **A.** Each new member makes the network more valuable to every other member — **correct.** The chapter attributes the leader's position to the network effect: as it grew it became more attractive to join, and a rival has to overcome that self-reinforcement rather than any particular feature.
- **B.** Members would lose the content they have already posted if they switched — Losing posted content is a real friction and a small one. The chapter's example of a collapsed network shows members leaving despite everything they had posted there.
- **C.** The incumbent can outspend a new entrant on advertising and acquisition — Spending follows the position rather than producing it, and the chapter's counter-example was prominent and well funded when it declined.
- **D.** The engineering required to run at that scale is difficult to replicate — Scale engineering is a solvable problem. What a competitor cannot buy is a reason for the first million members to be there.

3. What is the chapter's point about the review platform in its social-local-mobile example?

- **A.** It succeeded because it was early, and being early is what network effects reward — Timing helps a network effect start and does not explain why the position holds. Plenty of early networks were displaced, including the one the chapter names.
- **B.** It owns no reviews or photographs of its own, and holding what users wrote is enough — **correct.** The platform produced none of the content it profits from. It holds what its users wrote, which is what makes it the thing people consult and what lets it sell advertising to the businesses being reviewed.
- **C.** It is more accurate than its rivals because it verifies the reviews it publishes — The chapter records criticism of how reviews are handled rather than a claim of superior verification, so this reverses what it actually says.
- **D.** Its advantage is location awareness, which its competitors were slow to add — Location awareness is part of the shorthand and is available to every competitor with a phone app, so it cannot be what distinguishes this one.

Making it work inside a company

Every tool in the last three sections is available, cheap and proven somewhere. Most internal deployments of them fail anyway. This section is about why, and it is the part of the chapter that separates people who can name a technology from people who can introduce one.

The tool is necessary and nowhere near sufficient

The chapter's position is blunt: technology is a critical success factor, and it is not the only component. Change management decides the outcome. Providing the tools and hoping employees use them for the intended purpose is not a plan, and management has to make sure people know the tools exist, what they are for, and what the rules around them are.

Because these applications run on social interaction, information sharing and network effects, corporate culture matters more here than for almost any other class of system. A culture of knowledge sharing, trust and honest feedback makes them work. A culture without those three does not fail to adopt the tool; it adopts the tool and produces nothing.

What the web proves, and what it does not

The first trap the chapter names has a name of its own: the enterprise is not the web. Applications that succeeded publicly did so over many years and under conditions a firm does not have.

Two of those conditions are worth spelling out, because both are invisible from outside.

Public participation is voluntary, and people join because they want to, whereas nobody inside an organization can be made to contribute anything worth reading.

Seamless collaboration usually is not: good encyclopaedia articles rest on long arguments behind the scenes, and many open source projects let anyone suggest a change while letting only a few apply one.

Four factors that decide the outcome

The chapter lists a set of caveats to weigh before deploying. Four of them account for most real failures, and each one shows up as a recognisable complaint rather than as an announcement.

Culture is the first, because open communication does not sit comfortably inside rigid hierarchies built on control, and self-expression cannot be mandated into existence.

Critical mass is the second, because these tools depend on network effects and a small user base never reaches the point where participating repays the effort of participating.

Technological inertia is the third, because people do not switch tools until they see a real, tangible benefit, and familiarity is itself a benefit they already have.

Security is the fourth, because any application that increases sharing increases exposure, and a firm has to weigh collaboration against protecting intellectual property and meeting its reporting obligations.

Sort eight complaints into the factor each is evidence of. The complaints are the form these problems actually take when somebody reports them.

SORT

Which factor is this complaint evidence of?

Each line is a complaint somebody actually makes; sort it into the factor the chapter would attribute it to.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Culture — Open sharing sits badly with hierarchy and control, and participation cannot be ordered into existence.

- Nobody posts a question, because asking one in public looks like not knowing your job — This is culture in the chapter's exact sense: the tool works and the environment punishes the behaviour it depends on. No feature fixes it.
- Managers approve every post before it appears, so discussion has stopped — Open communication colliding with a control-based hierarchy. The chapter is explicit that social applications do not do well inside rigid top-down structures.

Critical mass — The tool needs enough participants before it repays the effort of participating.

- I looked for a colleague's expertise and only four people had filled in a profile — The value of a directory rises with how many people are in it. Below a threshold there is nothing to find, so nobody comes back, so there is still nothing to find.
- The forum has thirty members across a firm of two thousand, and nothing new this month — Straightforward critical mass. The chapter notes that smaller organizations often cannot reach the participation that makes these tools valuable at all.

Technological inertia — People will not move to something new without a benefit they can actually feel.

- We already have email for this, and email works fine for what I do — The chapter's definition of inertia: people do not switch unless they see a real, tangible benefit, and the incumbent tool has the advantage of being already understood.
- It duplicates three things I use, so learning it buys me nothing I do not have — Also inertia, and the chapter points out this is common with social applications precisely because they bundle chat, messaging and posting that people already do elsewhere.

Security — Wider sharing means wider exposure, and somebody has to carry that risk.

- A supplier's pricing was posted in an open channel that contractors can read — Exposure created by the sharing itself. The chapter's warning is that any application increasing data sharing necessarily increases the risk of a breach.
- Our auditors want to know how we would produce a record of who approved what — The compliance half of the same factor. The chapter names a financial-reporting statute for this reason: collaboration tools can quietly move decisions somewhere nobody can evidence them.

The other four, which arrive later

The remaining factors tend to appear after a pilot has gone well, which is exactly what makes them dangerous.

Organizational context means the work should choose the tool: ask what objective you are trying to meet before deciding what to install, rather than setting up a wiki and hoping.

Organizational hierarchies matter because a pilot runs inside one department, and an organization-wide deployment needs changes to culture, process and often the hierarchy itself.

The generation gap is a tendency rather than a rule: a workforce accustomed to interactive online environments raises the odds of adoption, and senior members may not see the point at all.

Technological integration decides whether the tool joins the firm's existing systems or simply becomes one more place somebody has to remember to look.

Scale changes which of these bites hardest. A small firm can carry a weak culture on personal relationships and will struggle to reach critical mass at all, because there are not enough people for participation to repay itself. A large firm has the numbers and has the hierarchy, the inertia and the compliance obligations that come with them.

Walk a rollout through four decisions, in the order they actually arrive rather than the order a plan puts them in.

WALKTHROUGH**Rolling out an internal social platform**

Choose at each step; every option shows what it would lead to, including the ones you did not take. The firm is hypothetical.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

You have been asked to introduce an enterprise social network at a firm of about two thousand people. Four decisions, in the order they actually arrive.

Decision 1. The chief executive has seen a demonstration and wants it deployed. What do you do first?

- Deploy it to everybody at once, so critical mass is reached immediately — Accounts are not participation. You reach the headcount without reaching critical mass, and you spend the one moment of attention the rollout gets on a tool nobody has a reason to open.
- Run a pilot in the most enthusiastic department and expand from there — A reasonable second step and a poor first one. A pilot with no stated objective produces a happy department and no evidence, and the chapter warns that a pilot's success rarely survives contact with the whole organization.
- Find the specific work it should serve, and name what would count as success — **the stronger call.** Right. The chapter calls this organizational context: the work-related need should drive the choice of tool. Naming the objective first is also what gives you something to measure later, instead of counting logins.
- Write the acceptable-use policy, so the rules exist before anybody posts — The policy is genuinely needed and it is not the first move. Written before you know what the tool is for, it tends to prohibit rather than enable, which sets the tone for everything after.

Decision 2. The pilot in one department worked well. Rolling out to the whole firm has stalled: two thousand accounts, and around one in twelve posts anything. What is happening?

- The pilot department was self-selected, and the firm as a whole has no such culture — **the stronger call.** Right. This is the chapter's warning in full: department-led pilots work, and organization-wide deployment needs changes in culture and process that a pilot never tested. What you measured was the department, not the tool.
- The tool is inadequate, so evaluate two alternatives and choose a better one — Replacing the tool restarts the same experiment with the same conditions. Nothing you have observed points at the software, and a second rollout after a failed one is much harder to get attention for.
- Participation is a training problem, so run hands-on sessions in every location and office — Training turns people who cannot use it into people who can, and the pilot already showed that people can. The stall is about reasons to post, which no session supplies.
- Set a posting target for each team, and report on it monthly — You will hit the target and learn nothing. The chapter's point that people cannot be forced to participate usefully is exactly this: a quota produces compliance posts that make the channel less worth reading.

Decision 3. You discover that most of the useful discussion is happening in a messaging tool a team installed themselves, which is not connected to anything.

- Block it, since unapproved tools create exposure the firm has not assessed — The exposure is real, and blocking it moves the conversation somewhere you cannot see rather than back to the platform. The chapter's observation about people routing around bans applies inside the firm as much as outside it.
- Ignore it, since the official platform is the one being measured — Then your measurement is now actively misleading, and the firm is running two systems of record with no connection between them, which is the compliance problem waiting to be discovered.
- Migrate the team onto the official platform and close the other tool — This is the answer that looks decisive. You would be moving a working conversation onto a platform that has already failed to attract one, without changing anything about why.
- Treat it as evidence of what the work needs, and integrate rather than ban it — **the stronger call.** Right. It is the clearest signal available about what people actually need, and the chapter's integration point applies: a tool that does not join the firm's existing systems becomes another place to look, which is what your official platform has become.

Decision 4. Legal asks how the firm would produce a record of a decision made in a channel, and who can read what.

- Restrict the platform to non-sensitive discussion, and say so in the policy — A rule nobody can apply in the moment. People do not classify a conversation before having it, and the sensitive material arrives inside an ordinary thread rather than announcing itself.
- Answer it now with retention, access and export rules, before wider use — **the stronger call.** Right. The chapter puts security and compliance on the list of factors precisely so this arrives before the material does. Retention and access decided early are policy; decided after a request, they are an incident.
- Point out that email carries the same exposure and has never been questioned — Comparably true and not an answer. Email retention is usually a solved problem at this kind of firm, which is the difference legal is pointing at rather than overlooking.
- Defer it until adoption improves, since the question is theoretical at low usage — Deferring it means the rules will be written after there is material to apply them to, which is the worst moment to decide what should have been kept.

Now the same problem seen from the other end, a year after the tool went in. The question is no longer what to do first; it is what the participation figures are actually telling you.

MINI CASE**A wiki nobody writes in**

Read the brief and the exhibit, then take the three decisions in order; the manufacturer and every figure attached to it are hypothetical and were invented for this exercise.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

A manufacturer of specialist pumps has run an internal wiki for a year, intended to hold the knowledge its engineers keep re-deriving. Fourteen months from now a third of its engineering staff reach retirement age. The wiki has plenty of readers and almost no writers, and the head of engineering wants to know whether to persist. The firm is hypothetical and every figure attached to it is invented.

The firm

A manufacturer of specialist pumps, around six hundred staff

The tool

An internal wiki, running for twelve months

Why it was built

To hold engineering knowledge that is currently in people's heads

What is coming

A third of engineering staff reach retirement age within fourteen months

What is observed

Many readers, very few writers, and almost nothing added by senior engineers

Exhibit A • Twelve months of the wiki — Participation broken down three ways. Every figure here is invented for practice and none of it is reported data.

GROUP	SHARE OF STAFF	READ AT LEAST MONTHLY	WROTE OR EDITED AT LEA
Engineers with under 5 years of service	31 percent	88 percent	44 percent
Engineers with 5 to 20 years	39 percent	71 percent	12 percent
Engineers with over 20 years	30 percent	52 percent	3 percent

GROUP	SHARE OF STAFF	READ AT LEAST MONTHLY	WROTE OR EDITED AT LEA
Pages created in the whole year	-	-	218
Pages edited by somebody other than the author	-	-	31
Pages describing a problem already solved once	-	-	9

1. Writing falls sharply with length of service. What is the weakest explanation of that pattern?

- **A.** Longer-serving engineers are less comfortable with this style of tool — **correct.** as the weakest. It is the first explanation people reach for, the chapter presents the generation gap as a tendency rather than a rule, and this group reads the wiki at fifty-two percent, so comfort with the tool is evidently not the barrier.
- **B.** Longer-serving engineers hold the knowledge that being asked for is their standing — This is the strongest of the four and the hardest to say aloud. Where being the person who knows something is a source of standing, writing it down is a loss, and no amount of usability work touches that.
- **C.** Longer-serving engineers have less unstructured time than newer staff do — Plausible and testable. Senior engineers are typically the most interrupted people in the building, and a contribution model that assumes spare time will select against them.
- **D.** Longer-serving engineers are unconvinced that writing it down will change anything — Also plausible, and it is the chapter's inertia argument: people do not adopt a new practice until they can see a tangible benefit from having done so.

2. Only thirty-one of two hundred and eighteen pages were edited by anybody other than their author. Why does that matter?

- **A.** It shows the pages were accurate enough that no correction was needed — Nothing establishes accuracy here. Unedited could mean correct, unread by anyone who knew better, or not worth the trouble of correcting, and the exhibit cannot distinguish them.

- **B.** It shows the wiki is being used as a filing cabinet, not peer production — **correct.** Peer production depends on incremental contributions that improve on each other, which is what makes a wiki more than a document store. Pages that nobody touches after publication are documents in a different container.
- **C.** It shows the version history is not being used, so the tool is misconfigured — History is what makes editing safe rather than what makes it happen. The tool is working; the behaviour it exists to support is not occurring.
- **D.** It shows most pages are outside the areas other engineers work in — Possible for some pages and it would not explain a rate this low across an engineering group whose work overlaps heavily by design.

3. Given the retirements coming, what should the head of engineering do?

- **A.** Persist and promote it harder, since awareness is what limits contribution — Promotion addresses awareness, and eighty-eight percent of the newest group already read it, so awareness is not the constraint. This is the response that feels active and changes nothing.
- **B.** Close it and commission written handover documents from each retiring engineer — Handover documents produce a burst of material written by people leaving, with nobody left to correct it and no habit that survives them. It also abandons the readership the wiki has genuinely built.
- **C.** Keep it, and make capture part of the work rather than something done afterwards — **correct.** Both the sort and the walkthrough point the same way: participation follows from the work rather than from exhortation. Capture attached to closing a job, and to the nine pages that recorded a re-solved problem, gives writing a reason that outlasts the campaign.
- **D.** Replace it with an enterprise social network, which senior staff may find easier — Changing the tool restarts the same experiment under the same conditions. Nothing observed here implicates the software, and the reading figures suggest the wiki is fine at being read.

What this case was really about. Reading is high and writing is not, which is the shape almost every internal knowledge tool takes. The exhibit points at incentives rather than at software: contribution falls exactly where knowledge is a source of standing and where interruptions are heaviest, and the thirty-one edited pages say peer production is not happening even among the people who do write. The nine pages recording an already-solved problem are the most useful line in the exhibit, because they are the only measured evidence of the benefit the wiki was built for. Fix the reason to write, attach it to work people are already doing, and measure the thing the tool exists to produce rather than the activity around it.

Ask three questions before any internal social tool is bought. What specific work does this serve? Will enough of the right people use it to pass critical mass? And what new exposure does it create that somebody is accountable for? A deployment that cannot answer all three has not been designed, only purchased.

CHECK**Why internal deployments fail**

Three questions on the conditions that decide adoption; every option explains itself, including the ones you did not pick.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. What does the chapter mean when it says the enterprise is not the web?

- **A.** Corporate networks cannot run the same software that public platforms run on — The software runs perfectly well inside a firm, and much of it is sold for exactly that. The obstacle is not technical.
- **B.** Public success took years under conditions of voluntary participation a firm lacks — **correct.** The chapter's argument is about conditions: public platforms grew over many years, people join because they want to, and behind apparently seamless collaboration sit long arguments and restricted commit rights.
- **C.** Public platforms are consumer products, and enterprises need enterprise products — Enterprise editions exist and buying one changes nothing about participation. The chapter's caution applies to enterprise products as much as to consumer ones.
- **D.** Public platforms are unmoderated, whereas a firm must moderate what employees post — Public platforms moderate heavily, and moderation is a real difference in degree rather than the point being made about why success does not transfer.

2. A firm installs a wiki and asks staff to use it for whatever they find useful. Which caveat does that violate?

- **A.** Critical mass, since a general-purpose tool spreads participation too thinly — Thin participation is a likely consequence rather than the error itself. Critical mass describes what happens next, not what was done wrong.
- **B.** Technological integration, since a tool with no stated purpose connects to nothing — Integration is about joining existing systems, and a tool with a clear purpose can be just as unintegrated as one without.
- **C.** Organizational context, since the work should determine the tool — **correct.** The chapter says implementation should be driven by a specific usage context, and warns in these words against setting up a wiki and hoping employees use it for the right purpose.
- **D.** Culture, since staff will not contribute without an explicit instruction to do so — Culture is about whether people can share safely, not about whether they were told to. An instruction produces posts rather than contribution.

3. Why does the chapter say organization-wide rollouts need senior sponsorship in a way pilots do not?

- **A.** Because they require changes to culture, process, and often the hierarchy itself — **correct.** The chapter says organization-wide implementations typically need changes in culture and processes and often the flattening of hierarchies, which is a magnitude of change that requires active senior involvement.
- **B.** Because a firm-wide licence costs more than a departmental one and needs approval — Budget approval is real and routine. The chapter is describing a change too large for a department to make rather than a purchase too large for one to fund.
- **C.** Because senior managers are the least likely group to adopt the tool voluntarily — Senior adoption is a separate observation the chapter also makes. It is a reason sponsorship is hard to obtain rather than the reason it is needed.
- **D.** Because the security and compliance questions only arise at organizational scale — Those questions arise as soon as sensitive material is posted, which can happen in a pilot of twenty people on its first day.

When it goes wrong, and how fast

Every capability in this chapter has an inverse. The reach that carries a campaign carries a complaint at the same speed, the reviews that help buyers choose can be manufactured, and the anonymity that protects a whistleblower protects somebody else entirely. This section is the inverse, and the response playbook that follows from it.

Two rules the chapter states outright

The first is that the internet never forgets: material that embarrassed a firm stays retrievable years later, and deleting the original rarely removes the copies. The second is that a firm has to watch continuously and react fast, because the alternative is finding out from a journalist.

Those two rules generate everything else in this section, including the parts that look like separate problems.

Reviews, posts, pages, and video

The chapter works through four failure points, and each one fails differently.

Reviews are consulted before buying and are not always what they seem: firms hire people to post favourable reviews or unfavourable ones about competitors, and analysis tools have found products where the large majority appear fabricated.

Microblogging fails in public and at speed. A police department invited photographs with its own hashtag and received pictures of alleged brutality instead.

Company pages are free to host and not free to run: somebody has to watch what appears, and decide where the line sits between content the firm dislikes and content it removes.

Video is where it becomes uncontrollable, because a complaint that is filmed rather than written travels to an audience no press office was built to answer.

Two of the chapter's examples set the scale. A tenant with a couple of dozen followers was sued over a post about mould, and the coverage of the lawsuit reached a national audience the post never would have.

Two employees posted a prank video from a restaurant kitchen that reached over a million people within days. The company decided not to respond, changed course after two days, opened a public channel and put its president on video, and a later survey found the response had largely restored trust.

The same visibility runs the other way through employment: employers check these networks before hiring, and people have been dismissed over posts they assumed a manager could not see.

MINI CASE **Four hundred reviews the firm did not know it bought**

Read the brief and the exhibit, then take the three decisions in order; the firm and every figure attached to it are hypothetical and were invented for this exercise.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

A kitchen appliance brand sells through a large online marketplace. A journalist has emailed asking about a report that flags most of the reviews on its best-selling blender as fabricated. Nobody at the brand commissioned reviews; the marketing agency it hired six months ago appears to have done so, under a contract clause promising rating improvement. The journalist wants a response by tomorrow. The firm is hypothetical and every figure attached to it is invented.

The firm

A kitchen appliance brand selling through a large online marketplace

What is alleged

That most reviews on its best-selling product are fabricated

Who commissioned them

An agency the brand hired, under a clause promising rating improvement

What the brand knew

Nobody inside the firm asked for reviews or approved any

The clock

A journalist has asked for a response by tomorrow

Exhibit A · Reviews on the blender, month by month — Reviews received before and after the agency was engaged, with the analysis tool's assessment. Every figure here is invented for practice.

PERIOD	REVIEWS RECEIVED	AVERAGE RATING	FLAGGED AS LIKELY FABRICATED
Six months before the agency	94	3.8	3
First month with the agency	61	4.6	44

PERIOD	REVIEWS RECEIVED	AVERAGE RATING	FLAGGED AS LIKELY FABRICATED
Second month	138	4.7	119
Third month	151	4.8	134
Fourth to sixth month	212	4.7	181
Sales change over the same six months	Up 31 percent	-	-

1. The chief executive wants to say the firm did nothing wrong, since it never commissioned a review. Why is that a poor response?

- **A.** It is inaccurate, since paying an agency for a rating improvement is commissioning reviews — The contractual point is arguable and would be argued somewhere else. Leading with it puts the firm's culpability at the centre of a story that was about whether the ratings can be trusted.
- **B.** It answers a question about blame while the public question is about the ratings — **correct.** Buyers want to know whether the rating means anything now. A statement about who is to blame leaves that unanswered, and the chapter's point about trust is that damage lands on the review system and the brand together.
- **C.** It will not be believed, so a firmer denial would serve the firm better — Escalating a denial makes matters worse, and the exhibit is a public artefact that any journalist can reproduce. Firmness is not the variable in play.
- **D.** It concedes the agency acted alone, which weakens any later claim against it — Preserving a claim against the agency is a real commercial interest, and it is a reason to word the statement carefully rather than a reason the response fails.

2. Sales rose thirty-one percent over the same period. What should the firm conclude from that line?

- **A.** The reviews worked, so the commercial damage of removing them will be significant — The reviews may well have worked and the exhibit does not establish it. Six months also covered whatever else the agency did, along with any pricing, seasonal or promotional change.

- **B.** The rise is unrelated, since a rating change cannot move sales by that much — A rating rise of nearly a full point on a marketplace listing plausibly moves sales substantially, so dismissing the connection is as unsupported as asserting it.
- **C.** The rise is now a liability, because it is evidence the firm benefited from this — It will certainly be read that way by others, and treating the firm's own inference as settled because an opponent would draw it is the wrong way round.
- **D.** Nothing yet, because the exhibit cannot separate this rise from other causes — **correct.** One line showing two things moving together, over a period when several things changed, supports no causal claim. Saying so is the analytically honest position and the one that survives being questioned.

3. What should the firm actually do in the next twenty-four hours?

- **A.** Respond publicly, say what it found and what it is doing, and fix the contract — **correct.** The chapter's playbook says the first day counts and that the response must be fast and appropriate. Acknowledging, stating what was found and naming the corrective action is the shape that a later survey found restored trust in its own example.
- **B.** Say nothing until the review analysis has been independently verified — Verification is prudent and takes longer than the story does. The chapter's case of a firm that waited two days before responding is the cautionary version of exactly this instinct.
- **C.** Ask the marketplace to remove the flagged reviews before responding publicly — Removing the evidence before commenting reads as concealment whatever the intent, and the flagged reviews are already documented in a report the journalist has.
- **D.** Publish the agency's name and the contract clause, and let the record speak — Naming the agency first makes the story about a contractual dispute between two companies, which is the least interesting version to everyone except the two companies.

What this case was really about. Two things are happening at once and they need separating. There is a commercial and legal matter with the agency, and there is a public question about whether the ratings mean anything, and only the second is urgent. The exhibit is deliberately seductive: a sales line placed beside a rating line invites a causal claim neither supports, and resisting that is the same discipline the analysis supplement asks for. What survives is the chapter's own playbook. Answer on the first day, say what you found, say what you are changing, and accept that the sentence about who is to blame is the one nobody outside the firm is waiting for.

Two more inversions, further from the marketing department

The chapter carries two cases that are not about a brand's reputation at all, and both run on capabilities earlier sections presented as benefits.

Cheap production and free distribution serve any message. The chapter's account of extremist propaganda turns on that and nothing else: portable cameras and simple editing software make high-quality video cheap, and social platforms distribute it globally at no cost, including to recruit teenagers.

A contribution buys a promise, not a product. Crowdfunding raises small amounts from many people for projects conventional finance would refuse, and the chapter's examples include campaigns that raised millions and shipped to a fraction of their backers, one that never produced a prototype, and one under investigation.

Neither is a failure of the technology. Both are the reach and the low cost this chapter has been recommending, pointed somewhere nobody intended.

The playbook

The chapter gives four preparations, and their value is that all of them are done before anything happens.

Assemble a crisis team in advance, drawn from inside the firm and outside it, so that nobody is deciding who has authority to speak while the clock is running.

Name your worst social media nightmare, and work out what its early signals look like, including the search terms an opponent would use.

Monitor the environment continuously and stay connected and responsive, because a complaint you have not seen is one you are not answering.

Act fast, because the first day decides the shape of it, and a response arriving after the story has settled is answering a version of events somebody else wrote.

Put the whole sequence in order, preparation and response together, and notice how much of it happens before there is anything to respond to.

SEQUENCE

The crisis playbook, preparation and response

Drag the steps into order; each one explains why it sits where it does, and most of this sequence happens before there is anything to respond to.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

The chapter's four preparations and the response that follows them. Put the whole sequence in the order it has to happen.

1. **Assemble a crisis team, from inside the firm and outside it** — First, and on an ordinary day. Deciding who has authority to speak, and which lawyer is on the call, is not something to work out while a story is spreading.
2. **Name the worst thing that could plausibly be said about you** — Second, because you cannot watch for signals until you know what you are watching for. The chapter asks firms to identify their own nightmare explicitly rather than generally.
3. **Work out the search terms an opponent would use, and watch them** — Third, and it follows directly from the previous step. Your own brand name is the term you would think of; the useful ones are the terms somebody attacking you would choose.
4. **Monitor continuously, and stay connected and responsive** — Fourth, and it is a standing activity rather than an event. A complaint nobody has seen is a complaint nobody is answering, and monitoring doubles as ordinary business intelligence.
5. **When something emerges, act inside the first day** — Fifth. The chapter is specific that the first twenty-four hours count, because a story left unanswered settles into the version somebody else wrote.
6. **Answer on the channel where it happened, not only on your own** — Sixth. A statement on a corporate site reaches people already looking for the firm; the audience that matters is the one on the platform the material spread across.
7. **Fix the underlying cause, and say specifically what you fixed** — Last, and it is what the chapter's most successful example actually did: found those responsible, closed and cleaned the site, and described the change rather than promising to do better.

Firms do this in practice through social media monitoring. The chapter's example is a manufacturer that opened a listening centre tracking mentions of itself in about a dozen languages, which lets it answer a customer before a problem spreads and doubles as business intelligence.

WALKTHROUGH

The first day of a viral complaint

Choose at each step; every option shows what it would lead to, including the ones you did not take.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

A customer's video about your product is spreading on a video platform. It is accurate. Four decisions across the first day, at a hypothetical firm.

Decision 1. The video has around forty thousand views and is climbing. Your monitoring caught it about an hour ago. What is the first move?

- Post a holding statement immediately, since the first day counts — Speed matters and a statement that says nothing spends the firm's one chance at a first impression. The chapter asks for a fast and appropriate response, and appropriate requires knowing what happened.
- Convene the crisis team and establish what actually happened — **the stronger call**. Right. The team exists so this takes minutes rather than a morning, and a response issued before anybody has established the facts is the one that has to be retracted, which starts a second story.
- Contact the platform and request the video be taken down — A takedown of accurate criticism is itself the story, and copies exist within hours. This is the move that converts a complaint about a product into a complaint about the company.
- Wait for the view count to plateau before deciding anything — This is the decision the chapter's restaurant example actually made, and it cost two days. By the time the count plateaus the account has been written by other people.

Decision 2. The facts hold up: the customer is right, and the fault is yours. Where do you respond?

- In a press release, which is the firm's formal channel for statements — A press release reaches journalists and reaches almost nobody watching the video. It also reads as a firm addressing a different audience about them rather than to them.
- On the firm's own site, so the wording stays under the firm's control — Control of wording is real and control of audience is not. People who saw the video have no reason to visit your site, so the response is published where the problem is not.
- Privately to the customer, offering to resolve it if the video comes down — A resolution conditional on silence is exactly the arrangement that becomes the next video. The private fix is worth doing, and attaching a condition to it is not.
- On the platform the video is on, and to the customer directly — **the stronger call**. Right. Answering where the audience already is, and reaching the person themselves, is the pattern the chapter's successful example follows. Both parts matter, and neither substitutes for the other.

Decision 3. A colleague notices the customer has a small following, and suggests the firm can wait it out.

- Point out that the audience is whoever shares it, not who follows them — **the stronger call**. Right, and the chapter has the case for it: a tenant with a couple of dozen followers reached a national audience once the dispute itself became the story. Follower counts predict very little about reach.
- Agree, and reduce the response to a single reply on the video — A minimal response is defensible on its merits and not for this reason. Sizing the reply to the poster's following is the reasoning error, whatever reply you end up with.
- Agree, but ask monitoring to re-check the numbers every hour — Watching more closely while doing nothing is still doing nothing, and it hands the initiative to whoever picks the story up next.

- Disagree, and escalate the response because the firm cannot risk it — Over-responding has its own costs: it draws attention the story had not yet earned, and it signals that pressure works, which shapes what arrives next.

Decision 4. The immediate response has gone out and the coverage is easing. What closes this properly?

- Publish an apology from the chief executive and consider the matter closed — An apology without a described change invites the same complaint again, and the second occurrence is materially worse because the record of the first is permanent.
- Increase advertising for a period to move attention onto other messages — Buying attention against an unresolved complaint is expensive and reads as exactly what it is. The chapter's rule that the internet never forgets means the material stays retrievable regardless.
- Fix the cause, then say specifically what changed and how it was checked — **the stronger call.** Right. The chapter's example named what had been found, what was closed and cleaned, and what would change in hiring. Specificity is what a survey later found had restored trust, not the apology.
- Add a review process so no employee can post about the firm publicly — This answers a problem the firm did not have, since the video came from a customer. It also imposes a control that people route around, which the chapter warns about directly.

What anonymity does, and what algorithms do

Anonymity online has legitimate uses and a well-documented cost. People who cannot be identified are more likely to enter **deindividuation**, a loss of self-awareness associated with antisocial behaviour.

An **internet troll** posts inflammatory content deliberately to provoke, and much of that behaviour is **cyberharassment**: false accusation, defamation, threats, identity theft and doxing, which is publishing somebody's private details. The chapter's account of one coordinated campaign against several women in the video game industry established a template that is still followed.

The other societal shift is quieter. Editorial judgement about what a public sees has largely been replaced by algorithms predicting what will hold attention, which filters out material that conflicts with a user's existing views. The chapter's balance is worth keeping: the same democratisation lowered the cost of being heard for candidates and causes that could not previously afford it.

A firm cannot prevent any of this. It can decide in advance who speaks, what it is watching for, and how fast it will answer — and those three decisions, made on an ordinary day, are the whole of what separates a bad week from a lasting one.

CHECK**Reach, reviews, and response**

Three questions on how this goes wrong and what to do about it; every option explains itself, including the ones you did not pick.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. What is the durable damage from fabricated product reviews, in the chapter's account?

- **A.** Trust in review systems erodes, so buyers make worse decisions everywhere — **correct.** The chapter's conclusion is that the practice undermines trust in review systems and leads customers to make suboptimal purchasing decisions, which is a cost borne by everybody rather than by the parties involved.
- **B.** The sellers who buy them gain an unfair advantage over honest competitors — The advantage to the buyer of fake reviews is real, and it is the local effect. Enforcement action exists precisely because that part is visible and addressable.
- **C.** Marketplaces face legal exposure for hosting reviews they know to be false — Marketplaces have pursued sellers and brokers, which shows exposure running the other way. Platform liability is not the harm the chapter identifies.
- **D.** Genuine reviewers stop writing once they realise their reviews are outnumbered — Plausible and not what the chapter argues. Its concern is with the reader's ability to rely on what is there, rather than with the supply of honest reviews drying up.

2. A firm's monitoring watches only for mentions of its own brand name. What does the chapter suggest it will miss?

- **A.** Complaints about a product where the brand name is not spelled correctly — Misspellings are a genuine gap and a technical one that most monitoring tools handle with fuzzy matching. It is not the gap the chapter is pointing at.
- **B.** Discussion using the terms an opponent would choose — **correct.** The chapter asks firms to identify their worst nightmare and to know the signs to look for, including the search terms their opposition could use, which are rarely the firm's own vocabulary.
- **C.** Anything posted on platforms the firm does not itself have an account on — Monitoring tools index public content regardless of whether the firm has an account. Absence from a platform limits where a firm can reply, not what it can see.
- **D.** Coverage in traditional media, which social monitoring tools do not index — Traditional coverage is usually indexed and is downstream anyway. By the time a story is in the press the early signal has already been missed.

3. Why does the chapter treat online anonymity as a genuine dilemma rather than a problem to remove?

- **A.** Because removing it is technically impossible on a network designed as this one was — The chapter does say the design makes anonymity hard to remove, and difficulty of removal makes it a practical problem. A dilemma requires that removing it would also cost something worth having.
- **B.** Because most harassment is committed by identifiable people rather than anonymous ones — The chapter's account runs the other way: most trolls operate anonymously, which is part of why prosecuting the behaviour is so difficult.
- **C.** Because platforms have a commercial incentive to keep anonymous accounts active — Platform incentives are worth examining and are not the chapter's argument here, which is about the value anonymity has for the people using it.
- **D.** Because it protects privacy and dissent as well as enabling harassment — **correct.** The chapter opens by noting the legitimate need for anonymity, naming privacy protection and freedom from oppression, before describing deindividuation and the harassment that follows from it. Both sides are real, which is what makes it a dilemma.

Collaboration read through the five forces and the value chain

You now know what every tool in this chapter does. That is worth almost nothing on its own. A manager who can list collaboration technologies and a manager who can say which competitive pressure each one relieves are doing different jobs, and only the second one is doing analysis.

What this section is. It is an application supplement rather than a textbook objective. It teaches no new framework and assumes you have already met the five competitive forces and the value chain in the strategy chapter. Every firm in it is hypothetical, and it solves no graded case for you.

The two frameworks, in one list each

The five forces describe the pressure on a firm's profitability from outside it. A force is not good or bad; it is high or low, and the analyst's job is to say which and on what evidence.

Rivalry among existing competitors is pressure from the firms already selling what you sell, into the market you are already in.

Threat of new entrants is how easily somebody who is not yet competing could start, which is a question about barriers rather than about anybody in particular.

Threat of substitutes is a customer meeting the same need another way entirely, without buying from anybody in your industry.

Bargaining power of buyers is your customers' leverage over your price and your terms, including how cheaply they can leave.

Bargaining power of suppliers is the leverage held by whoever supplies your inputs, and access to customers is an input like any other.

The value chain answers the other half of the question: not what is pressing on the firm, but where inside it value is actually created. Five primary activities carry the product, and four support activities carry the firm.

Inbound logistics, operations and outbound logistics are the physical spine: getting inputs in, turning them into something, and getting the result to the customer.

Marketing and sales, and service are the customer-facing pair, covering everything that persuades somebody to buy and everything that happens once they have.

Firm infrastructure is the shared capability every other activity draws on, including planning, finance, legal and the systems that hold the firm together.

Human resource management, technology development and procurement are the three remaining supports: finding and keeping people, improving how the firm makes things, and buying what it needs.

A recommendation that does not land in one of those nine is a recommendation nobody can act on.

Both are worth practising outside this module too. A companion page, [Five Forces and the Value Chain — a working review](#), turns them into a rater, an evidence sort, an investment simulator and three case studies.

Which force does each tool move?

Take the forces one at a time. For each, the chain runs the same way: what the force is, what in this chapter moves it, what evidence you would need to claim that, and what a firm can actually do about it.

FRAMEWORK

The five forces, one at a time, with this chapter's evidence

Step through each force; the chain reads as what the force is, what moves it, what evidence would support the claim, and what a firm can do. The situations are generic rather than any particular company.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Threat of new entrants — How hard is it for somebody new to start competing with you?

What it is (The cost and difficulty of entering) → What moves it here (Cloud tools, and network effects) → Evidence to look for (Entrants launching without capital) → What a firm can do (Build what an entrant cannot copy)

- Cloud-based collaboration tools raise this threat against you. A competitor now assembles a distributed team with free software and no premises, so the coordination capability you spent years building is available on a subscription.

- Network effects push the other way and are the strongest barrier in this chapter. A community, a review corpus or an internal knowledge base grows more valuable with every participant, and a new entrant starts at zero on all three.
- The evidence is concrete: how many competitors appeared in the last two years, what they needed to start, and how long a new customer waits before your accumulated reviews or community become the reason they choose you.

Bargaining power of buyers — How much can your customers push you on price and terms?

What it is (Customers' leverage over you) → What moves it here (Reviews, comparison, public complaint) → Evidence to look for (Price sensitivity and switching) → What a firm can do (Compete on what is not comparable)

- Social media raises buyer power, and this is the force this chapter moves furthest. Buyers arrive having read what other buyers said, and a complaint from one of them reaches an audience that used to require a broadcaster.
- The subtler shift is that the buyer's information is now better than the seller's account of itself. A product page written by the firm competes with a hundred owners describing what actually happened.
- The evidence is in the returns log, the abandonment rate, what proportion of buyers arrive from a comparison or a review, and how quickly a public complaint changes anybody's behaviour.

Bargaining power of suppliers — How much leverage do the people you depend on have over you?

What it is (Leverage held by your inputs) → What moves it here (Platforms, and the labour market) → Evidence to look for (Share of business through one channel) → What a firm can do (Own a relationship you can keep)

- The platform that carries your customers is a supplier, and an unusually powerful one. It lends you its audience, keeps the relationship, sets the commission, and can change either without negotiating.
- Crowdsourcing and open source cut the other way by reducing what a specialist supplier can charge. A labelling agency, a stock photography library or a proprietary software vendor each faces an alternative that did not exist.
- The evidence is the share of revenue arriving through any single channel, what that channel keeps, whether the firm holds the customer's contact details, and what the switching cost would be if terms changed tomorrow.

Threat of substitutes — What else could your customer do instead of buying from anybody in your industry?

What it is (A different way to meet the need) → What moves it here (Peer production and free tools) → Evidence to look for (Customers meeting the need elsewhere) → What a firm can do (Sell what the substitute cannot)

- This is the force people most often misidentify. A cheaper rival is rivalry; a customer meeting the same need by another route entirely is a substitute, and peer production is very good at creating those.
- A community that answers each other's questions substitutes for paid support. A free encyclopaedia substituted for a purchased one. A crowd that labels images substitutes for a service firm that employed people to do it.
- The evidence is where the need is being met now: forum threads answering what your support desk used to, or a free tool covering the work a purchase used to cover, counted rather than asserted.

Rivalry among existing competitors — How hard are the firms already in your industry competing, and on what?

What it is (Pressure from firms like yours) → What moves it here (Speed, visibility, and response) → Evidence to look for (How fast rivals answer, and where) → What a firm can do (Change the basis, not just the pace)

- Collaboration tools raise rivalry by compressing cycles. When a competitor can convene the right people in an hour and ship a change in a week, matching them is now part of competing rather than an ambition.
- Visibility is the other half. Every rival's response time, tone and failures are public, so customers compare service on evidence they can read rather than on reputation.
- The evidence is a comparison anybody can gather in an afternoon: how long each competitor takes to answer a public complaint, on which channels, and what a customer sees when they look for each firm by name.

The move that matters is the last one in each chain, and it is where most analyses stop too early. Naming a force is a diagnosis. Naming the evidence is what makes the diagnosis contestable, which is the only reason anybody should believe it.

MATCH

The observation, and the force it is evidence of

Pair each observation with the force it belongs to; every pair explains itself once matched, including why the neighbouring force is the wrong answer.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

A competitor launched last year with a distributed team, free collaboration tools and no premises

Threat of new entrants: the cost of entering fell The capability that used to require capital is now a subscription, so the barrier protecting incumbents has come down. Note this is about somebody arriving, not about how hard the firms already present are competing.

Buyers arrive having read other customers' reviews and compared prices before they contact you

Bargaining power of buyers: information moved to their side The customer now knows the market before the conversation starts, which is leverage. It would be rivalry only if the pressure came from what a competitor did rather than from what the buyer knows.

The marketplace carrying most of your orders raised its commission again this year

Bargaining power of suppliers: a channel you depend on The platform supplies your access to customers, which makes it an input, and its power grows with the share of your revenue it carries. Concentration in one channel is the whole of the problem.

Two competitors answer public complaints within the hour, and yours take three days

Rivalry among existing competitors: the basis of competition shifted Firms already in the industry are competing on a dimension that used to be invisible. Nobody new arrived and nothing was substituted; the same players changed what they compete on.

A free package maintained by volunteers now does most of what your paid product does

Threat of substitutes: another way to meet the same need The customer's need is met without buying from anybody in your industry, which is what makes it a substitute rather than a rival. Peer production is unusually good at producing these.

Every additional member makes the incumbent platform more useful, and almost nobody leaves

A barrier to entry the incumbent never had to build The network effect is doing the work of a moat. It was not purchased or engineered, it strengthens with use, and a challenger has to overcome it before its product is even compared.

Where does the work actually land?

A force tells you what is pressing on the firm. It does not tell you where to do anything about it. That is the value chain's job, and this chapter's technologies land in a surprisingly narrow part of it.

Marketing and sales and service take most of the customer-facing tools, because reviews, public replies, viral campaigns and listening centres all act on somebody who is buying or has bought.

Technology development takes the internal knowledge tools, because a wiki, a shared design workspace and an engineering forum all change how fast the firm can produce the next thing.

Human resource management takes the portal, the self-service form and the professional network, because all three change what employing somebody costs and how the firm finds them.

Firm infrastructure takes the intranet, enterprise search and the retention rules, because those are the shared capabilities every other activity draws on rather than work anybody does.

Place ten of the chapter's tools yourself, and pay attention to the two that could plausibly sit in more than one activity.

Sort

Where in the value chain does each tool do its work?

Place each tool in the activity where the work actually happens; two of these could argue for a second home, and the reason says which and why.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Marketing and sales — Everything that acts on somebody deciding whether to buy.

- Customer reviews published on your own product pages — They act before the purchase, on somebody deciding. A review also helps an existing owner, and the value the firm captures is the next sale, which is what places it here.
- A viral campaign built so that people pass it on — Brand awareness spread through a network is promotion by another mechanism. The network effect is how it travels; the activity it belongs to is unchanged.

Service — Everything that acts on somebody who has already bought.

- A live chat window staffed by support agents — It engages somebody who already owns the product and has a problem with it, which is the definition of the service activity.
- A social media listening centre that catches complaints early — Its purpose is answering a customer before a problem spreads. It also produces business intelligence, which is a genuine second use and a by-product rather than the reason it exists.

Technology development — Everything that changes how fast or how well the firm can produce the next thing.

- An engineering wiki holding problems the firm has already solved — It changes how quickly the firm can produce the next design, which is technology development. Nobody buys anything because of it and no customer ever sees it.
- A shared workspace where engineers revise the same design file — The chapter's own example cut product development cycle time, which is the clearest statement of what this activity is for.

Human resource management — Everything that changes what employing people costs, or how the firm finds them.

- An employee portal for benefits, leave and payroll self-service — It changes the cost of administering employment. The saving is real and lands in a support activity, which is why it never shows up in the price of the product.
- A professional network used to find and approach candidates — Recruiting is human resource management, and the network changes both the reach of a search and what it costs to run one.

Firm infrastructure — Shared capability every other activity draws on, rather than work anybody does.

- An intranet with enterprise search across every internal system — It is not work anybody does; it is a capability every other activity draws on, which is exactly what firm infrastructure means.
- Retention and access rules for collaboration channels, for compliance — Governance sits in infrastructure alongside legal, finance and planning. It enables the other activities to operate safely rather than producing anything itself.

The whole move, on one firm

Diagnosis, location and recommendation together, on a hypothetical firm facing a real kind of pressure. This is the shape the written analysis takes, compressed into three decisions.

MINI CASE

The forum turned out to be the moat

Read the brief and the exhibit, then take the three decisions in order; the firm and every figure attached to it are hypothetical and were invented for this exercise.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

A small firm sells electronics kits to hobbyists and has run a busy user forum for nine years, where owners answer each other's questions and publish their own projects. A large consumer electronics manufacturer has just launched a near-identical kit at two-thirds the price, with a support telephone line and no community. The small firm's owners are deciding how to respond. The firm is hypothetical and every figure attached to it is invented.

The firm

A small maker of electronics kits for hobbyists, twenty-two staff

What it also runs

A user forum, nine years old, where owners answer each other

What changed

A large manufacturer launched a near-identical kit at two-thirds the price

What the competitor has

Scale, distribution, and a support telephone line

What the competitor lacks

Any community, and nine years of accumulated answers

Exhibit A • What the forum does, counted — A year of forum activity beside the firm's own support and sales figures. Every figure here is invented for practice and none of it is reported data.

WHAT WAS COUNTED	FIGURE	WHAT IT WOULD HAVE COST OTHERWISE
Questions answered by other owners, not by staff	11,400	About US\$228,000 in support time
Questions that reached the firm's own support desk	1,900	Already paid for in staff time
Project write-ups published by owners	640	Content the firm did not write
New buyers who cite the forum as the reason they chose the kit	38 percent	Acquisition the firm did not pay for
Threads answering a question asked more than five times	2,050	A support cost that never recurs
Owners who have posted at least once	17 percent of buyers	The participation the rest depends on

1. The competitor is larger, cheaper and better distributed. Which force is the forum actually acting on?

- **A.** The threat of new entrants, because it is a barrier a challenger has to overcome — **correct.** The forum is a network-effect asset that grows with participation and cannot be bought or built quickly. That is a barrier to entry, and it explains why a larger, cheaper competitor is not automatically going to win.
- **B.** Rivalry, because it is how the firm differentiates against a direct competitor — Differentiation against this competitor is what the forum does today, and framing it as rivalry misses why it works. A rival can match a feature; nine years of accumulated answers is not a feature.
- **C.** Buyer power, because owners who post are less price-sensitive than those who do not — Nothing in the exhibit measures the price sensitivity of contributors, and inventing that relationship to reach buyer power is exactly the unsupported move this supplement warns against.
- **D.** The threat of substitutes, because the forum substitutes for the firm's own support — The forum does absorb support work, and a substitute is something the customer uses instead of buying from the industry at all. Here it is part of what they bought.

2. Where in the value chain does the forum do its work? The exhibit supports more than one answer.

- **A.** Service only, since almost all of the counted activity is answering questions — Service is where most of the volume sits and it accounts for only part of the value. Stopping there would miss the acquisition line entirely, which is the larger commercial number.
- **B.** Marketing and sales only, since it is what thirty-eight percent of buyers cite — The acquisition effect is real and it is the smaller half of what the exhibit shows. Eleven thousand answered questions is service work by any definition.
- **C.** Service and marketing and sales, because it both resolves and persuades — **correct.** The exhibit deliberately holds two things: support work the firm did not pay for, and acquisition it did not pay for. An honest analysis names both, and the assessment asks for at least two areas for this reason.
- **D.** Firm infrastructure, since it is a shared capability the whole firm draws on — Infrastructure is capability every activity draws on, such as the intranet or the governance rules. The forum does specific customer-facing work, so it belongs with the activities that do it.

3. The owners can fund one initiative. Which recommendation follows from the analysis rather than from the panic?

- **A.** Cut the kit price to match the competitor, and absorb the margin loss — Price is the dimension the competitor chose because it wins there. Meeting it spends the firm's margin on a fight it is structurally placed to lose, and it does nothing to the asset the analysis identified.
- **B.** Invest in the forum's participation, since seventeen percent carries everything — **correct.** Every number in the exhibit rests on the seventeen percent who post, which makes participation the single point of failure and the only input that compounds. It answers the force the analysis named rather than the competitor's move.
- **C.** Add a support telephone line, so the firm matches the competitor's offer — A telephone line copies the competitor's weaker substitute for what the firm already has, adds fixed cost, and would reduce forum participation by removing the reason to post.
- **D.** Advertise the community heavily, so more buyers hear about it before choosing — Promotion converts an existing asset into more sales and does nothing to strengthen it. It is also the recommendation most likely to be chosen because it feels active, which is why it is worth naming as declined.

What this case was really about. Three moves, in order, and none of them optional. The observation is a cheaper competitor. The force is not rivalry, which is where instinct goes, but the threat of new entrants, because what protects this firm is a network-effect asset a challenger cannot buy. The location is two value-chain activities rather than one, since the exhibit measures both support absorbed and buyers acquired. And the recommendation follows from the force: strengthen the thing that compounds, rather than matching a price on a dimension chosen by somebody with more scale. Notice that the analysis also names what it declined and why, which is an analytical move rather than a hedge.

One warning before the next section. Every claim in this supplement is an argument rather than a fact about the chapter.

That a wiki sits in technology development, or that a network effect raises an entry barrier, is a reading you are being taught to construct and should be prepared to defend. The evidence line in each chain exists because a reading is only as good as what supports it.

A technology is not a strategy, and neither is a framework. The analysis is the join: this observation, therefore this force, therefore this place in the value chain, therefore this one thing to do. Everything in this module has been building the middle two steps, because they are the ones people skip.

CHECK

Forces, activities, and the join between them

Three questions on using the frameworks rather than reciting them; every option explains itself, including the ones you did not pick.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. A firm writes: our customers are very active on social media, so buyer power is high. What is wrong with it?

- **A.** Nothing, since customer visibility on social media is what raises buyer power — Visibility contributes to buyer power and does not constitute it. The sentence as written would be equally true of a firm whose customers cannot switch, which is where the reasoning fails.
- **B.** Social media activity is a substitutes question rather than a buyer-power question — The observation is about the firm's own customers, so it is not a substitute question. Misfiling it would replace one unsupported claim with another.
- **C.** Activity is not leverage; the claim needs evidence that buyers can act on the firm — **correct.** Buyer power is leverage: the ability to push on price or terms, or to leave. Evidence would be switching rates, price sensitivity, or a public complaint that visibly changed the firm's behaviour, none of which is in that sentence.
- **D.** Buyer power should be rated against industry averages rather than in absolute terms — Comparison against an industry is useful context and does not fix an unsupported claim. A claim with no evidence is unsupported at any benchmark.

2. Which pairing of force and value-chain activity is the strongest fit?

- **A.** High buyer power, answered by better inbound logistics from key suppliers — Inbound logistics governs how materials reach the firm and does nothing about customers' leverage. This is the shape of error where the recommendation lands in an unrelated activity.

- **B.** High threat of substitutes, answered by more spending on human resource management — Substitutes describe customers meeting a need elsewhere, and hiring more people does not address that. A response would have to change what the firm offers or what it costs.
- **C.** High supplier power from one platform, answered by owning the customer relationship — **correct.** A platform holding your customers is a powerful supplier, and the response is to build a relationship the platform does not intermediate, which is marketing and sales work aimed squarely at the force identified.
- **D.** High rivalry, answered by improving procurement terms on components — Cheaper components help margin and can support a price move. As a general answer to rivalry it names an activity without saying what dimension of competition is being contested.

3. Why does the analysis insist on locating the fix in a specific value-chain activity?

- **A.** Because the framework requires every recommendation to reference a named activity — Satisfying a framework is not a reason to do anything. If naming the activity added nothing, the requirement would be ceremony.
- **B.** Because it converts a diagnosis into a place where work can be assigned and measured — **correct.** A force says what is pressing on the firm; an activity says where somebody can be given the work, what it will cost and what to measure afterwards. Without that a recommendation cannot be acted on or judged.
- **C.** Because forces describe the industry, and activities describe the individual firm — The distinction between industry-level and firm-level analysis is real and is a description of what the two frameworks cover rather than a reason to combine them.
- **D.** Because a recommendation in a support activity is cheaper than one in a primary activity — Cost varies enormously in both, and this chapter contains cheap primary-activity changes and expensive infrastructure ones. Nothing about the category predicts the price.

Before moving to the written deliverable, check honestly whether you could make each of these moves unaided.

READY?

Can you make the three moves unaided?

Rate each statement honestly; anything you cannot do yet has a pointer back to the part of this module to reread.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

- I can name all five forces and say what each one is pressure from, without looking. Go back to the two-paragraph refresher at the top of this section, and to the framework walkthrough that follows it.

- I can name the nine value-chain activities and tell a primary one from a support one. Go back to the second paragraph of this section, and to the sort that places ten tools across five of the activities.
- I can explain why a network effect is an entry barrier rather than a form of rivalry. Go back to the first panel of the framework walkthrough, and to the last pair in the matching exercise.
- I can explain why a platform that carries your customers counts as a supplier. Go back to the supplier panel of the framework walkthrough, and to the third pair in the matching exercise.
- I can tell an observation from a force, and say what evidence would turn one into the other. Go back to the evidence line in each panel of the walkthrough, and to the first question of the closing check.
- I can take one observation and carry it through force, activity and recommendation in order. Go back to the mini case, whose three decisions are exactly those three moves in sequence.

Turning this chapter into the written analysis

An observation is not a diagnosis. A firm can count forum threads, reply times and marketplace commission and still hold nothing anybody can act on, because a figure never says which pressure produced it. This supplement practises the move from counted evidence to one recommendation, in the order a written analysis has to make it.

What this section is. It is an application supplement rather than a textbook objective. It teaches no new framework, and it assumes you have worked through the previous section. Everything here runs on a hypothetical firm, and it solves no graded case for you.

The shape of the deliverable

The written analysis this course asks for is fixed, and knowing its shape in advance is half the work. Four things have to be in it, and each one fails in a predictable way.

- 1 **Rate all five forces on the firm.** All five, including the ones that turn out to be weak, and each with the evidence that made you rate it that way.
- 2 **Find at least two places in the value chain where technology could create an advantage.** Two genuinely different activities, not one activity described twice.
- 3 **Recommend exactly one initiative.** One. With what it changes, what it costs, and what you would measure afterwards to know whether it worked.
- 4 **Write a comparison of 150 to 200 words against a company you choose.** Not a profile of that company: a lesson drawn back to the firm you are advising.

The firm below is hypothetical and every figure attached to it is invented. Take the case first; the explanation after it reads as the debrief of what you have just tried.

MINI CASE

Four offices, one retiring generation, and a public that reviews you

Read the brief and the exhibit, then take the three decisions in order; the practice and every figure attached to it are hypothetical and were invented for this exercise.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

A regional architecture and engineering practice of seventy staff works from four offices, deliberately, because that is where its specialists live. Project teams are assembled across offices for every job. Three things arrived at once this year: a national practice opened an office in its region, a third of its senior engineers reach retirement age within two years, and clients have begun choosing consultants from a public directory that carries reviews. The practice is hypothetical and every figure attached to it is invented.

The firm

A regional architecture and engineering practice, seventy staff, four offices

How work is done

Project teams assembled across offices for every job

What arrived this year

A national competitor in the region, and a reviewed public directory

What arrives next

A third of senior engineers reaching retirement age within two years

The ask

One initiative, with what it changes, what it costs and what to measure

Exhibit A · Five things the partners can count — What a month of measurement produced, with what ignoring each one would cost. Every figure here is invented for practice and none of it is reported data.

WHAT THE PARTNERS COUNTED	THIS YEAR	WHAT IGNORING IT COSTS
Median days to answer a cross-office technical question	4.5 days, against 0.5 within an office	Project time bought and not used
Detailing decisions re-derived because nobody found the earlier one	62 in twelve months	Senior time spent twice on one answer
Share of new enquiries citing the public directory	41 percent, up from 6 percent	Selection happening before any contact
Median days to respond to a directory review	11 days; the national competitor takes 1	A visible comparison on a public page
Senior engineers reaching retirement age within two years	9 of 27	Knowledge leaving with no copy of it

1. A national competitor opened in the region and clients now choose from a reviewed directory. Which is the more significant force change, and why?

- **A.** The competitor, since a national practice can undercut a regional one on price — A larger competitor is real pressure and is the familiar kind. Price is also not established anywhere in the exhibit, so treating it as the mechanism is an assumption rather than a finding.
- **B.** The directory, since it moved the selection decision to before any contact happens — **correct.** The directory raises buyer power by moving information to the client's side: forty-one percent of enquiries now arrive after a comparison the practice was not present for. That is a structural change, and the competitor is one more entry on a page the client already reads.
- **C.** Neither; both are rivalry, and the practice should treat them as one problem — Collapsing the two loses the distinction that makes the analysis useful. One is a firm competing with you; the other changed how every firm gets chosen, including the new one.
- **D.** The competitor, since rivalry tends to dominate the other forces in professional services — Nothing establishes a dominant force in advance, and asserting one is the move this module warns against. The rating has to come from the evidence in front of you.

2. The four-and-a-half day answer time and the sixty-two re-derived decisions. Where do those land, and are they one problem or two?

- **A.** One problem in operations: both are project delivery running slower than it should — Calling both operations captures the delay and loses the second one entirely. Re-deriving a decision is not slow delivery; it is work being done that had already been done.
- **B.** Two problems in one activity: both of them sit somewhere in technology development, at different scales — Both do touch how well the firm produces the next thing, and treating them as one activity hides that they need different fixes: one is about reaching a person, the other about reaching a record.
- **C.** Two problems in two activities: coordination in operations, knowledge in technology development — **correct.** Waiting four and a half days for a colleague is a coordination cost in delivery, and re-deriving a decision sixty-two times is knowledge the firm holds and cannot reach. Two activities, two figures, and the analysis needs at least two areas anyway.
- **D.** One problem in human resource management, since both are about how staff work — Human resource management covers recruiting, developing and administering people. Neither figure is about employment terms, and filing both here would put the recommendation where nobody could act on it.

3. The partners can fund one initiative. Which recommendation is best supported by this exhibit?

- **A.** Answer directory reviews within a day, since the competitor comparison is visible — This is cheap, fast, visibly behind a competitor, and worth doing. It is also a service fix that changes nothing about the two largest structural risks, and the brief asks for the one initiative rather than the easiest.
- **B.** Capture detailing decisions as they are made, before nine seniors retire — **correct.** It answers the sixty-two re-derived decisions and the nine retirements at once, it is the only option addressing something the practice cannot recover once it is gone, and it lands in technology development where the firm's ability to produce the next project is decided.

- **C.** Open a fifth office in the competitor's city, to defend the region directly — A fifth office is a large capital commitment answering rivalry with scale, which is the dimension a national practice is better placed to win. Nothing in the exhibit supports it.
- **D.** Run all cross-office questions through a daily scheduled videoconference — A daily meeting compresses the four-and-a-half day wait and spends seventy people's time on questions that mostly concern two of them, and it does nothing about the answers being lost afterwards.

What this case was really about. The exhibit is arranged so the loudest event is not the most important one. A national competitor is visible and rivalrous; the directory quietly moved selection to before the practice is ever contacted, which is buyer power and the larger change. The two internal figures are deliberately separable, because the deliverable asks for at least two value-chain areas and they are genuinely different: reaching a person, and reaching a record. And the recommendation follows from what is irreversible. Review response time can be fixed next month and any time after that; nine retirements cannot be undone, which is what makes knowledge capture the one initiative this evidence supports.

What this chapter supplies to the frameworks

Each decision asked for the same two moves: name the pressure an observation belongs to, then say where in the firm a fix would land. A symptom is not a force. Five ideas from this chapter make that translation for you.

Network effects and accumulated participation raise a barrier to entry that a challenger cannot buy, which is why a community, a review corpus or a knowledge base is a strategic asset rather than a marketing channel.

A platform that carries your customers is a supplier whose power grows with every order it handles, because it lends you its audience and keeps the relationship attached to itself.

Reviews and public complaint move information to the buyer, so buyer power rises even when nothing about the product or the price has changed at all.

Internal knowledge tools sit in technology development, because what they change is how quickly and how well the firm can produce the next thing rather than how it sells the last one.

Coordination cost is a value-chain cost, so nine days of waiting for an answer is an operations and technology development problem with a number attached, not a complaint about meetings.

Working the evidence yourself

Naming the mapping is the point, and the naming has to start from something counted. That is the step readers skip, and the only one that produces facts somebody else can check. The sheet below holds one year of the firm's support channels.

SPREADSHEET

Turn raw channel counts into evidence

Type one formula for row 2 in each column and run it; it is applied down every row the way a filled-down formula behaves in a real sheet. The channels and figures belong to the hypothetical practice and are invented.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Columns: Channel · Enquiries received · Resolved by other customers · Resolved by staff · Staff minutes per resolution · Community resolution rate · Staff hours used · Priority?

1. Column F: what share of each channel's enquiries were resolved by other customers rather than by staff? Give it as a percentage, rounded to one decimal place.

```
=ROUND(C2/B2*100,1)
```

One channel resolves most of its own volume and the other three resolve none. That single ratio is the whole argument for treating a community as an asset rather than as a channel, and it is the number a competitor without one cannot produce.

2. Column G: how many staff hours did each channel actually consume? Multiply the staff resolutions by the minutes each one took, and convert to hours.

```
=ROUND(D2*E2/60,1)
```

Now the two columns disagree, and the disagreement is the finding. The forum takes by far the most enquiries and consumes the fewest staff hours of any channel here, while the page the firm controls least consumes the most.

3. Column H: mark any channel that consumes more than eight hundred staff hours for review, and mark the rest as held. Column G holds your own answer rather than sheet data, so work the hours out again inside this formula.

```
=IF(D2*E2/60>800,"Review","Hold")
```

A flag is a decision rule written down, which is the honest way to use a threshold. Notice what it selects: three channels for review, and the one that carries the most enquiries is not among them, which is the opposite of what enquiry volume alone would have told you.

A resolution rate and a staff-hour total tell you where the effort is going. They do not tell you which channel makes customers wait, which leaves work unfinished, or which one the firm does not control. Those are questions for the records themselves.

DATABASE

Ask the records this chapter's questions

Write a SELECT for each question and run it; any query returning the right answer is accepted. These twelve rows are one week pulled from the log, not the year the spreadsheet totals, so the counts are small and the ratios will not match it. The practice and its channels are hypothetical.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

posts — id, channel, topic, hours_to_reply, staff_minutes, resolved (12 rows)

channels — channel, owns_relationship, staff_rate (4 rows)

1. List the topic and the hours to first reply for every post on the forum.

```
SELECT topic, hours_to_reply FROM posts WHERE channel = 'forum'
```

Choosing columns and choosing rows are separate decisions. Read the result and the pattern is already there: forum replies arrive in under two hours, and three of the four consumed no staff time at all.

2. How many staff minutes did each channel consume this week? Show the largest first.

```
SELECT channel, SUM(staff_minutes) AS minutes FROM posts GROUP BY channel  
ORDER BY minutes DESC
```

Grouping turns a list of posts into a description of the channels. The channel with the most posts is not the one consuming the most staff time, which is the first sign that post counts are a poor way to judge a channel.

3. Which channels left more than one post unresolved? Show the channel and how many.

```
SELECT channel, COUNT(*) AS unresolved FROM posts WHERE resolved = 'no'  
GROUP BY channel HAVING COUNT(*) > 1
```

WHERE filters rows and HAVING filters groups, because at the WHERE stage nothing has been counted yet. The answer also says where unfinished work accumulates, which is on a channel somebody else owns.

4. What did each channel cost the practice in staff time? Show the channel, the minutes, and the cost, most expensive first.

```
SELECT p.channel, SUM(p.staff_minutes) AS minutes, SUM(p.staff_minutes *
c.staff_rate / 60) AS staff_cost FROM posts p JOIN channels c ON p.channel =
c.channel GROUP BY p.channel ORDER BY staff_cost DESC
```

This answer exists in neither table. Minutes sit on one and the rate sits on the other, and only the join shows that the most expensive channel is also the slowest to reply and the one where the practice does not own the relationship. The same table records who does own it, which is the supplier-power half of the story.

What separates a strong version from a weak one

The difference at each step is small, repeatable, and visible to anybody reading the two side by side.

STEP	WEAK VERSION	STRONGER VERSION
Gather	"Our social media presence needs work."	Median hours to first reply on each channel, beside two named competitors.
Rate	"Customers are very active online."	"Buyer power is high: buyers arrive having read reviews, and switching costs them nothing."
Locate	"We need a better digital strategy."	"The loss is in service and in technology development, and here is the figure for each."
Recommend	Three initiatives, none of them costed.	One initiative, with what it changes, what it costs and how you would tell.

WALKTHROUGH

Writing the analysis, decision by decision

Choose at each step; every option shows what it would lead to, including the ones you did not take. The firm is hypothetical.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

You have the practice's records and a week before the partners expect a document. Four decisions, in the order they arrive.

Decision 1. You have a week with the practice's records before you write anything. What do you come back with?

- A written summary of what the four office heads believe is going wrong — Their judgement is worth having and will help you read the numbers later. On its own a list of beliefs cannot be checked, and it tends to describe the month rather than locate a pressure.

- A review of the national competitor's website and brochures, noting every service the practice lacks — This produces a feature list, and a feature list quietly becomes the recommendation. You end up proposing work because a rival advertises it rather than because a force demands it.
- Every project file exported, so nothing at all is left out of the analysis — Completeness is not evidence. Without a question in mind you cannot tell which figures matter, and the work stalls in the data rather than in the argument you were asked for.
- Reply times by channel, re-derived decisions, directory share, and the retirement profile — **the stronger call**. Right. Each of these can be counted, each attaches to a force or to a value-chain activity, and each can be checked by somebody who doubts you. Evidence is what separates an analysis from an opinion.

Decision 2. Your first draft says: cross-office questions take four and a half days, and clients now choose from a directory. What is missing?

- Nothing. It states two observed facts, and an analysis should stay close to its evidence — Staying close to the evidence is right, and an analysis is more than the evidence. A description that names no pressure gives the reader nowhere to go and no way to test whether you are correct.
- It names no force. Say which pressure each observation is evidence of — **the stronger call**. Right. As drafted the sentence describes the year. Naming buyer power and placing the coordination cost inside the value chain is what lets a reader argue with you and what makes a recommendation follow from something.
- A stronger opening claim, such as calling this an existential threat to the practice — Force of language is not force in the framework sense. Escalating the wording adds urgency without adding a diagnosis, and it makes the whole piece harder to trust.
- More figures, so add the re-derived decisions and the retirement count to that sentence — More numbers in the same sentence still leave it a description. The figures belong in the evidence; the sentence exists to say which pressure they are evidence of.

Decision 3. You have three candidate initiatives you like, and the brief asks for one. What do you write?

- All three, ranked in order, so the partners can pick whichever suits the budget — Ranking looks generous and quietly hands the decision back to the reader. It also hides which pressure you were answering, and no single option ever gets costed properly.
- The most ambitious of the three, since a bolder call shows a firmer grasp — Ambition is not a criterion. The recommendation has to answer the pressure the evidence showed, and a bold initiative aimed at the wrong force is worse than a modest one aimed at the right one.
- One initiative, with the force it answers, what it costs, and what you would watch — **the stronger call**. Right. A single recommendation can be costed, argued and later judged. The other two can be named in a sentence as deliberately not chosen, which is itself an analytical move rather than a hedge.

- A new collaboration platform, because the national competitor is known to use one — This is technology with no force behind it. If you cannot say which pressure the platform relieves and where it sits in the value chain, the reader has no reason to accept it.

Decision 4. The last piece is a comparison of 150 to 200 words against a company you choose. Which draft does the work?

- Name what that company does differently, then what the practice should do or refuse — **the stronger call**. Right. The comparison exists to produce a lesson for the firm you are advising. Naming the difference in the vocabulary of the frameworks is what makes it a lesson rather than a story about somebody else.
- A short profile of the company, its history, its scale and its best-known work — This is the commonest failure of the whole piece. It reads well and says nothing about the practice, and by the end the reader knows more about a company nobody asked you to advise.
- A list of every difference between the two firms, so nothing relevant is left out — Length is not coverage. Most differences do not bear on the pressure you diagnosed, and burying the one that matters among twenty that do not is how a reader misses your point.
- An argument that the practice should copy the company as closely as its budget allows — Copying ignores what your own evidence said about this firm, which holds ground the comparison company may not hold at all. A lesson is not an instruction to imitate.

The comparison summary

The last piece is short and is where most of the marks are quietly lost. Three things make it work: pick a comparison with a real similarity or a real contrast, name what that company does differently in the vocabulary of the frameworks, and end on what your firm should therefore do or deliberately refuse to do.

A summary that spends its two hundred words admiring somebody else has answered a different question. Every sentence should be recoverable as advice to the firm in front of you.

READY?

Could you write the analysis now?

Rate each statement honestly; anything you cannot do yet has a pointer back to the part of this module to reread.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

- I can rate all five forces on a firm, including the ones that turn out to be weak, with evidence for each. Go back to the framework walkthrough in the previous section, and to the first decision of this section's mini case.

- I can name two genuinely different value-chain activities rather than describing one activity twice. Go back to the second decision of the mini case, and to the sort that places ten tools across five activities.
- I can explain why a platform that carries a firm's customers counts as a powerful supplier. Go back to the list of what this chapter supplies to the frameworks, and to the last database exercise, whose join shows who owns each relationship.
- I can turn raw channel counts into the two or three ratios an analysis actually argues from. Go back to the spreadsheet exercise under working the evidence yourself.
- I can state one recommendation with what it changes, what it costs, and how I would know whether it worked. Go back to the third decision of the walkthrough, and to the strong-against-weak table above it.
- I can write a comparison that ends in advice to my own firm rather than in a profile of somebody else. Go back to the closing paragraphs on the comparison summary, and to the fourth decision of the walkthrough.

REFERENCE

Chapter vocabulary

Every term the chapter defines, written so it makes sense on its own, with an example beside it. Search by word, or filter by the objective the term belongs to. Watch the pairs that sound alike and are not: cooperation and collaboration, a tag and a geotag, a wiki and a content management system, collective intelligence and crowdsourcing, a network effect and critical mass.

collaboration

5.1

Two or more people working together to achieve a common goal, inside an organization or across its boundaries.

For example: Three engineers in different countries producing one specification between them, rather than three opinions about it.

task force

5.1

A temporary work group with a finite task and a defined life cycle, formed to solve a problem that existing departments cannot.

For example: A group pulled from four departments for six months to fix a recurring delivery failure, then disbanded.

virtual team

5.1

A group whose members are drawn from different geographic areas and assembled as needed for a particular project, with membership that forms, grows, shrinks and disbands as the work requires.

For example: Programmers in one country, project managers in a second and testers in a third, working as one team on one release.

synchronous communication

5.1

Exchange that is coordinated in time, so the participants are present together and answers arrive immediately.

For example: A telephone call, a live chat window, or a videoconference where somebody can interrupt to disagree.

asynchronous communication

5.1

Exchange that is not coordinated in time, so a message waits until its recipient chooses to attend to it.

For example: Email, a text message, or a discussion board thread that collects replies over three days.

virtual meeting

5.1

A meeting held in an online environment rather than a room, which may run at the same time for everybody or spread across days.

For example: A teleconference for the same-time version, and a moderated discussion board for the different-times version.

collaboration software

5.1

Software that lets people work together, usually classified by whether it supports same-time or different-time work and whether the participants share a location.

For example: A shared document editor, a videoconferencing tool, and a project board are three points in the same category.

groupware

5.1

The older class of software from which collaboration tools evolved, built to let people work together more effectively, with the first mainstream product appearing in 1989.

For example: A shared mail, calendar and discussion system installed on a company's own servers in the early 1990s.

electronic communication tools

5.1

The family of tools that convey verbal and written information and move files, documents or other content between people.

For example: Instant messaging, email, voice mail, blogs, wikis and ordinary websites all belong here.

electronic conferencing tools

5.1

The family of tools that support information sharing and rich interaction, where the value comes from participants doing something together in the moment.

For example: Screen sharing during a design review, or a videoconference where tone and interruption matter.

collaboration management tools

5.1

The family of tools that facilitate meetings and manage a group's activity rather than carrying the conversation itself.

For example: Shared calendars, knowledge management systems, intranets and online document systems.

desktop videoconferencing

5.1

Video meetings run from an ordinary computer or mobile device, needing only a camera, a microphone, software and a fast connection.

For example: Joining a client meeting from a laptop in a hotel room using software that costs nothing.

webcam

5.1

A small video camera connected directly to a computer or built into a laptop screen, which is one of the four things desktop videoconferencing requires.

For example: The camera above a laptop display that most people never think about until it is covered with tape.

intranet

5.1

A private network using web technologies, used to move proprietary information securely inside one organization, with its pages behind the company firewall.

For example: An internal site where a human resources employee sees payroll figures and a warehouse supervisor does not.

virtual private network (VPN)

5.1

An encrypted connection carried across the public internet that lets a remote device use an organization's internal resources as though it were inside the building.

For example: A salesperson opening the company benefits page from an airport, through a tunnel nothing in between can read.

enterprise search

5.1

Search built to retrieve content from an organization's own documents, databases, applications and cloud storage, which is a different problem from searching linked public pages.

For example: Finding the one specification a colleague wrote three years ago, in a language you do not read, on a server you did not know existed.

social intranet

5.1

An internal site where every user can create and update content and connect to whoever wrote it, rather than only a small group holding editorial rights.

For example: A workshop supervisor correcting a procedure page directly, instead of emailing the communications team about it.

employee portal

5.1

An intranet-based entry point that gives each employee the tools and information their own job requires, including self-service for administrative tasks.

For example: One page that shows a nurse their roster, their payslip and the forms they are entitled to file.

employee self-service

5.1

The practice of letting employees file and manage their own administrative transactions rather than sending paper to a central team.

For example: Changing a pension contribution through a form that checks the entry as it is typed, at a cost of a few cents rather than tens of dollars.

Web 2.0

5.2

The dynamic web applications that let people collaborate and share content online, turning a user from a consumer of content into a creator of it.

For example: A video site whose entire catalogue was uploaded by the people watching it.

user-generated content

5.2

Material created and uploaded by a site's own visitors rather than by the organization running the site.

For example: Product reviews, uploaded videos, and encyclopaedia articles written by the people who read them.

social media

5.2

Applications embodying the collaborative and sharing values of the later web, which let people communicate, interact and collaborate; also called social software.

For example: Blogs, wikis, media sharing, social networks and bookmarking services are all in the category.

semantic web

5.2

A set of design principles intended to let computers index pages by meaning rather than by keywords, so a search returns an answer instead of pages containing the right words.

For example: Marking up the parts of an address so a machine can tell a street from a postcode, rather than seeing one string of text.

webmaster

5.2

The person responsible for creating and maintaining a site's pages, and the person search engines ask to add structured markup.

For example: Whoever actually edits the page templates, which in a small firm is often somebody with another job title entirely.

Web 3.0

5.2

The contested label for the next wave of internet technologies, framed variously around mobility, around context, or around artificial intelligence extending what the web can do.

For example: Content filtered by where you are, what time it is and what you were doing an hour ago, rather than by what you typed.

blog

5.2

An online diary made up of dated entries, published on a site the author controls, usually with comments or a linked discussion.

For example: A company's official site where it explains a decision at length before anybody else frames it.

blogging

5.2

The practice of keeping such a diary, which began as a way for people with no technical skill to publish and became a way for firms to sound like people.

For example: A software company posting the reasoning behind a change, rather than a press release announcing it.

amateurization of journalism

5.2

The criticism that publishing without editorial process trades accuracy for speed, producing unverified sources and visible bias alongside genuinely faster news.

For example: A breaking story that reaches readers hours before a newsroom could confirm it, and is wrong in a way nobody catches for a week.

microblogging

5.2

Broadcasting short status updates in near real time to anyone who has chosen to follow, in contrast with a social network where the author chooses the audience.

For example: A company account answering a customer complaint in public, where every other customer can see the answer.

hashtag

5.2

A word or phrase marked with a leading symbol to indicate a post's topic and relate it to other posts about the same thing.

For example: A tag a firm creates for a campaign, which its critics can use just as easily as its customers.

trending

5.2

The state of a word, phrase or topic being tagged at a greater rate than others, whether through a concerted effort or because something happened.

For example: A topic climbing within minutes of a sporting result, then disappearing by the next morning.

instant messaging

5.2

Real-time written conversation between two people or a group, with immediate feedback from the other participants; also called online chat.

For example: A team platform organised into channels, or a live chat window on a shop's own website.

cooperation

5.2

An arrangement creating a win-win, where one participant's success improves the chances of the others without anybody working on the same thing at the same time.

For example: A video embedded in somebody else's article, which improves the article and finds the video an audience.

webcast

5.2

Digital audio or video distributed for on-demand playback whenever the audience chooses; also called a podcast, despite having nothing to do with any particular device.

For example: A recorded lecture a student plays on the way to work, or a shareholder meeting made available afterwards.

RSS

5.2

A feed format publishers provide so that subscribers are notified of new content automatically, instead of checking several sources by hand.

For example: A reader that collects new posts from forty sites and presents them in one list.

metadata

5.2

Data about data, describing content in terms of who, where, when and why, some captured automatically and some added by a person.

For example: A document's author and save time, or a photograph's date, shutter speed and coordinates.

tag

5.2

A keyword added by a person to digital content so that it can be categorised and found later, as distinct from the metadata a device records by itself.

For example: The names of the people in a photograph, which no camera can record and no search can find without.

tag cloud

5.2

A display of tags in which each word's size represents its frequency or importance, so the shape of a collection is visible without reading it.

For example: A view of a site's topics where the largest words show what it is mostly about.

geotag

5.2

Geospatial metadata such as latitude, longitude or altitude attached to a photograph, video or post, allowing it to be placed on a map.

For example: Coordinates a phone attaches to every picture, which is how a photograph revealed where a man in hiding was staying.

social bookmarking

5.2

Sharing the pages people found worth keeping, and building shared categories for them, so the collection grows more useful as more people take part.

For example: Inside a firm, a way of finding who already knows about a topic rather than which document mentions it.

folksonomy

5.2

A categorisation system that emerges from the labels many users apply, rather than one imposed by a librarian or an administrator.

For example: The set of topic labels a bookmarking community settles into over a few years, without anybody designing it.

social cataloging

5.2

The building of shared catalogues of items rather than links, on specific topics such as books, music or academic citations.

For example: A shared citation library researchers draw on to assemble reference lists instead of each building one alone.

cloud-based collaboration tools

5.2

Services that store shared files and often the productivity software itself remotely, so several people work the current version from any device with a browser.

For example: A document three people edit at once, which none of them has ever emailed to the others.

content management system (CMS)

5.2

Software that lets people publish, edit, version and retrieve digital content without going through a technical department, with roles controlling who may do what.

For example: A product manager correcting a description on the company website, without raising a request with anybody.

learning management system (LMS)

5.2

Software that carries teaching material, administers assessment, and handles self-service tasks such as enrolment and viewing grades.

For example: The system a course runs on, increasingly carrying discussion and group work alongside the content.

collective intelligence

5.2

The idea that a distributed group holding divergent information and expertise can outperform the capabilities of individual experts.

For example: An encyclopaedia article whose quality rests on hundreds of small corrections rather than on one commissioned author.

peer production

5.2

The creation of goods or services by self-organizing communities, where the outcome depends on incremental contributions and anyone can improve it.

For example: An open encyclopaedia and an open source project are the chapter's two examples of the same mechanism.

open source software

5.2

Software created, maintained and updated by volunteers worldwide, whose source code anyone may read, use and propose changes to.

For example: A firm contributing to a project it depends on, to influence its direction and to audit code it is trusting.

wiki

5.2

A site allowing people to create, edit or delete content and to discuss changes, backed by a database that keeps every prior version so anything can be reverted.

For example: An internal knowledge repository written by the engineers who know things rather than by whoever was assigned to document them.

crowdsourcing

5.2

Distributing small, self-contained tasks to whoever will take them, usually through a marketplace and usually for a few cents each.

For example: Tagging images, fixing product titles, or transcribing short audio clips, posted by a firm and completed by strangers.

human intelligence task (HIT)

5.2

A small, self-contained piece of work that a person can do easily and a computer cannot, which is the unit crowdsourcing marketplaces trade in.

For example: Deciding whether a photograph contains a receipt, which takes a person two seconds and a program a great deal of effort.

open innovation

5.2

The practice of bringing external stakeholders into a firm's innovation process, forming research and development networks beyond its own staff.

For example: A manufacturer running a public challenge for a materials problem its own engineers have not solved.

social networking

5.2

Building online communities where individuals with diverse interests meet, communicate and collaborate, whether generally or around a profession.

For example: A professional network where people post career profiles and are contacted about opportunities.

social online community

5.2

The group that forms on such a site, held together by shared interest rather than by employment or geography.

For example: A network built around checking in at places, or one built around a single profession.

consumerization of information technology

5.2

The pattern where technologies designed first for consumers make their way into organizations, because that is what people already know how to use.

For example: An internal social network deliberately built to resemble the consumer apps its users are accustomed to.

network effect

5.2

The property whereby each additional participant makes a service more valuable to every existing participant, so growth reinforces itself.

For example: A social network that became more attractive to join precisely because so many people had already joined.

clickbait

5.2

An incomplete or tantalising headline written to be clicked rather than to describe accurately, because the revenue comes from the visit rather than the reading.

For example: A headline promising a revelation that the article then never quite delivers.

viral marketing

5.2

Using the network effect to increase brand awareness, by making content that entices viewers to pass it to other people.

For example: An email service adding a short line about itself to every message its users sent, which brought in millions of subscribers.

social-local-mobile

5.2

The intersection of social recommendation, location awareness and mobile devices, where a phone that knows where it is meets reviews written by other customers.

For example: Opening an app to find a well-reviewed restaurant within walking distance, and being shown an advertisement from one of them.

crowdfunding

5.3

Funding a project or venture by raising small amounts from a large number of people, usually online, for projects traditional finance would not support.

For example: A product that raised millions from tens of thousands of backers and shipped to a fraction of them years later.

critical mass

5.3

The level of participation a social application has to reach before taking part repays the effort of taking part.

For example: A staff directory with four completed profiles, which nobody consults, so nobody completes a fifth.

technological inertia

5.3

The reluctance to move to a new application until a real and tangible benefit is visible, with familiarity counting as a benefit already held.

For example: A team that keeps using email for something the new platform does better, because email is already understood.

deindividuation

5.3

A loss of self-awareness that people are more likely to enter when they cannot be identified, and which is associated with antisocial behaviour.

For example: Behaviour in an anonymous forum that the same person would not sign their own name to.

internet troll

5.3

Somebody who creates discord online by starting arguments or posting inflammatory content with the deliberate intent of provoking an emotional response.

For example: An anonymous account that appears in a discussion only to make it unpleasant, and then moves on.

cyberharassment

5.3

Use of the internet to stalk or harass a person, group or organization, covering false accusation, defamation, threats, vandalism and identity theft.

For example: A coordinated campaign against several women in one industry, which established a template still followed.

doxing

5.3

Researching and publishing somebody's personally identifiable information, which converts online hostility into a physical safety problem.

For example: Posting a target's home address and workplace so that others can act on it.

filter bubble

5.3

The result of feeds assembled by algorithms that predict what will hold attention, which filters out material conflicting with a user's existing views.

For example: Two people searching the same thing on the same day and being shown two different pictures of what is happening.

PUT IT TOGETHER

Final challenge

Situations rather than definitions. Each one puts you in front of a decision somebody actually has to make — adopt it or refuse it, answer the post or stay quiet, name the force or keep describing the symptom — and asks what is really going on. Every organization described is hypothetical. Answer all of them, then score: the result breaks out by objective, so a weak area points at a section to reread instead of disappearing into one number.

FINAL CHALLENGE

Final challenge

Answer key — the interactive, self-scoring version loads when scripting is available.

1. **5.1** At a hypothetical instrument maker, Larchfield Controls, a calibration question needs a metallurgist in one country, a production lead in a second and a compliance officer in a third. Handled by email it takes eight days and about two hours of actual work. The operations director proposes a rule that internal email must be answered within four hours. What is wrong with that proposal?
 - **A.** It treats a turnaround problem as an effort problem rather than removing the delay — **correct.** Two hours of work spread across eight days is a mode problem: each round of clarification costs a full turnaround. A faster reply shortens each wait and leaves the number of rounds untouched.
 - **B.** It applies to all email rather than only to the messages that belong to a project — Scoping the rule more narrowly would make it more palatable and would not change what is producing the delay, which is the number of exchanges rather than which messages they are.

- **C.** It cannot be enforced, because nobody can be required to read email at a particular time — Enforceability is a fair practical objection and a secondary one. Even a perfectly obeyed rule would leave a question needing four exchanges taking most of a day.
- **D.** It will slow other work, because staff will interrupt what they are doing to answer messages — Interruption cost is real and is an argument about the price of the rule rather than about whether it addresses the cause of the delay.

2. **5.1** A hypothetical logistics firm, Pellworth Freight, buys videoconferencing systems for all five of its depots and reports a year later that collaboration has not improved. Most of the work its planners do involves handing over partly finished schedules between shifts and time zones. What best explains the disappointment?

- **A.** The equipment was too expensive to be used casually, so staff booked it only for formal meetings — Formality does discourage casual use, and that would still leave the systems well matched to the meetings that were held. It does not explain why the work itself did not improve.
- **B.** Five depots are too few for a network effect, so the systems went largely unused — Network effects govern tools whose value rises with participation. A meeting between two depots works perfectly well with two depots taking part.
- **C.** The purchase serves scheduled same-time meetings, and most of that work is different-time work — **correct.** Handing partly finished work between shifts is the different-times mode. Equipment built for a same-time meeting between fixed locations does not touch it, however good the equipment is.
- **D.** Videoconferencing is a conferencing tool, and planning needs collaboration management tools — The three families are not ranked, and a planning function draws on all of them. Naming the family is closer to the problem than the others and still misses the time dimension that explains it.

3. **5.1** A hypothetical hospital group is told by its ward managers that clinical staff will not use the proposed employee portal, because there is no machine available to them mid-shift. The project manager records this as resistance to change and proceeds. What has been misread?

- **A.** The objection is about access rather than attitude, and it is the best information available — **correct.** The managers are reporting a physical condition of the work. Filing that under resistance discards a solvable design constraint and converts a fixable problem into a grievance later.
- **B.** The objection is about training, which the project should budget for before proceeding — Training answers a capability gap. Nobody here has said staff cannot use a browser; they have said there is nothing to use one on.
- **C.** The objection is about seniority, since ward managers speak for a group they do not represent — Whether managers speak for their staff is worth checking and does not change the content of what they said, which is a claim about equipment rather than about feelings.
- **D.** The objection is about timing, and the portal should launch after the next roster change — A launch date does not create a machine on a ward. Delay postpones the same failure to a later month.

4. 5.1 A hypothetical manufacturer's intranet has grown to about ninety thousand documents. Employees increasingly stop searching it and ask a colleague instead. What does the chapter identify as the underlying difficulty?

- **A.** Employees prefer human answers, so any search tool will be underused whatever its quality — There is a genuine preference for asking somebody, and the chapter treats the search failure as the cause rather than as an unavoidable human trait.
- **B.** Internal content sits in systems with no linking signal for a search engine to rank by — **correct.** The chapter contrasts enterprise search with web search precisely here: internal content lives in documents, databases and applications rather than in pages that link to one another, so the signal a web engine ranks by does not exist.
- **C.** Ninety thousand documents is beyond what an internal search index can practically hold — Index size is not a limit at this scale. Public engines index vastly more, and the difficulty the chapter names is about ranking rather than capacity.
- **D.** The documents were never tagged, so a search engine has nothing to match a query against — Full-text search matches untagged documents perfectly well. Tags help, and their absence is not what makes the enterprise problem different in kind.

5. 5.1 A hypothetical charity wants its four regional offices to stop keeping separate copies of its safeguarding procedure, which currently exists in three versions. Which capability addresses the actual cause?

- **A.** Enterprise search, so staff can find whichever version is nearest to hand — Finding a version faster is not the same as there being one version. Search over three copies returns three copies, which is the problem restated.
- **B.** A virtual private network, so every office reaches the same server — A private network governs who can reach the network and from where. Four offices already connected can still keep four separate documents.
- **C.** Real-time access to one authoritative document held in a single place — **correct.** The chapter's first intranet benefit exists for this: a paper or filed document has to be updated, reprinted and redistributed, so older copies keep circulating, and a single updated page has no older copies.
- **D.** A social intranet, so any member of staff can correct the procedure directly — Open editing raises how much knowledge reaches the intranet and would let four people edit four copies just as easily. It does not by itself establish a single source.

6. 5.1 A hypothetical engineering consultancy assembles a project team from three offices. The project manager schedules a daily meeting at an hour that suits two of the three. Six weeks in, attendance from the third office has quietly stopped. What does this illustrate?

- **A.** Technological inertia, since the third office prefers the tools it was already using — Inertia describes refusing to move to a new tool. Nobody here has refused anything; they attended for six weeks and then stopped.

- **B.** A generation gap, since the third office has a longer-serving workforce — Length of service might correlate with anything and nothing here measures it. Attributing the drop-off to age is an assumption rather than a reading of the situation.
- **C.** A critical mass problem, since three offices are too few to sustain a daily meeting — Critical mass governs tools whose value depends on how many people take part. A scheduled meeting between three offices does not need a threshold to be useful.
- **D.** The uneven cost of synchronous work across time zones — **correct.** A same-time meeting across time zones has no hour convenient everywhere, so the cost falls on one group. Attendance decaying rather than being refused is exactly how that cost shows up.

7. **5.2** A hypothetical retailer, Ashmere Home Supply, adds customer reviews to its product pages and its returns fall by a fifth. Its chief executive describes this as a technology improvement. What is a more accurate description?

- **A.** A shift in who creates the content, which supplied information the firm could not write — **correct.** This is the Web 2.0 shift stated exactly: the audience became the authors, and owners know things about fit, bulk and suitability that no staff writer working from supplier data can supply.
- **B.** An improvement to product descriptions, achieved at lower cost than hiring more writers — The cost comparison is real and describes the saving rather than the mechanism. Ten more writers with the same source material would not have produced the same information.
- **C.** A marketing improvement, since reviews increase the persuasiveness of a product page — Persuasion is one effect, and a returns fall is evidence that buyers chose more accurately rather than more enthusiastically, which is a different thing.
- **D.** A service improvement, since reviews answer questions that would have reached support — Deflecting support contacts is a genuine secondary benefit. The returns figure measures decisions made before purchase, not questions answered after it.

8. **5.2** A hypothetical trade publisher argues that it adopted Web 2.0 when it launched a daily news feed written by its own editors. What is wrong with the claim?

- **A.** A feed is a communication tool, and the later web is defined by conferencing tools — The three families of collaboration tools classify tools, not eras, and a conferencing tool used one-way would be just as one-way.
- **B.** The strict line between content creator and content consumer is exactly where it was — **correct.** The change the chapter describes is in who creates content. A feed written by editors and read by everybody else preserves precisely the arrangement the later web dissolved.
- **C.** Daily publication is not frequent enough to count as a dynamic web application — Frequency was never the distinction. A daily bulletin and a monthly one are both the organization talking while the audience listens.
- **D.** The publisher would need a social network presence before it could make the claim — Presence on a particular platform is not the test. A publisher could run reviews, a forum and a wiki entirely on its own site and have changed the relationship completely.

9. 5.2 A hypothetical outdoor brand posts progress photographs from a client site. Nobody types a location, yet a reader identifies the site within hours. Which distinction explains this?

- A. Between public and private posting, since the account's privacy settings were never adjusted — Privacy settings govern who can see a post. Everyone who could see this one could also read what was inside the files, whatever the setting.
- B. Between a tag and a hashtag, since hashtags make content findable to people outside the audience — Hashtags do widen reach and they are typed deliberately, so they are visible to anybody reviewing the post before it goes out.
- C. Between metadata a person adds and metadata a device attaches without being asked — **correct.** Coordinates, timestamps and device identifiers are attached automatically, which is why reviewing what somebody wrote catches nothing. The chapter's own example is a photograph that revealed where a man in hiding was.
- D. Between synchronous and asynchronous publishing, since immediate posting leaves no time to review — Timing affects how much review is possible and changes nothing about what is inside the file, which would be there a week later too.

10. 5.2 A hypothetical software firm embeds a video it hosted on a sharing service into a partner's blog post. Both parties benefit and neither negotiated anything. What is this an instance of?

- A. Collaboration, since two organizations produced a single piece of content together — Collaboration requires working the same problem together. Here two parties each did their own thing and both gained, which the chapter treats as a separate category.
- B. Cooperation, where one participant's success improves the chances of the other — **correct.** The chapter's definition is exactly this: a win-win where one participant's success improves the chances of the other, and the embedded video is its own example of it.
- C. Viral marketing, since the arrangement spreads the content through a network — Viral marketing depends on people choosing to pass content on. One deliberate embed between two known parties is not that mechanism.
- D. Open innovation, since an external party contributed to the firm's own output — Open innovation brings outside parties into a firm's research and development. Nothing was invented here; existing content was placed somewhere useful.

11. 5.2 A hypothetical news site's editor argues that because a complainant has only twenty-eight followers, a hostile post about the site can be ignored. What does the chapter's evidence suggest?

- A. Follower counts predict reach poorly, and the response to a post often outgrows the post — **correct.** The chapter's case is a tenant with a couple of dozen followers whose landlord sued her, at which point the dispute itself reached a national audience the original post never would have.
- B. Ignoring it is correct, since responding would draw attention the post had not earned — Proportionate response is a real consideration and it has to be decided on the merits rather than on the poster's following, which is the reasoning error the chapter illustrates.

- **C.** The post should be reported to the platform, since a small account is easier to have removed — Attempting removal is what converted the chapter's example from a small complaint into national coverage, so this is closer to the cause than to the cure.
- **D.** Follower counts matter, but only on microblogging services rather than on other platforms — Nothing in the chapter limits the effect to one kind of platform. The mechanism is that other people amplify, and that works everywhere.

12. **5.2** A hypothetical museum wants tags added to forty thousand digitised photographs so they can be found. Its curator proposes a crowdsourcing marketplace. Under what condition is that a good fit?

- **A.** When the budget is limited, since marketplace tasks cost a fraction of staff time — Cost is why people reach for it and not what makes it work. A cheap answer that is wrong is not cheaper than a correct one.
- **B.** When speed matters, since a marketplace completes in weeks what staff take months over — Speed is a genuine advantage of the arrangement, and it is available regardless of whether the task is one a stranger can actually do.
- **C.** When the tags describe what is visible rather than what a local would know — **correct.** The chapter's rule is that these tasks must be easy for any person and hard for a computer. Naming a street from 1974 is easy only for somebody who was there, which no fee can supply.
- **D.** When accuracy is not critical, since a marketplace trades quality for cost and speed — Accepting poor quality is not a condition for using crowdsourcing; it is what happens when the task was a bad fit. Well-matched tasks return accurate work.

13. **5.2** A hypothetical utility adopts a cloud collaboration suite. Eight months later a feature its field teams depended on is changed by the provider without notice, and the workflow around it breaks. Which of the chapter's four domains does this belong to?

- **A.** Information technology, since control of the tooling has moved to the provider — Loss of control over data and service quality is the chapter's technology risk, and it is about where things live rather than about when they change.
- **B.** Organization, since staff had no documentation or support when the change landed — Thin documentation and support is the organizational risk and it is a consequence here rather than the cause. The change happened first.
- **C.** Competition, since the disruption slowed the utility relative to its rivals — The competitive risk in the chapter's table concerns security and compliance enforcement. Being slowed down is an effect rather than the category.
- **D.** Upgrade cycles, since the provider now decides when and how the software changes — **correct.** The chapter's upgrade-cycle risk is stated in these terms: tools and features can change without notice, causing problems for users and for the firm's technology strategy.

14. **5.2** A hypothetical trade association runs a members' wiki. In a year it accumulates four hundred pages, of which nineteen have ever been edited by anybody other than their original author. What does that figure indicate?

- **A.** The pages are accurate, since incorrect content attracts correction quickly — Nothing establishes accuracy. Unedited could mean correct, unread by anybody who knew better, or not worth the trouble, and the figure cannot distinguish them.
- **B.** Members are reading rather than writing, which is normal and needs no response — The reading-to-writing ratio is indeed skewed everywhere, and this figure measures something narrower: whether the people who do write ever build on each other.
- **C.** Peer production is not occurring, so the wiki is functioning as a document store — **correct.** What makes a wiki more than a shared folder is that contributions improve on one another. Pages nobody touches after publication are documents in a different container.
- **D.** The version history is not being used, which suggests the wiki is misconfigured — Retained history makes editing safe rather than making it happen. The mechanism is available and the behaviour is not occurring.

15. **5.2** A hypothetical bank's internal social network is described by staff as clumsy compared with the apps they use outside work. What does the chapter call this pressure?

- **A.** Consumerization, where consumer products set the usability benchmark for internal tools — **correct.** The chapter names this directly: applications designed for the consumer market make their way into organizations, and users judge an internal tool against what they already use.
- **B.** Technological inertia, since people will not adopt a tool without a tangible benefit — Inertia is refusing to move at all. These staff have moved and are reporting that the destination compares badly, which is a different complaint.
- **C.** A generation gap, since younger staff hold higher expectations than older colleagues — A gap describes differing comfort across a workforce. Here the staff agree with each other, which is the opposite of a gap.
- **D.** Technological integration, since the tool has not been joined to existing systems — Integration governs whether a tool connects to other systems. A perfectly integrated tool can still be unpleasant to use.

16. **5.2** A hypothetical food producer runs a campaign whose video can only be watched after registering an email address. Sharing is minimal. Which of the chapter's viral marketing principles was broken?

- **A.** Do something unexpected, since a registration form is a familiar and tiresome device — Familiarity is a weakness of the campaign and not the mechanism that stopped it. An unexpected video behind a form would also fail to spread.
- **B.** Make sequels, since a single video gives an audience nothing to come back for — Sequels sustain a campaign over time. This campaign failed at the first hand-off rather than at the second piece of content.
- **C.** Make people feel something, since a form interrupts whatever the video established — Emotional content travels further and the form sits before the video rather than inside it, so it prevents the feeling rather than diluting it.

- **D.** Never restrict access, since a gate stops the sharing the whole mechanism depends on — **correct.** The mechanism is that each viewer passes the content to the next one. A gate breaks that at the first hand-off, whatever the content is like.

17. **5.2** A hypothetical firm asks why a review platform it does not own can influence which restaurants near it succeed, given that the platform writes no reviews and takes no photographs. What is the answer?

- **A.** It holds the reviews, and each new participant makes the collection more useful to everyone — **correct.** This is the network effect doing the work. The platform's asset is participation it did not create, and each contribution makes the whole more valuable to every user, which is what a competitor cannot buy.
- **B.** It advertises heavily, which is why consumers consult it rather than a competitor — Advertising follows the position rather than producing it, and a well-advertised platform with no reviews would have nothing to consult.
- **C.** It is preinstalled on most devices, so consumers reach it before any alternative — Integration with device software is real and downstream. It happened because the data was already worth reaching, not the other way round.
- **D.** It is more accurate than the alternatives, having invested heavily in verifying that its reviews come from real customers — Nothing establishes superior accuracy, and the chapter records criticism about how reviews are handled, so this claim is unsupported.

18. **5.3** A hypothetical charity introduces an internal discussion platform. Every post is approved by a manager before it appears, and after four months there is almost no discussion. Which factor is this?

- **A.** Critical mass, since too few people joined for the platform to feel worth using — Thin participation is the symptom being observed. The question is what produced it, and an approval queue is a cause rather than a count.
- **B.** Technological integration, since the approval step was not built into the workflow — The approval step is working as designed. Building it more smoothly into the workflow would make a discouraging arrangement more efficient.
- **C.** Culture, since open communication collides with a structure built on control — **correct.** The chapter says social applications do not do well within traditional top-down structures based on rigid hierarchies and control, and pre-approval is that structure applied to conversation.
- **D.** Technological inertia, since staff continue to use the tools they already know — Inertia would show as people staying with email. Here they came to the platform and found their posts held, which is a different failure.

19. **5.3** A hypothetical insurer's pilot of an enterprise social network in one department is judged a success. The organization-wide rollout stalls at low participation. What did the pilot fail to test?

- **A.** The software's ability to handle the whole organization's load rather than one department's — Capacity is testable and rarely the issue for this class of tool. Nothing described points at performance.

- **B.** Whether the wider organization has the culture and processes a department volunteered — **correct.** The chapter warns that department-led pilots often work while organization-wide implementation needs changes in culture and processes and often the flattening of hierarchies. The pilot measured a self-selected department.
- **C.** Whether the tool integrates with the systems used outside the pilot department — Integration gaps are a real risk and would show as friction rather than as absence. People would complain about the tool rather than not appearing.
- **D.** Whether the training approach would scale to every location the insurer operates in — Training scale matters when capability is the constraint, and the pilot showed that people are capable. What is missing is a reason to participate.

20. **5.3** A hypothetical distributor discovers that most useful discussion has moved to a messaging tool one team installed itself, connected to nothing. What is the strongest response?

- **A.** Block it, since an unapproved tool creates exposure the firm has not assessed — The exposure is genuine, and blocking moves the conversation somewhere the firm cannot see rather than back onto the official platform.
- **B.** Ignore it, since the official platform is the one the firm is measuring — Ignoring it makes the firm's own measurement misleading and leaves two disconnected records of how decisions were made.
- **C.** Migrate the team onto the official platform and close the unapproved tool — Moving a working conversation onto a platform that has already failed to attract one changes the location and nothing about why.
- **D.** Treat it as evidence of what the work needs, and integrate rather than prohibit — **correct.** It is the clearest signal available about what people actually need, and the chapter's integration point applies: a tool joined to nothing becomes another place to look, which is what the official platform has become.

21. **5.3** A hypothetical brand discovers its agency bought favourable reviews for its product. A journalist has asked for comment by tomorrow. Which response best matches the chapter's playbook?

- **A.** State that the firm commissioned no reviews, since that is the accurate position — The blame point is arguable and answers a question the public is not asking, which is whether the ratings mean anything now.
- **B.** Respond publicly today, say what was found and what is being changed — **correct.** The chapter's playbook says the first day counts and asks for a response that is fast and appropriate, and its most successful example named what was found and what was being changed.
- **C.** Wait until an independent analysis has verified the review flagging — Verification is prudent and slower than the story. The chapter's cautionary case is a company that waited two days and had to recover from its own silence.
- **D.** Ask the marketplace to remove the flagged reviews before commenting — Removing the evidence before commenting reads as concealment whatever the intent, and the flagged reviews already appear in a report the journalist holds.

22. 5.3 A hypothetical airline monitors social media for mentions of its own brand name and nothing else. What does the chapter say it will miss?

- **A.** Complaints where the brand name is misspelled or abbreviated by the poster — Misspellings are a genuine gap and a technical one that fuzzy matching handles routinely. It is not the gap the chapter identifies.
- **B.** Coverage in traditional media, which social monitoring tools do not index — Traditional coverage is generally indexed and is downstream in any case. By the time a story is in the press the early signal was already missed.
- **C.** Discussion using the terms an opponent would choose rather than the airline's own — **correct.** The chapter asks firms to identify their worst nightmare and know the signs to look for, including the search terms their opposition could use, which are rarely the firm's own vocabulary.
- **D.** Anything posted on platforms where the airline does not itself hold an account or a published page — Monitoring tools read public content regardless of whether the firm has an account. Absence limits where a firm can reply rather than what it can see.

23. 5.3 A hypothetical restaurant group's board asks why it should assemble a crisis team now, when nothing has happened. What is the strongest answer?

- **A.** Because deciding who may speak, and with which adviser, cannot be done while a story is running — **correct.** The chapter puts the crisis team first among its preparations precisely because the first day decides the shape of a story, and authority questions settled in advance take minutes rather than a morning.
- **B.** Because the team will identify risks in the business that nobody has yet noticed — Risk identification is the second preparation, naming your worst nightmare, and it is a different activity from establishing who speaks.
- **C.** Because insurers and regulators increasingly expect a documented crisis process to be in place before an incident — Insurance and governance expectations may well exist and are external reasons rather than the operational one the chapter gives.
- **D.** Because the team can monitor social media more cheaply than an external service — Monitoring is the third preparation and is usually a tool or a service rather than something a crisis team does by hand.

24. 5.3 A hypothetical council debates whether anonymous accounts should be permitted on its public consultation forum. Which framing matches the chapter's treatment?

- **A.** Anonymity should be removed, since it is the main cause of abusive behaviour online — The chapter documents deindividuation and the harassment that follows it and opens by naming legitimate needs for anonymity, so a one-sided removal misstates its position.
- **B.** Anonymity should be preserved, since the harms are outweighed by convenience for users — Convenience is not the value the chapter identifies. It names privacy protection and freedom from oppression, which is a much stronger claim than convenience.

- **C.** It is a dilemma: anonymity protects privacy and dissent while enabling harassment — **correct.** The chapter presents both sides: legitimate need for privacy and freedom from oppression on one, and deindividuation, trolling and cyberharassment on the other, which is what makes it a dilemma rather than a decision.
- **D.** It is a technical question, since anonymity cannot be removed on this network anyway — The chapter does note that the design makes anonymity hard to remove, and difficulty is a practical constraint rather than a reason the question is finely balanced.

25. **5.3** A hypothetical software vendor is deciding what to say after resolving a widely shared complaint. Which closing action does the chapter's most successful example support?

- **A.** Publish an apology from the chief executive and treat the matter as closed — An apology without a described change invites the same complaint again, and a second occurrence is worse because the record of the first is permanent.
- **B.** Increase advertising for a period so that attention moves to other messages — Buying attention against an unresolved complaint is expensive and reads as exactly what it is, and the material stays retrievable regardless.
- **C.** Introduce a review process so no employee can post about the firm publicly — That answers a problem the firm did not have when the complaint came from outside, and it imposes a control people route around.
- **D.** Fix the cause and describe specifically what changed and how it was verified — **correct.** The chapter's example named what had been found, said the site had been closed and cleaned, and described what would change, and a survey afterwards found the response had largely restored trust.

26. **5.4** A hypothetical consultancy writes: our clients are very active on review sites, therefore buyer power is high. What does the claim lack?

- **A.** Evidence that clients can act on the firm, such as switching, price pressure, or changed behaviour — **correct.** Buyer power is leverage. Activity is not leverage, and the sentence would be equally true of clients who cannot switch, which is where the reasoning fails.
- **B.** A comparison against the industry average level of buyer power in professional services — Benchmarking adds useful context and does not repair an unsupported claim. A claim with no evidence is unsupported at any benchmark.
- **C.** An acknowledgement that review activity is a substitutes question rather than a buyer-power one — The observation concerns the firm's own clients, so filing it under substitutes would replace one unsupported claim with a misclassified one.
- **D.** A distinction between clients who post reviews and clients who only read them — That distinction would sharpen the evidence and is a refinement of a claim that currently has none rather than the missing element itself.

27. **5.4** A hypothetical kit maker's nine-year user forum absorbs most of its support questions and is cited by four in ten new buyers. A larger competitor launches a similar product with no community. Which force does the forum principally act on?

- **A. Rivalry**, since it is how the firm differentiates against this particular competitor — Differentiation is what the forum does today and misses why it works. A rival can copy a feature and cannot copy nine years of accumulated answers.
- **B. The threat of new entrants**, since it is a barrier money cannot buy — **correct**. A network-effect asset that grows with participation and cannot be purchased quickly is a barrier to entry, which is what explains a larger, cheaper competitor not automatically winning.
- **C. Bargaining power of buyers**, since forum members are less sensitive to price — Nothing establishes a relationship between posting and price sensitivity, and inventing one to reach a force is the unsupported move the supplement warns against.
- **D. The threat of substitutes**, since the forum substitutes for the firm's paid support — The forum does absorb support work, and a substitute is something a customer uses instead of buying from the industry at all rather than a part of what they bought.

28. 5.4 A hypothetical wholesaler wants to place an internal engineering wiki in the value chain. Where does it belong, and why?

- **A. Firm infrastructure**, since it is a shared capability every part of the business draws on — Infrastructure covers capabilities like the intranet, governance and finance. A wiki holding engineering decisions serves one function rather than all of them.
- **B. Operations**, since it makes the production of each order more efficient — Operations transforms inputs into outputs. A wiki changes the design that operations then executes, which is a step earlier.
- **C. Human resource management**, since it captures what employees know before they leave — Retention of knowledge sounds like a people question and is about employment terms and staffing rather than about how the firm designs its products.
- **D. Technology development**, since it changes how fast and how well the next thing is produced — **correct**. What an engineering wiki changes is the firm's ability to produce the next design well and quickly, which is what the technology development activity covers.

29. 5.4 A hypothetical publisher takes forty percent of its orders through one marketplace, which has raised its commission twice. Which force is this, and what follows?

- **A. Supplier power**: the channel is an input, and the answer is to build a relationship it does not intermediate — **correct**. The platform supplies access to customers, which makes it an input, and its power grows with the share of revenue it carries. The response is to own a customer relationship the platform does not sit inside.
- **B. Rivalry**: other publishers on the same marketplace are competing the margin away — Rivalry on the platform may well be intense and would show as price pressure rather than as the platform itself taking more.
- **C. Buyer power**: the marketplace's customers are extracting better terms from the publisher — The marketplace's shoppers are not negotiating anything. It is the intermediary rather than the buyer that changed its terms.

- **D.** New entrants: the marketplace has lowered the cost of entry for competing publishers that hold no distribution of their own — The marketplace does lower entry costs for competitors, which is a separate and real force. It does not explain a commission increase on this publisher's own orders.

30. **5.5** A hypothetical analyst is asked why a recommendation must name a specific value-chain activity rather than saying the firm needs a better digital strategy. What is the substantive reason?

- **A.** The framework requires every recommendation to reference one of its nine activities — Satisfying a framework is ceremony. If naming the activity added nothing, the requirement would be worth dropping.
- **B.** It turns a diagnosis into work that can be assigned, costed and measured afterwards — **correct.** A force says what is pressing on the firm; an activity says who can be given the work, what it will cost and what to watch afterwards, without which nothing can be acted on or judged.
- **C.** Support activities are cheaper to change than primary activities, so the choice matters — Cost varies enormously within both categories, and this chapter contains cheap primary-activity changes and expensive infrastructure ones.
- **D.** Forces describe the industry while activities describe the individual firm being advised — The distinction between industry-level and firm-level analysis is accurate as a description of the two frameworks and is not why a recommendation needs a location.

31. **5.5** A hypothetical practice has four candidate initiatives and is asked for one. Its analyst submits all four, ranked, so the partners can choose by budget. What is the analytical objection?

- **A.** Four is too many to cost properly in the time available for the analysis — Time is a practical constraint and would be solved by working longer. The objection stands even with unlimited time.
- **B.** Ranking implies a scoring method the analyst has not made explicit — An unexplained ranking is a fair criticism and a fixable one. Making the method explicit would not repair the deeper problem.
- **C.** It hands the decision back and hides which pressure was being answered — **correct.** A single recommendation can be costed, argued and later judged. A ranked list looks generous, hands the decision back, and leaves no option properly costed or tied to a diagnosed force.
- **D.** The partners will choose the cheapest, which is rarely the strongest option — Budget-driven selection is a plausible outcome and describes what the partners might do rather than what is wrong with the submission.

32. **5.5** A hypothetical analyst's comparison summary spends most of its two hundred words describing the history and scale of the company chosen for comparison. What has gone wrong?

- **A.** The comparison company is too large, so the parallel with the advised firm is weak — Size mismatch can weaken a comparison and a well-drawn lesson can survive it. A small firm can learn something specific from a very large one.

- **B.** The summary is descriptive where it should end in advice to the firm being advised — **correct.** The comparison exists to produce a lesson for the firm in front of you, and every sentence should be recoverable as advice to it. A profile answers a question nobody asked.
- **C.** Two hundred words is insufficient to profile a company and draw a lesson from it — The length is a constraint that forces the choice rather than causing the error, and a summary of this length is ample for a lesson if the profile is left out.
- **D.** The company should have been a direct competitor rather than one from another sector — A contrast from another sector is often the more useful comparison. What matters is the lesson, not the sector.

33. **5.5** A hypothetical practice measures a median of four and a half days to answer a cross-office technical question and sixty-two design decisions re-derived in a year because nobody found the earlier one. How should the analysis treat these two figures?

- **A.** As one operations problem, since both describe project delivery running slower than it should — Calling both operations captures the delay and loses the second figure. Re-deriving a decision is not slow delivery; it is work being done that had already been done once.
- **B.** As one technology development problem measured at two different scales — Both touch how well the firm produces the next thing, and merging them hides that they need different remedies: reaching a person, and reaching a record.
- **C.** As two distinct problems: a coordination cost in delivery, and knowledge the firm cannot reach — **correct.** They are genuinely different activities with different fixes, and the written analysis asks for at least two value-creating areas, which these two supply honestly rather than by describing one twice.
- **D.** As one human resource problem, since both describe how staff work with one another — Human resource management covers recruiting, developing and administering people. Neither figure concerns employment terms, and filing them here puts the recommendation where nobody could act on it.

34. **5.4** A hypothetical manufacturer, Denmoor Castings, tells its board that its internal engineering wiki gives it competitive advantage because none of its rivals has one. What does that claim still need?

- **A.** A count of how many rivals have something comparable, gathered from their published material — Counting rivals establishes that the firm has something unusual. Unusual and advantageous are different claims, and plenty of things no competitor has are worth nothing.
- **B.** A statement of which pressure the wiki relieves, and the evidence that it does — **correct.** An asset becomes an advantage only when it answers a pressure. Naming the force, and the evidence that the wiki moves it, is what turns a possession into an argument somebody can test.
- **C.** A costing of the wiki, so the advantage can be set against what it was built for — A cost belongs in the recommendation and says nothing about whether an advantage exists. A cheap thing that relieves no pressure is still worth nothing.

- **D.** A comparison of the wiki against the commercial products the firm could have bought — Product comparisons decide what to build. They are a procurement question, and they arrive after somebody has established that the capability is worth having.

35. 5.4 A hypothetical recruiter-heavy consultancy runs a professional-network campaign that halves its cost of hiring an experienced engineer. Where does that land in the value chain, and does it matter which activity you name?

- **A.** Marketing and sales, and naming the activity matters mainly for presentation — Marketing and sales acts on somebody deciding whether to buy. A candidate is not buying anything, and treating the naming as cosmetic gives up the reason for doing it.
- **B.** Operations, because a shorter hiring cycle means projects start sooner — Projects do start sooner, which is a downstream effect. Operations is where inputs are turned into outputs, and hiring sits before that rather than inside it.
- **C.** Human resource management, and naming it decides who owns the work and what is measured — **correct.** Recruiting is human resource management, and naming it puts the work with the people who do it and fixes what gets measured, which is what makes a recommendation actionable rather than decorative.
- **D.** Firm infrastructure, because recruiting is a shared capability the whole firm draws on — Infrastructure covers planning, finance, legal and the systems everything draws on. Recruiting is a distinct support activity with its own name in the framework.

36. 5.5 A hypothetical analyst rates four of the five forces on a firm and skips supplier power, on the grounds that a professional services firm has no real suppliers. What is wrong with that?

- **A.** Nothing, provided the omission is stated; a force that does not apply need not be rated — Declaring an omission is better than hiding one and still leaves the analysis incomplete. A force rated low with evidence is a finding; a force skipped is a gap.
- **B.** The firm does have suppliers, since it buys software, premises and professional insurance — Those purchases are genuine supplier relationships and they are the least interesting ones, because none of them has leverage over the firm worth writing about.
- **C.** All five are asked for, and the platforms and labour markets it depends on are suppliers — **correct.** The deliverable asks for all five, and in a services firm the powerful inputs are the directory or platform that brings enquiries and the labour market for scarce specialists, both of which can dictate terms.
- **D.** Supplier power should be inferred from buyer power, which the analyst has already rated — The two forces are independent. Strong buyers tell you nothing about whether your inputs are concentrated, and inferring one from the other invents a finding.

37. 5.5 A hypothetical analyst offers marketing and sales as the two value-chain areas where technology could create an advantage, describing an advertising change for one and a lead-handling change for the other. What is the objection?

- **A.** The two changes are too similar in cost, so the comparison between them is uninformative — Similar costs would make the choice between them harder and would not affect whether two distinct areas had been found, which is the requirement in question.

- **B.** Marketing and sales is a single activity, so only one area has actually been identified — **correct.** Marketing and sales is one primary activity in the framework, so describing two changes inside it identifies one area twice. The deliverable asks for at least two genuinely different activities.
- **C.** Neither change is technological, so the areas do not answer what was asked — Both changes could be delivered by technology, and whether they are is a separate question from whether they sit in two different places in the value chain.
- **D.** Advertising belongs to marketing and lead handling belongs to service, so both are misfiled — Lead handling belongs with marketing and sales rather than with service, which acts on somebody who has already bought. Misfiling is not the problem here.

What to do next. Spend a week watching one organization you already belong to — an employer, a team, a club, this course — and write down where its coordination actually breaks. Not the tool it uses; the moment somebody could not find what somebody else already knew, or waited a day for an answer that took ten seconds to give. Then name the force that moment is evidence of, put it somewhere in the value chain, and say what one initiative would change and how you would tell whether it worked. That is the whole method, and it is far easier to practise on an organization you can actually see.